SEAGIS

Database Architecture, Data Lineage and Contracts

English

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Code and historical-snapshot audit. Current runtime is unverified.

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Status: code and historical-snapshot audit; current runtime UNVERIFIED.

Label	Meaning
VERIFIED	Supported by code/inspection; not automatically runtime-tested
INFERRED	Reasonable interpretation, not explicit
UNVERIFIED	Not established with available access
NOT FOUND	Searched within the stated scope, not found
NOT APPLICABLE	Does not apply to component

1 Executive Summary

This audit documents inspectable backend and local consumer implementations as of 27 October 2026. One application PostgreSQL database is identified through migrations and the 22 October catalog snapshot: six schemas, 28 tables, one view, 268 columns including nine view columns, and 47 foreign keys. No live database connection was made in this audit; row counts, current deployment, backups, role membership and post-snapshot drift remain UNVERIFIED.

A1 accepts temperature packets through an Edge Function, derives identity from credentials, atomically stores raw provenance, observations and QC, and recognizes semantic retries. A2 retrieves four regional datasets, preserves grids and native cadence, and persists immutable vintages. A3 builds hourly contexts using availability cutoffs and saves review snapshots. A3 is not a production feature builder or disease model.

The Seagis app and AppPOC prototype were inspected as separate consumers. MockAssessmentRepository produces synthetic assessments; no production assessment-to-backend-database path was found. QC state, regional model data and demo content must not be treated as ice-ice diagnosis or probability. This document is a versioned code-audit reference, not production-system certification.

Main implementation evidence: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

2 System and Data Architecture Overview

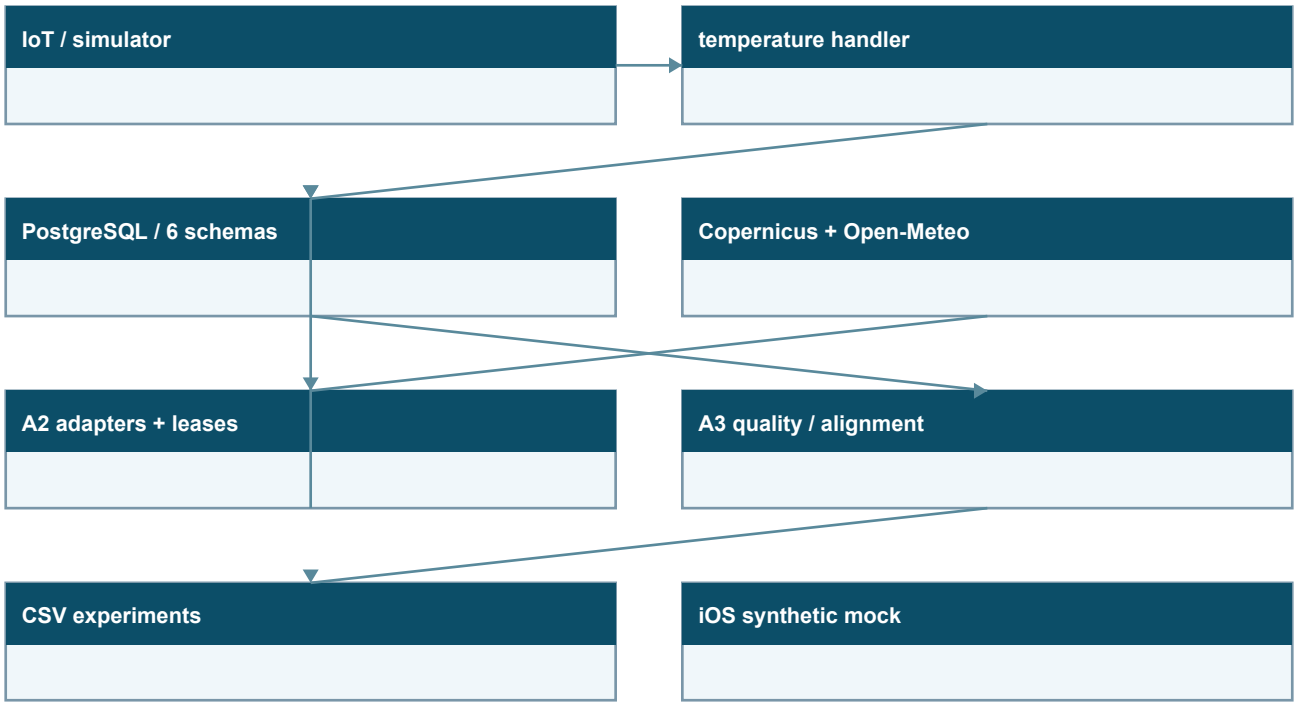
PostgreSQL/Supabase is the backend persistence defined by code. Schema seagis owns registries, observations, forecasts and provenance; seagis_raw holds raw manifests; seagis_api bounds the IoT gateway; seagis_environment manages regional workers; seagis_quality stores QC reviews; seagis_features supplies containers not connected to a production feature builder. Supabase internal schemas such as auth/storage are not inventoried as application-owned: current catalog access is unavailable and they must not be assumed empty.

Python retrieves provider data outside the database. A2/A3 workers call stored functions through psycopg; the TypeScript handler invokes only ingest/history/status functions. No ORM was found on these paths. Mutation transactions live in SQL functions; fetch claiming is a separate transaction from persistence so network calls do not hold a database transaction.

The diagram contains disconnected components: the iOS mock has no backend edge. Future integration remains UNVERIFIED. Production Core ML, distributed caches, warehouses, queues and search indexes were NOT FOUND in inspected implementation files; that does not establish absence of services outside these checkouts.

Main implementation evidence: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Reproducible diagram: assets/diagrams/architecture.mmd.



3 Database and Schema Inventory

PostgreSQL/Supabase is the backend persistence defined by code. Schema seagis owns registries, observations, forecasts and provenance; seagis_raw holds raw manifests; seagis_api bounds the IoT gateway; seagis_environment manages regional workers; seagis_quality stores QC reviews; seagis_features supplies containers not connected to a production feature builder. Supabase internal schemas such as auth/storage are not inventoried as application-owned: current catalog access is unavailable and they must not be assumed empty.

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Store	PostgreSQL / Supabase
Evidence status	VERIFIED (repository + historical catalog)
Purpose/environment	Backend; current deployment environment UNVERIFIED
Owner	Seagis-Model
Scope	Registry, sensor, regional, QC, features
Access	read/write via stored functions
Authority	Authoritative for persisted records; not a diagnosis
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1

Store	Supabase Storage
Evidence status	configured / UNVERIFIED usage
Purpose/environment	Local storage enabled; object_uri supported, application bucket NOT FOUND
Owner	Supabase
Scope	objects / metadata_only
Access	configured
Authority	Not proven to be active raw storage
Evidence	Seagis-Model/supabase/config.toml:116

Store	UserDefaults / launch arguments
Evidence status	VERIFIED code
Purpose/environment	Preference/debug UI access; not plot persistence
Owner	AppPOC
Scope	initialPlot / initialFarmer / skipCaptureSteps / detailDay
Access	read preference/argument values
Authority	Not authoritative for backend data
Evidence	AppPOC/POCApp/Features/FarmMap/ViewModels/FarmMapViewModel.swift:1

Store	legacy AR files
Evidence status	VERIFIED cleanup code
Purpose/environment	ar-site.json/ar-site.worldmap are only deleted at new AR session start
Owner	AppPOC
Scope	legacy local AR files

Access	best-effort delete; active writer NOT FOUND
Authority	Not active persistence
Evidence	AppPOC/POCApp/Features/PlotCapture/ViewModels/ARCaptureViewModel.swift:1

Store	JSON / CSV files
Evidence status	VERIFIED code
Purpose/environment	Synthetic fixtures, policies, experimental exports and manifests
Owner	Python scripts
Scope	fixtures/config/export
Access	file read/write
Authority	Authoritative only for that capture/fixture version
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

Store	in-memory Swift arrays
Evidence status	VERIFIED code
Purpose/environment	Prototype state; lost at process termination
Owner	AppPOC
Scope	Farmer / CultivationPlot
Access	array append/remove
Authority	Local demo state
Evidence	AppPOC/POCApp/Shared/Data/FarmStore.swift:1

Supabase auth, vector and analytics service defaults appear in local configuration, but active application buckets/indexes/datasets are not established. These are service settings, not additional application databases. All table row counts are UNVERIFIED; fixtures are not used as runtime estimates.

Schema	seagis
Objects	dataset, dataset_parameter, forecast_run, forecast_value, ingestion_run, observation, parameter, plot, plot_grid_mapping, quality_event, sensor, source, source_grid
Tables	13
Views	0
Access boundary	Access through backend routines; live membership UNVERIFIED

Schema	seagis_api
Objects	credential, packet_receipt, reader_plot, request_event, temperature_policy
Tables	5
Views	0
Access boundary	Access through backend routines; live membership UNVERIFIED

Schema	seagis_environment
Objects	attempt, current_speed, demo_grid_mapping, policy, schedule_state
Tables	4
Views	1
Access boundary	Access through backend routines; live membership UNVERIFIED

Schema	seagis_features

Objects	feature_input, feature_snapshot
Tables	2
Views	0
Access boundary	Access through backend routines; live membership UNVERIFIED

Schema	seagis_quality
Objects	alignment_snapshot, policy, snapshot_input
Tables	3
Views	0
Access boundary	Access through backend routines; live membership UNVERIFIED

Schema	seagis_raw
Objects	raw_payload
Tables	1
Views	0
Access boundary	Access through backend routines; live membership UNVERIFIED

Physical object	seagis.dataset
Purpose	Dataset identity, version, cadence, licence and ingestion gates.
Columns	15
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Physical object	seagis.dataset_parameter
Purpose	Source-variable to parameter contract and conversion rules.
Columns	9
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Physical object	seagis.forecast_run
Purpose	Regional model capture/vintage; unknown publication time remains null.
Columns	13
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Physical object	seagis.forecast_value
Purpose	Parameter value per run, grid and valid time; not duplicated per plot.
Columns	20
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Physical object	seagis.ingestion_run
Purpose	Audit of a logical ingestion and its completion state.
Columns	9
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Physical object	seagis.observation
Purpose	Local sensor water-temperature readings, retaining native values, QC and raw lineage.
Columns	19
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Physical object	seagis.parameter
Purpose	Registry of parameters and normalized units.
Columns	3
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Physical object	seagis.plot
Purpose	Plot identity and survey status; nullable coordinates are not evidence of a surveyed location.
Columns	8
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Physical object	seagis.plot_grid_mapping
Purpose	Operational plot-to-grid mapping with effective and recorded times.
Columns	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Physical object	seagis.quality_event
Purpose	Append-only QC log for ingestion or one measurement.
Columns	11
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Physical object	seagis.sensor
Purpose	Temperature sensor deployment identity, plot, depth and active state.
Columns	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Physical object	seagis.source
Purpose	Registry of dataset providers.
Columns	3
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Physical object	seagis.source_grid
Purpose	Regional cell and depth-layer identity; distinct from cultivation plots.
Columns	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Physical object	seagis_api.credential
Purpose	API actor identity using token hashes, expiry and revocation.
Columns	8
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Physical object	seagis_api.packet_receipt
Purpose	Sensor-packet idempotency key and semantic digest.
Columns	6
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Physical object	seagis_api.reader_plot
Purpose	Plot allowlist for a reader credential.

Columns	2
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Physical object	seagis_api.request_event
Purpose	Authenticated API response audit and rate-limit basis.
Columns	6
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Physical object	seagis_api.temperature_policy
Purpose	Pilot engineering QC settings and request limit; not disease thresholds.
Columns	8
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Physical object	seagis_environment.attempt
Purpose	Audit of a fetch attempt, including duplicate and abandoned outcomes.
Columns	7
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Physical object	seagis_environment.current_speed
Purpose	Current-magnitude view joining components from the same run, grid and time.
Columns	9
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Physical object	seagis_environment.demo_grid_mapping
Purpose	Bridge from two fictional plots to real grids for preview, not training authorization.
Columns	11
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Physical object	seagis_environment.policy
Purpose	Regional fetch, licence, scheduler cadence, staleness and retention policy.
Columns	11
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Physical object	seagis_environment.schedule_state
Purpose	Mutable lease, due time, backoff and last-success state.
Columns	9
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Physical object	seagis_features.feature_input
Purpose	Foreign-key input lineage and cutoff guards for feature snapshots.
Columns	4
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Physical object	seagis_features.feature_snapshot
Purpose	Draft/sealed feature container with cutoff and hashes; no production builder found.
Columns	12
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Physical object	seagis_quality.alignment_snapshot
Purpose	Immutable hourly QC/alignment document; always training_authorized=false.
Columns	11
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Physical object	seagis_quality.policy
Purpose	Immutable registry of QC policy versions and builder code hashes.
Columns	5
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Physical object	seagis_quality.snapshot_input
Purpose	Per-hour measurement lineage of an alignment snapshot.
Columns	5
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Physical object	seagis_raw.raw_payload
Purpose	Immutable raw manifest: digest, hash basis, licence and retention mode.
Columns	14
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

4 Complete Data Dictionary

The seagis.finite_float domain is double precision rejecting NaN and infinity; column nullability remains separate. value_status present requires numbers; nonpresent statuses require null; valid_value/valid_support and SQL guards enforce cross-field rules. Precipitation mm is a one-hour total, salinity 1e-3 is not automatically PSU/ppt, and wind direction is the direction from in degrees. Type/null/default/constraint/index metadata comes from the 22 October catalog snapshot; SQL definitions were checked against the local 0001–0005 migration chain. View rows have no physical PK/FK even when fields originate from measurements. Examples are synthetic illustrations, not necessarily a valid complete INSERT row. NOT APPLICABLE means an attribute does not apply; UNVERIFIED means a specific meaning or derivation is not established.

4.1 seagis.dataset

Dataset identity, version, cadence, licence and ingestion gates.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	UNIQUE (dataset_key, scope)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

seagis.dataset.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'sensor_temperature_v1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.dataset_key

seagis.dataset.source_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (source_key) REFERENCES seagis.source(source_key)
Business / technical meaning	Source-provider key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'sensor' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.source_key

seagis.dataset.provider_product_id

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Provider product identity when available
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.provider_product_id

seagis.dataset.provider_dataset_id

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Original provider dataset identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'seagis-water-temperature' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.provider_dataset_id

seagis.dataset.dataset_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Pinned dataset version
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'v1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.dataset_version

seagis.dataset.scope

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Origin: local_sensor or regional_model
Valid values / checks	CHECK (scope = ANY (ARRAY['local_sensor'::text, 'regional_model'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'local_sensor' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.scope

seagis.dataset.native_cadence_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Native sample cadence, distinct from fetch interval
Valid values / checks	CHECK (native_cadence_seconds > 0)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 3600 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.native_cadence_seconds

seagis.dataset.resolution_lat_deg

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Latitude resolution in degrees when known
Valid values / checks	CHECK (resolution_lat_deg::double precision > 0::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.resolution_lat_deg

seagis.dataset.resolution_lon_deg

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Longitude resolution in degrees when known
Valid values / checks	CHECK (resolution_lon_deg::double precision > 0::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.resolution_lon_deg

seagis.dataset.source_timezone

Attribute	Definition
Type / nullable	text / false
Default / generated	'UTC'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Original source timezone context
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'UTC'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.source_timezone

seagis.dataset.licence_url

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Source licence reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'project-owned://sensor-data' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.licence_url

seagis.dataset.attribution

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Attribution text to preserve
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'SEAGIS; simulated until hardware commissioning' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.attribution

seagis.dataset.raw_storage_permitted

Attribute	Definition
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Raw-retention permission gate, not proof of stored objects
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE

Attribute	Definition
Example	true
Origin / transformation	caller / SQL default → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.raw_storage_permitted

seagis.dataset.real_ingestion_enabled

Attribute	Definition
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Gate for nonsynthetic ingestion
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	caller / SQL default → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.real_ingestion_enabled

seagis.dataset.metadata_checked_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Time source metadata was checked
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → '2026-10-21T00:00:00Z' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.metadata_checked_at

Constraints, indexes and triggers

constraints

```
UNIQUE (dataset_key, scope);
CHECK (native_cadence_seconds > 0);
PRIMARY KEY (dataset_key);
CHECK (resolution_lat_deg::double precision > 0::double precision);
CHECK (resolution_lon_deg::double precision > 0::double precision);
CHECK (scope = ANY (ARRAY['local_sensor'::text, 'regional_model'::text]));
FOREIGN KEY (source_key) REFERENCES seagis.source(source_key);
```

indexes

CREATE UNIQUE INDEX dataset_dataset_key_scope_key ON seagis.dataset USING btree (dataset_key, scope);
CREATE UNIQUE INDEX dataset_pkey ON seagis.dataset USING btree (dataset_key);

triggers

CREATE TRIGGER guard_dataset_identity BEFORE UPDATE ON seagis.dataset FOR EACH ROW EXECUTE FUNCTION seagis.guard_dataset_identity();

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.2 seagis.dataset_parameter

Source-variable to parameter contract and conversion rules.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (dataset_key, parameter_code)
Unique	UNIQUE (dataset_key, parameter_code, native_unit, unit, conversion_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

seagis.dataset_parameter.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'sensor_temperature_v1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.dataset_key

seagis.dataset_parameter.parameter_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit)
Business / technical meaning	Environmental parameter identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'water_temperature' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.parameter_code

seagis.dataset_parameter.source_variable

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Original provider variable name
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'water_temperature' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.source_variable

seagis.dataset_parameter.native_unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Original unit before normalization
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'degC' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.native_unit

seagis.dataset_parameter.unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit)
Business / technical meaning	Normalized value unit from registry
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'degC' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.unit

seagis.dataset_parameter.conversion_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Unit-conversion method identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'identity' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_code

seagis.dataset_parameter.conversion_factor

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	1
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Multiplier in value=native_value*factor+offset
Valid values / checks	CHECK (conversion_factor::double precision <> 0::double precision)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	caller / SQL default → 1 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_factor

seagis.dataset_parameter.conversion_offset

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	0
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Offset after multiplying by factor
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	caller / SQL default → 0 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_offset

seagis.dataset_parameter.temporal_support

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Temporal support: instantaneous, mean or sum
Valid values / checks	CHECK (temporal_support = ANY (ARRAY['instantaneous'::text, 'mean'::text, 'sum'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'instantaneous' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.temporal_support

Constraints, indexes and triggers

constraints

```
CHECK (conversion_factor::double precision <> 0::double precision);
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
UNIQUE (dataset_key, parameter_code, native_unit, unit, conversion_code);
FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit);
PRIMARY KEY (dataset_key, parameter_code);
CHECK (temporal_support = ANY (ARRAY['instantaneous'::text, 'mean'::text, 'sum'::text]));
```

indexes

```
CREATE UNIQUE INDEX dataset_parameter_dataset_key_parameter_code_native_unit_un_key ON
seagis.dataset_parameter USING btree (dataset_key, parameter_code, native_unit, unit,
```

```
conversion_code);
CREATE UNIQUE INDEX dataset_parameter_pkey ON seagis.dataset_parameter USING btree (dataset_key,
parameter_code);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.dataset_parameter FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.3 seagis.forecast_run

Regional model capture/vintage; unknown publication time remains null.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, run_key); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

seagis.forecast_run.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.id

seagis.forecast_run.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope); FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.dataset_key

seagis.forecast_run.scope

Attribute	Definition
Type / nullable	text / false
Default / generated	'regional_model'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Business / technical meaning	Origin: local_sensor or regional_model
Valid values / checks	CHECK (scope = 'regional_model'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'regional_model'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.scope

seagis.forecast_run.ingestion_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Source ingestion reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → ing (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.ingestion_run_id

seagis.forecast_run.raw_payload_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Source raw-manifest reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → raw_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.raw_payload_id

seagis.forecast_run.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.is_synthetic

seagis.forecast_run.run_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Semantic capture identity per dataset
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid

Attribute	Definition
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → batch->>'run_key' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.run_key

seagis.forecast_run.run_kind

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	forecast, analysis or analysis_forecast_unspecified
Valid values / checks	CHECK (run_kind = ANY (ARRAY['forecast'::text, 'analysis'::text, 'analysis_forecast_unspecified'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → CASE WHEN d.source_key='copernicus' THEN 'analysis_forecast_unspecified' ELSE 'forecast' END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.run_kind

seagis.forecast_run.provider_run_id

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Provider run ID only when actually available
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.provider_run_id

seagis.forecast_run.model_initialized_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Model initialization instant when known
Valid values / checks	CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.model_initialized_at

seagis.forecast_run.issued_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Verified provider publication instant; null when unknown
Valid values / checks	CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL)); CHECK (issued_at IS NULL OR issued_at <= ingested_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.issued_at

seagis.forecast_run.issue_time_basis

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Known/unknown publication-time basis
Valid values / checks	CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL)); CHECK (issue_time_basis = ANY (ARRAY['provider_verified'::text, 'unknown'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → 'unknown' (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.issue_time_basis

seagis.forecast_run.ingested_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant data became available to the system
Valid values / checks	CHECK (issued_at IS NULL OR issued_at <= ingested_at); CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → t (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.ingested_at

Constraints, indexes and triggers

constraints

```
CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL));
CHECK (issued_at IS NULL OR issued_at <= ingested_at);
CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at);
UNIQUE (dataset_key, run_key);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
UNIQUE (id, dataset_key, is_synthetic);
CHECK (issue_time_basis = ANY (ARRAY['provider_verified'::text, 'unknown'::text]));
PRIMARY KEY (id);
FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES
seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);
CHECK (run_kind = ANY (ARRAY['forecast'::text, 'analysis'::text,
'analysis_forecast_unspecified'::text]));
CHECK (scope = 'regional_model'::text);
```

indexes

```
CREATE INDEX forecast_run_asof ON seagis.forecast_run USING btree (dataset_key, ingested_at DESC);
CREATE UNIQUE INDEX forecast_run_dataset_key_run_key_key ON seagis.forecast_run USING btree
(dataset_key, run_key);
CREATE UNIQUE INDEX forecast_run_id_dataset_key_is_synthetic_key ON seagis.forecast_run USING btree
(id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX forecast_run_pkey ON seagis.forecast_run USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.forecast_run FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_run BEFORE INSERT ON seagis.forecast_run FOR EACH ROW EXECUTE FUNCTION
seagis.guard_run();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.4 seagis.forecast_value

Parameter value per run, grid and valid time; not duplicated per plot.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (forecast_run_id, source_grid_id, parameter_code, valid_at); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

seagis.forecast_value.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.id

seagis.forecast_value.forecast_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic)
Business / technical meaning	Regional model capture/vintage reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A2 adapter batch → persist run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.forecast_run_id

seagis.forecast_value.source_grid_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Regional cell and layer reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A2 adapter batch → grid lookup by dataset/key/synthetic (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.source_grid_id

seagis.forecast_value.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code); FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic); FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A2 adapter batch → leased dataset (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.dataset_key

seagis.forecast_value.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic); FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	A2 adapter batch → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.is_synthetic

seagis.forecast_value.parameter_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Environmental parameter identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A2 adapter batch → batch.values[].parameter (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.parameter_code

seagis.forecast_value.native_value

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Physical value in source units
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A2 adapter batch → batch.values[].value (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.native_value

seagis.forecast_value.native_unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Original unit before normalization
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	registered seagis.dataset_parameter par → par.native_unit (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.native_unit

seagis.forecast_value.value

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Normalized value; null follows value_status
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A2 adapter batch → batch.values[].value (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.value

seagis.forecast_value.unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Normalized value unit from registry
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	registered seagis.dataset_parameter par → par.unit (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.unit

seagis.forecast_value.conversion_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Unit-conversion method identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	registered seagis.dataset_parameter par → par.conversion_code (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.conversion_code

seagis.forecast_value.value_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Numeric availability, separate from quality
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A2 adapter batch → v.value_status (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.value_status

seagis.forecast_value.detection_limit

Attribute	Definition
Type / nullable	seagis.finite_float / true

Attribute	Definition
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Detection limit in normalized units when applicable
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A2 adapter batch → omitted -> SQL NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.detection_limit

seagis.forecast_value.valid_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Representative time of model value
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A2 adapter batch → v.valid_at::timestamptz (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.valid_at

seagis.forecast_value.ingested_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant data became available to the system
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A2 adapter batch → persist t = clock_timestamp() (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.ingested_at

seagis.forecast_value.temporal_support

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Temporal support: instantaneous, mean or sum
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	registered seagis.dataset_parameter par → par.temporal_support (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.temporal_support

seagis.forecast_value.interval_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	known, unknown or not_applicable interval status
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A2 adapter batch → v.interval_status (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.interval_status

seagis.forecast_value.period_start_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Mean/sum interval start when known

Attribute	Definition
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A2 adapter batch → v.period_start_at::timestamptz (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.period_start_at

seagis.forecast_value.period_end_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Mean/sum interval end when known
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A2 adapter batch → v.period_end_at::timestamptz (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.period_end_at

seagis.forecast_value.quality_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	QC state unreviewed, accepted, suspect or rejected; not a diagnosis
Valid values / checks	CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A2 adapter batch → default unreviewed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.quality_status

Constraints, indexes and triggers

constraints

CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision));
CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at));
FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);
FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic);
UNIQUE (forecast_run_id, source_grid_id, parameter_code, valid_at);
UNIQUE (id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]));
FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic);

indexes

CREATE INDEX forecast_grid_time ON seagis.forecast_value USING btree (source_grid_id, parameter_code, valid_at DESC);
CREATE UNIQUE INDEX forecast_value_forecast_run_id_source_grid_id_parameter_cod_key ON seagis.forecast_value USING btree (forecast_run_id, source_grid_id, parameter_code, valid_at);
CREATE UNIQUE INDEX forecast_value_id_dataset_key_is_synthetic_key ON seagis.forecast_value USING btree (id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX forecast_value_pkey ON seagis.forecast_value USING btree (id);

triggers

CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.forecast_value FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_forecast_value BEFORE INSERT ON seagis.forecast_value FOR EACH ROW EXECUTE FUNCTION seagis.guard_measurement();

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.5 seagis.ingestion_run

Audit of a logical ingestion and its completion state.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, request_key); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Reader	A3 source_bundle; provenance guards
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

seagis.ingestion_run.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key

Attribute	Definition
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.id

seagis.ingestion_run.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.dataset_key

seagis.ingestion_run.request_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Logical ingestion retry key per dataset
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'a1:'||s.id||':'||key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.request_key

seagis.ingestion_run.adapter_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Ingestion adapter implementation version
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'temperature-1.0.0' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.adapter_version

seagis.ingestion_run.started_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Ingestion or attempt start instant
Valid values / checks	CHECK (completed_at IS NULL OR completed_at >= started_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → received (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.started_at

seagis.ingestion_run.completed_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Completion instant; null while running
Valid values / checks	CHECK (completed_at IS NULL OR completed_at >= started_at); CHECK ((status = 'running'::text) = (completed_at IS NULL))
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → UPDATE at end of A1/A2: clock_timestamp(); NULL while running (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.completed_at

seagis.ingestion_run.status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	running/succeeded/failed/partial together with completed_at
Valid values / checks	CHECK ((status = 'running'::text) = (completed_at IS NULL)); CHECK (status = ANY (ARRAY['running'::text, ' succeeded'::text, 'partial'::text, 'failed'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'running' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.status

seagis.ingestion_run.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → s.is_synthetic (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.is_synthetic

seagis.ingestion_run.error_code

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sanitized failure code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → A1/A2 successful rows omit -> NULL; failure codes tracked in environment.attempt (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.error_code

Constraints, indexes and triggers

constraints

```
CHECK (completed_at IS NULL OR completed_at >= started_at);
CHECK ((status = 'running'::text) = (completed_at IS NULL));
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
UNIQUE (dataset_key, request_key);
UNIQUE (id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (status = ANY (ARRAY['running'::text, 'succeeded'::text, 'partial'::text, 'failed'::text]));
```

indexes

```
CREATE UNIQUE INDEX ingestion_run_dataset_key_request_key_key ON seagis.ingestion_run USING btree
(dataset_key, request_key);
CREATE UNIQUE INDEX ingestion_run_id_dataset_key_is_synthetic_key ON seagis.ingestion_run USING
btree (id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX ingestion_run_pkey ON seagis.ingestion_run USING btree (id);
```

triggers

```
CREATE TRIGGER guard_ingestion BEFORE INSERT OR UPDATE ON seagis.ingestion_run FOR EACH ROW EXECUTE
FUNCTION seagis.guard_ingestion();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.6 seagis.observation

Local sensor water-temperature readings, retaining native values, QC and raw lineage.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic); UNIQUE (sensor_id, parameter_code, source_record_key)
Rows	UNVERIFIED
Writer	seagis_api.ingest; A0 fixtures
Reader	seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

seagis.observation.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.id

seagis.observation.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A1 packet + registered identity + server → registered sensor.plot_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.plot_id

seagis.observation.sensor_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Business / technical meaning	Sensor deployment reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A1 packet + registered identity + server → authenticated credential.sensor_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.sensor_id

seagis.observation.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code); FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic); FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → registered sensor.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.dataset_key

seagis.observation.ingestion_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Source ingestion reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A1 packet + registered identity + server → ingest-generated run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.ingestion_run_id

seagis.observation.raw_payload_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Source raw-manifest reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE

Attribute	Definition
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A1 packet + registered identity + server → ingest-generated raw_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.raw_payload_id

seagis.observation.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic); FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	A1 packet + registered identity + server → registered sensor.is_synthetic (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.is_synthetic

seagis.observation.source_record_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	boot_id:sequence for A1 packets
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → boot_id || : || sequence (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.source_record_key

seagis.observation.parameter_code

Attribute	Definition
Type / nullable	text / false

Attribute	Definition
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Environmental parameter identity
Valid values / checks	CHECK (parameter_code = 'water_temperature'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → constant water_temperature (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.parameter_code

seagis.observation.native_value

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Physical value in source units
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A1 packet + registered identity + server → packet.water_temperature (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.native_value

seagis.observation.native_unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Original unit before normalization
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → constant degC (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.native_unit

seagis.observation.value

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Normalized value; null follows value_status
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A1 packet + registered identity + server → packet.water_temperature (identity) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.value

seagis.observation.unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Normalized value unit from registry
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → constant degC (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.unit

seagis.observation.conversion_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Business / technical meaning	Unit-conversion method identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → constant identity (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.conversion_code

seagis.observation.value_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Numeric availability, separate from quality
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A1 packet + registered identity + server → packet.value_status (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.value_status

seagis.observation.detection_limit

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Detection limit in normalized units when applicable
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Defined by unit/native_unit and dataset_parameter; NOT APPLICABLE
Example	1
Origin / transformation	A1 packet + registered identity + server → omitted -> SQL NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.detection_limit

seagis.observation.observed_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sensor measurement instant, distinct from receipt
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A1 packet + registered identity + server → packet.observed_at cast timestamptz (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.observed_at

seagis.observation.ingested_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant data became available to the system
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A1 packet + registered identity + server → server received = clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.ingested_at

seagis.observation.quality_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	QC state unreviewed, accepted, suspect or rejected; not a diagnosis
Valid values / checks	CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid

Attribute	Definition
Origin / transformation	A1 packet + registered identity + server → pilot policy + flags (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.quality_status

Constraints, indexes and triggers

constraints

```
CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status,
detection_limit::double precision));
FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES
seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);
UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic);
CHECK (parameter_code = 'water_temperature'::text);
PRIMARY KEY (id);
CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text,
'rejected'::text]));
FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES
seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);
UNIQUE (sensor_id, parameter_code, source_record_key);
FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id,
dataset_key, is_synthetic);
```

indexes

```
CREATE UNIQUE INDEX observation_id_ingestion_run_id_dataset_key_is_synthetic_key ON
seagis.observation USING btree (id, ingestion_run_id, dataset_key, is_synthetic);
CREATE INDEX observation_ingestion ON seagis.observation USING btree (ingestion_run_id);
CREATE UNIQUE INDEX observation_pkey ON seagis.observation USING btree (id);
CREATE INDEX observation_plot_time ON seagis.observation USING btree (plot_id, parameter_code,
observed_at DESC);
CREATE UNIQUE INDEX observation_sensor_id_parameter_code_source_record_key_key ON seagis.observation
 USING btree (sensor_id, parameter_code, source_record_key);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.observation FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_observation BEFORE INSERT ON seagis.observation FOR EACH ROW EXECUTE FUNCTION
seagis.guard_measurement();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.7 seagis.parameter

Registry of parameters and normalized units.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (parameter_code)
Unique	UNIQUE (parameter_code, unit)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

seagis.parameter.parameter_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Environmental parameter identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'water_temperature' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.parameter_code

seagis.parameter.unit

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Normalized value unit from registry
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'degC' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.unit

seagis.parameter.description

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Parameter registry description
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'Submerged local sensor temperature; distinct from air temperature' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.description

Constraints, indexes and triggers

constraints

```
UNIQUE (parameter_code, unit);
PRIMARY KEY (parameter_code);
```

indexes

```
CREATE UNIQUE INDEX parameter_parameter_code_unit_key ON seagis.parameter USING btree
(parameter_code, unit);
CREATE UNIQUE INDEX parameter_pkey ON seagis.parameter USING btree (parameter_code);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.8 seagis.plot

Plot identity and survey status; nullable coordinates are not evidence of a surveyed location.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, is_synthetic); UNIQUE (plot_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

seagis.plot.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.id

seagis.plot.plot_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Unique human-readable plot code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:2 → 'plot-a' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.plot_code

seagis.plot.name

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Plot display name
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:2 → 'Plot A (synthetic)' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.name

seagis.plot.latitude

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	WGS84 latitude in degrees
Valid values / checks	CHECK ((latitude IS NULL) = (longitude IS NULL)); CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL); CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.latitude

seagis.plot.longitude

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	WGS84 longitude in degrees
Valid values / checks	CHECK ((latitude IS NULL) = (longitude IS NULL)); CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.longitude

seagis.plot.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	CHECK ((location_status = 'synthetic'::text) = is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/seed.sql:2 → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.is_synthetic

seagis.plot.location_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Location state: synthetic, not_surveyed or surveyed

Attribute	Definition
Valid values / checks	CHECK ((location_status = 'synthetic'::text) = is_synthetic); CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL); CHECK (location_status = ANY (ARRAY['not_surveyed'::text, 'surveyed'::text, ' synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:2 → 'synthetic' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.location_status

seagis.plot.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.created_at

Constraints, indexes and triggers

constraints

```
CHECK ((latitude IS NULL) = (longitude IS NULL));
CHECK ((location_status = 'synthetic'::text) = is_synthetic);
CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL);
UNIQUE (id, is_synthetic);
CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision
 <= 90::double precision);
CHECK (location_status = ANY (ARRAY['not_surveyed'::text, 'surveyed'::text, 'synthetic'::text]));
CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double
precision <= 180::double precision);
PRIMARY KEY (id);
UNIQUE (plot_code);
```

indexes

```
CREATE UNIQUE INDEX plot_id_is_synthetic_key ON seagis.plot USING btree (id, is_synthetic);
CREATE UNIQUE INDEX plot_pkey ON seagis.plot USING btree (id);
CREATE UNIQUE INDEX plot_plot_code_key ON seagis.plot USING btree (plot_code);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.9 seagis.plot_grid_mapping

Operational plot-to-grid mapping with effective and recorded times.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, source_grid_id, effective_from)
Rows	UNVERIFIED
Writer	A0 fixture/admin SQL
Reader	A3 source_bundle; feature_input guards
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

seagis.plot_grid_mapping.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.id

seagis.plot_grid_mapping.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied plot_id subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.plot_id

seagis.plot_grid_mapping.source_grid_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id)
Business / technical meaning	Regional cell and layer reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied source_grid_id subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.source_grid_id

seagis.plot_grid_mapping.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied is_synthetic subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.is_synthetic

seagis.plot_grid_mapping.sampling_method

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Grid selection method for plot
Valid values / checks	CHECK (sampling_method = ANY (ARRAY['nearest_wet_cell'::text, 'nearest_cell'::text, ' provider_returned_coordinate'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid

Attribute	Definition
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied sampling_method subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.sampling_method

seagis.plot_grid_mapping.distance_m

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Plot-to-cell distance in metres
Valid values / checks	CHECK (distance_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied distance_m subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.distance_m

seagis.plot_grid_mapping.representativeness

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Regional representativeness status, not proof of local validity
Valid values / checks	CHECK (representativeness = ANY (ARRAY['unverified'::text, 'regional_only'::text, 'land_mask_issue'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied representativeness subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.representativeness

seagis.plot_grid_mapping.effective_from

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Inclusive mapping interval start
Valid values / checks	CHECK (effective_until IS NULL OR effective_until > effective_from)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied effective_from subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.effective_from

seagis.plot_grid_mapping.effective_until

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Exclusive mapping end; null means open-ended
Valid values / checks	CHECK (effective_until IS NULL OR effective_until > effective_from)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied effective_until subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.effective_until

seagis.plot_grid_mapping.recorded_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant mapping first became known to the system
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.recorded_at

Constraints, indexes and triggers

constraints

```
CHECK (effective_until IS NULL OR effective_until > effective_from);
CHECK (distance_m::double precision >= 0::double precision);
PRIMARY KEY (id);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
UNIQUE (plot_id, source_grid_id, effective_from);
CHECK (representativeness = ANY (ARRAY['unverified'::text, 'regional_only'::text,
'land_mask_issue'::text]));
CHECK (sampling_method = ANY (ARRAY['nearest_wet_cell'::text, 'nearest_cell'::text,
'provider_returned_coordinate'::text, 'synthetic'::text]));
FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id);
```

indexes

```
CREATE INDEX mapping_grid_plot ON seagis.plot_grid_mapping USING btree (source_grid_id, plot_id);
CREATE UNIQUE INDEX plot_grid_mapping_pkey ON seagis.plot_grid_mapping USING btree (id);
CREATE UNIQUE INDEX plot_grid_mapping_plot_id_source_grid_id_effective_from_key ON
seagis.plot_grid_mapping USING btree (plot_id, source_grid_id, effective_from);
```

triggers

```
CREATE TRIGGER guard_mapping BEFORE INSERT OR UPDATE ON seagis.plot_grid_mapping FOR EACH ROW
EXECUTE FUNCTION seagis.guard_mapping();
CREATE TRIGGER immutable_alignment_mapping BEFORE DELETE OR UPDATE ON seagis.plot_grid_mapping FOR
EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.10 seagis.quality_event

Append-only QC log for ingestion or one measurement.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (ingestion_run_id, event_key)
Rows	UNVERIFIED
Writer	seagis_api.ingest; A0 fixtures
Reader	A3 source_bundle/alignment
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

seagis.quality_event.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.id

seagis.quality_event.ingestion_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Source ingestion reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.ingestion_run_id

seagis.quality_event.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic); FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.dataset_key

seagis.quality_event.is_synthetic

Attribute	Definition
Type / nullable	boolean / false

Attribute	Definition
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic); FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → s.is_synthetic (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.is_synthetic

seagis.quality_event.event_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Unique event identity per ingestion
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → flag (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.event_key

seagis.quality_event.observation_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Business / technical meaning	Local sensor reading reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → observation_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.observation_id

seagis.quality_event.forecast_value_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic)
Business / technical meaning	Single regional model value reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.forecast_value_id

seagis.quality_event.event_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	QC event-type code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → flag (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.event_code

seagis.quality_event.severity

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Event level info, warning or error; not disease risk
Valid values / checks	CHECK (severity = ANY (ARRAY['info'::text, 'warning'::text, 'error'::text]))

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → CASE WHEN quality='rejected' THEN ' error' WHEN flag IN ('gap_before','out_of_order') THEN 'info' ELSE 'warning' END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.severity

seagis.quality_event.details

Attribute	Definition
Type / nullable	jsonb / false
Default / generated	'{}'::jsonb
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	QC JSON details without secrets
Valid values / checks	CHECK (jsonb_typeof(details) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	caller / SQL default → '{}'::jsonb (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.details

seagis.quality_event.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.created_at

Constraints, indexes and triggers

constraints

CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1);
CHECK (jsonb_typeof(details) = 'object'::text);
FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic);

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);
UNIQUE (ingestion_run_id, event_key);
FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (severity = ANY (ARRAY['info'::text, 'warning'::text, 'error'::text]));

indexes

CREATE INDEX quality_event_forecast ON seagis.quality_event USING btree (forecast_value_id) WHERE (forecast_value_id IS NOT NULL);
CREATE UNIQUE INDEX quality_event_ingestion_run_id_event_key_key ON seagis.quality_event USING btree (ingestion_run_id, event_key);
CREATE INDEX quality_event_observation ON seagis.quality_event USING btree (observation_id) WHERE (observation_id IS NOT NULL);
CREATE UNIQUE INDEX quality_event_pkey ON seagis.quality_event USING btree (id);

triggers

CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.quality_event FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_quality BEFORE INSERT ON seagis.quality_event FOR EACH ROW EXECUTE FUNCTION seagis.guard_quality_event();

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.11 seagis.sensor

Temperature sensor deployment identity, plot, depth and active state.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, plot_id, dataset_key, is_synthetic); UNIQUE (sensor_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

seagis.sensor.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.id

seagis.sensor.sensor_code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Unique sensor-deployment code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:7 → 'simulator-a' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.sensor_code

seagis.sensor.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/seed.sql:7 → '00000000-0000-4000-8000-000000000001' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.plot_id

seagis.sensor.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code); FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:7 → 'sensor_temperature_v1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.dataset_key

seagis.sensor.scope

Attribute	Definition
Type / nullable	text / false
Default / generated	'local_sensor'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Business / technical meaning	Origin: local_sensor or regional_model
Valid values / checks	CHECK (scope = 'local_sensor'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'local_sensor'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.scope

seagis.sensor.parameter_code

Attribute	Definition
Type / nullable	text / false
Default / generated	'water_temperature'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code)
Business / technical meaning	Environmental parameter identity
Valid values / checks	CHECK (parameter_code = 'water_temperature'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'water_temperature'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.parameter_code

seagis.sensor.depth_m

Attribute	Definition
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	Depth positive downward in metres
Valid values / checks	CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL); CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL); CHECK (depth_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.depth_m

seagis.sensor.depth_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sensor-depth knowledge status
Valid values / checks	CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL); CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL); CHECK (depth_status <> 'synthetic'::text OR is_synthetic); CHECK (depth_status = ANY (ARRAY['unknown'::text, 'measured'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/seed.sql:7 → 'unknown' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.depth_status

seagis.sensor.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	CHECK (depth_status <> 'synthetic'::text OR is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/seed.sql:7 → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.is_synthetic

seagis.sensor.active

Attribute	Definition
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Whether the sensor deployment remains active
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	caller / SQL default → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.active

Constraints, indexes and triggers

constraints

```
CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL);
CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL);
CHECK (depth_status <> 'synthetic'::text OR is_synthetic);
FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key,
parameter_code);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
CHECK (depth_m::double precision >= 0::double precision);
CHECK (depth_status = ANY (ARRAY['unknown'::text, 'measured'::text, 'synthetic'::text]));
UNIQUE (id, plot_id, dataset_key, is_synthetic);
CHECK (parameter_code = 'water_temperature'::text);
PRIMARY KEY (id);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (scope = 'local_sensor'::text);
UNIQUE (sensor_code);
```

indexes

```
CREATE UNIQUE INDEX one_active_temperature_sensor_per_plot ON seagis.sensor USING btree (plot_id)
WHERE active;
CREATE UNIQUE INDEX sensor_id_plot_id_dataset_key_is_synthetic_key ON seagis.sensor USING btree (id,
 plot_id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX sensor_pkey ON seagis.sensor USING btree (id);
CREATE UNIQUE INDEX sensor_sensor_code_key ON seagis.sensor USING btree (sensor_code);
```

triggers

```
CREATE TRIGGER guard_sensor_deployment BEFORE UPDATE ON seagis.sensor FOR EACH ROW EXECUTE FUNCTION
seagis.guard_sensor_deployment();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.12 seagis.source

Registry of dataset providers.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (source_key)
Unique	NOT FOUND

Attribute	Definition
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

seagis.source.source_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Source-provider key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'sensor' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.source_key

seagis.source.provider

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Dataset provider name
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'SEAGIS local sensor / simulator' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.provider

seagis.source.documentation_url

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Provider documentation URL
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'https://github.com/De-Voeux/Seagis-Model' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.documentation_url

Constraints, indexes and triggers

constraints

```
PRIMARY KEY (source_key);
```

indexes

```
CREATE UNIQUE INDEX source_pkey ON seagis.source USING btree (source_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.13 seagis.source_grid

Regional cell and depth-layer identity; distinct from cultivation plots.

Attribute	Definition
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, grid_key, is_synthetic); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

seagis.source_grid.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.id

seagis.source_grid.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.dataset_key

seagis.source_grid.scope

Attribute	Definition
Type / nullable	text / false
Default / generated	'regional_model'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Business / technical meaning	Origin: local_sensor or regional_model
Valid values / checks	CHECK (scope = 'regional_model'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'regional_model'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.scope

seagis.source_grid.grid_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Cell key based on coordinates/depth or fixture
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → g->>'key' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.grid_key

seagis.source_grid.grid_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Grid identity version
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → d.dataset_version (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.grid_version

seagis.source_grid.identity_basis

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Identity basis: native cell, response coordinate or synthetic
Valid values / checks	CHECK (identity_basis <> 'synthetic'::text OR is_synthetic); CHECK (identity_basis = ANY (ARRAY['provider_cell'::text, 'provider_returned_coordinate'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → g->>'identity_basis' (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.identity_basis

seagis.source_grid.latitude

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	WGS84 latitude in degrees
Valid values / checks	CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'latitude')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.latitude

seagis.source_grid.longitude

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	WGS84 longitude in degrees
Valid values / checks	CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'longitude')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.longitude

seagis.source_grid.depth_m

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	Depth positive downward in metres
Valid values / checks	CHECK (depth_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'depth_m')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.depth_m

seagis.source_grid.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	CHECK (identity_basis <> 'synthetic'::text OR is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.is_synthetic

Constraints, indexes and triggers

constraints

```
CHECK (identity_basis <> 'synthetic'::text OR is_synthetic);
UNIQUE (dataset_key, grid_key, is_synthetic);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
CHECK (depth_m::double precision >= 0::double precision);
UNIQUE (id, dataset_key, is_synthetic);
CHECK (identity_basis = ANY (ARRAY['provider_cell'::text, 'provider_returned_coordinate'::text,
'synthetic'::text]));
CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision
 <= 90::double precision);
CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double
precision <= 180::double precision);
PRIMARY KEY (id);
CHECK (scope = 'regional_model'::text);
```

indexes

```
CREATE UNIQUE INDEX source_grid_dataset_key_grid_key_is_synthetic_key ON seagis.source_grid USING
btree (dataset_key, grid_key, is_synthetic);
CREATE UNIQUE INDEX source_grid_id_dataset_key_is_synthetic_key ON seagis.source_grid USING btree
(id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX source_grid_pkey ON seagis.source_grid USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.source_grid FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.14 seagis_api.credential

API actor identity using token hashes, expiry and revocation.

Attribute	Definition
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (token_sha256)
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

seagis_api.credential.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.id

seagis_api.credential.label

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Administrative credential label
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid

Attribute	Definition
Origin / transformation	owning routine / caller → provision_a1.create(label) CLI --label (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.label

seagis_api.credential.token_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Bearer token hash; original token not stored in column
Valid values / checks	CHECK (token_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → sha256((sgi_ + secrets.token_urlsafe(32)).encode()).hexdigest(); original token excluded (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.token_sha256

seagis_api.credential.kind

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Actor kind device or reader
Valid values / checks	CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL)); CHECK (kind = ANY (ARRAY['device'::text, 'reader'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → provision_a1.create(kind), CLI device/reader (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.kind

seagis_api.credential.sensor_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)

Attribute	Definition
Business / technical meaning	Sensor deployment reference
Valid values / checks	CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → validated CLI sensor UUID for device, NULL for reader (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.sensor_id

seagis_api.credential.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	CHECK (expires_at > created_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.created_at

seagis_api.credential.expires_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant credential expires
Valid values / checks	CHECK (expires_at > created_at)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → now()+make_interval(days=>days); CLI 1..365, default 30 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.expires_at

seagis_api.credential.revoked_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Revocation instant; null if not revoked
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → NULL initially; revoke command UPDATE revoked_at=now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.revoked_at

Constraints, indexes and triggers

constraints

```
CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL));
CHECK (expires_at > created_at);
CHECK (kind = ANY (ARRAY['device'::text, 'reader'::text]));
PRIMARY KEY (id);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
CHECK (token_sha256 ~ '^[0-9a-f]{64}$'::text);
UNIQUE (token_sha256);
```

indexes

```
CREATE UNIQUE INDEX credential_pkey ON seagis_api.credential USING btree (id);
CREATE UNIQUE INDEX credential_token_sha256_key ON seagis_api.credential USING btree (token_sha256);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.15 seagis_api.packet_receipt

Sensor-packet idempotency key and semantic digest.

Attribute	Definition
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (sensor_id, source_record_key)
Unique	UNIQUE (observation_id)
Rows	UNVERIFIED
Writer	seagis_api.ingest/reply
Reader	seagis_api.actor/rate_ok/ingest/history/status
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

seagis_api.packet_receipt.sensor_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)
Business / technical meaning	Sensor deployment reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → s.id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.sensor_id

seagis_api.packet_receipt.source_record_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	boot_id:sequence for A1 packets
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.source_record_key

seagis_api.packet_receipt.semantic_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Packet hash after retry-safe semantic normalization
Valid values / checks	CHECK (semantic_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → digest (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.semantic_sha256

seagis_api.packet_receipt.observation_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / true (single-column uniqueness; composites below)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Business / technical meaning	Local sensor reading reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → observation_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.observation_id

seagis_api.packet_receipt.ingestion_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id)
Business / technical meaning	Source ingestion reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.ingestion_run_id

seagis_api.packet_receipt.flags

Attribute	Definition
Type / nullable	text[] / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	QC codes from packet acceptance
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	[]
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → flags (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.flags

Constraints, indexes and triggers

constraints

```
FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
UNIQUE (observation_id);
PRIMARY KEY (sensor_id, source_record_key);
CHECK (semantic_sha256 ~ '^[0-9a-f]{64}$'::text);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
```

indexes

```
CREATE UNIQUE INDEX packet_receipt_observation_id_key ON seagis_api.packet_receipt USING btree
(observation_id);
CREATE UNIQUE INDEX packet_receipt_pkey ON seagis_api.packet_receipt USING btree (sensor_id,
source_record_key);
```

triggers

```
CREATE TRIGGER receipt_append_only BEFORE DELETE OR UPDATE ON seagis_api.packet_receipt FOR EACH ROW
 EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.16 seagis_api.reader_plot

Plot allowlist for a reader credential.

Attribute	Definition
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (credential_id, plot_id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

seagis_api.reader_plot.credential_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id)
Business / technical meaning	Registered API actor reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → provision_a1.create generated credential UUID (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39; catalog seagis_api.reader_plot.credential_id

seagis_api.reader_plot.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id) REFERENCES seagis.plot(id)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → CLI plots validated with UUID(), FK checked by database (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39; catalog seagis_api.reader_plot.plot_id

Constraints, indexes and triggers

constraints

```
FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);
PRIMARY KEY (credential_id, plot_id);
FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);
```

indexes

```
CREATE UNIQUE INDEX reader_plot_pkey ON seagis_api.reader_plot USING btree (credential_id, plot_id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.17 seagis_api.request_event

Authenticated API response audit and rate-limit basis.

Attribute	Definition
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND

Attribute	Definition
Rows	UNVERIFIED
Writer	seagis_api.ingest/reply
Reader	seagis_api.actor/rate_ok/ingest/history/status
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

seagis_api.request_event.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.id

seagis_api.request_event.credential_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id)
Business / technical meaning	Registered API actor reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → actor_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.credential_id

seagis_api.request_event.sensor_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)
Business / technical meaning	Sensor deployment reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → sensor (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.sensor_id

seagis_api.request_event.http_status

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Audited response HTTP status
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → status (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.http_status

seagis_api.request_event.code

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	API response code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → code (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.code

seagis_api.request_event.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.created_at

Constraints, indexes and triggers

constraints

```
FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);
PRIMARY KEY (id);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
```

indexes

```
CREATE INDEX request_event_actor_time ON seagis_api.request_event USING btree (credential_id,
created_at DESC);
CREATE UNIQUE INDEX request_event_pkey ON seagis_api.request_event USING btree (id);
```

triggers

```
CREATE TRIGGER request_event_append_only BEFORE DELETE OR UPDATE ON seagis_api.request_event FOR
EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.18 seagis_api.temperature_policy

Pilot engineering QC settings and request limit; not disease thresholds.

Attribute	Definition
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

seagis_api.temperature_policy.id

Attribute	Definition
Type / nullable	boolean / false
Default / generated	true
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Singleton boolean must be true; not a UUID
Valid values / checks	CHECK (id)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	DB default / explicit UUID → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.id

seagis_api.temperature_policy.version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Policy version identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 'pilot-qc-1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.version

seagis_api.temperature_policy.min_degc

Attribute	Definition
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Pilot engineering QC lower envelope in degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → -5 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.min_degc

seagis_api.temperature_policy.max_degc

Attribute	Definition
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Pilot engineering QC upper envelope in degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 60 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.max_degc

seagis_api.temperature_policy.max_step_degc

Attribute	Definition
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Pilot temperature-step limit in degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 3 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.max_step_degc

seagis_api.temperature_policy.future_tolerance_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Future sensor-clock tolerance in seconds
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 300 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.future_tolerance_seconds

seagis_api.temperature_policy.stale_after_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sensor delay boundary in seconds
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 7200 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.stale_after_seconds

seagis_api.temperature_policy.requests_per_minute

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Request limit per actor per one-minute window
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 120 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.requests_per_minute

Constraints, indexes and triggers

constraints

```
CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0
AND stale_after_seconds > 0 AND requests_per_minute > 0);
CHECK (id);
PRIMARY KEY (id);
```

indexes

```
CREATE UNIQUE INDEX temperature_policy_pkey ON seagis_api.temperature_policy USING btree (id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.19 seagis_environment.attempt

Audit of a fetch attempt, including duplicate and abandoned outcomes.

Attribute	Definition
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

seagis_environment.attempt.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → lease (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.id

seagis_environment.attempt.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → k (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.dataset_key

seagis_environment.attempt.started_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Ingestion or attempt start instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.started_at

seagis_environment.attempt.completed_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Completion instant; null while running
Valid values / checks	CHECK ((completed_at IS NULL) = (status = 'running'::text))
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → clock_timestamp() at persist/failure/abandon; initially NULL (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.completed_at

seagis_environment.attempt.status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	running/succeeded/duplicate/failed/abandoned fetch-attempt state
Valid values / checks	CHECK ((completed_at IS NULL) = (status = 'running'::text)); CHECK (status = ANY (ARRAY['running'::text, ' succeeded'::text, 'duplicate'::text, 'failed'::text, 'abandoned'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → 'running' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.status

seagis_environment.attempt.error_code

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sanitized failure code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → finish_failure sanitized code or lease_expired on abandoned; otherwise NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.error_code

seagis_environment.attempt.forecast_run_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id)
Business / technical meaning	Regional model capture/vintage reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → persist run_id, also on duplicate; NULL before completion (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.forecast_run_id

Constraints, indexes and triggers

constraints

```
CHECK ((completed_at IS NULL) = (status = 'running'::text));
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id);
PRIMARY KEY (id);
CHECK (status = ANY (ARRAY['running'::text, 'succeeded'::text, 'duplicate'::text, 'failed'::text,
'abandoned'::text]));
```

indexes

```
CREATE UNIQUE INDEX attempt_pkey ON seagis_environment.attempt USING btree (id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.20 seagis_environment.current_speed

Current-magnitude view joining components from the same run, grid and time.

Attribute	Definition
Schema / domain	seagis_environment
Kind	view
Primary key	NOT APPLICABLE: view
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	NOT APPLICABLE: view
Reader	A2 status/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

seagis_environment.current_speed.eastward_value_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	current_eastward component ID in view
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	seagis.forecast_value u/v → u.id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.eastward_value_id

seagis_environment.current_speed.northward_value_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	current_northward component ID in view
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	seagis.forecast_value u/v → v.id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.northward_value_id

seagis_environment.current_speed.forecast_run_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Regional model capture/vintage reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	seagis.forecast_value u/v → u.forecast_run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.forecast_run_id

seagis_environment.current_speed.source_grid_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Regional cell and layer reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	seagis.forecast_value u/v → u.source_grid_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.source_grid_id

seagis_environment.current_speed.valid_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Representative time of model value
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	seagis.forecast_value u/v → u.valid_at (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.valid_at

seagis_environment.current_speed.ingested_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Instant data became available to the system
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	seagis.forecast_value u/v → u.ingested_at (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.ingested_at

seagis_environment.current_speed.available_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Conservative input availability time
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	seagis.forecast_value u/v → greatest(u.ingested_at,v.ingested_at) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.available_at

seagis_environment.current_speed.speed_m_s

Attribute	Definition
Type / nullable	double precision / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	sqrt(uo²+vo²), null unless both components are present
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	m/s; NOT APPLICABLE
Example	1
Origin / transformation	seagis.forecast_value u/v → CASE WHEN both value_status='present' THEN sqrt(u.value*u.value+v.value*v.value) END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.speed_m_s

seagis_environment.current_speed.derivation

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	Current-magnitude formula and version label
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	seagis.forecast_value u/v → 'sqrt(uo^2+vo^2); a2-vector-magnitude-v1' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.derivation

Constraints, indexes and triggers

constraints NOT APPLICABLE / NOT FOUND

indexes NOT APPLICABLE / NOT FOUND

triggers NOT APPLICABLE / NOT FOUND

```
SELECT u.id AS eastward_value_id,
    v.id AS northward_value_id,
    u.forecast_run_id,
    u.source_grid_id,
    u.valid_at,
    u.ingested_at,
    GREATEST(u.ingested_at, v.ingested_at) AS available_at,
        CASE
            WHEN ((u.value_status = 'present'::text) AND (v.value_status = 'present'::text)) THEN
sqrt((((u.value)::double precision * (u.value)::double precision) + ((v.value)::double precision *
(v.value)::double precision)))
            ELSE NULL::double precision
        END AS speed_m_s,
    'sqrt(uo^2+vo^2); a2-vector-magnitude-v1'::text AS derivation
   FROM (seagis.forecast_value u
     JOIN seagis.forecast_value v ON (((u.forecast_run_id = v.forecast_run_id) AND (u.source_grid_id
 = v.source_grid_id) AND (u.valid_at = v.valid_at))))
  WHERE ((u.parameter_code = 'current_eastward'::text) AND (v.parameter_code =
'current_northward'::text));
```

4.21 seagis_environment.demo_grid_mapping

Bridge from two fictional plots to real grids for preview, not training authorization.

Attribute	Definition
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (plot_id, source_grid_id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

seagis_environment.demo_grid_mapping.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Cultivation plot reference
Valid values / checks	CHECK (plot_id = ANY (ARRAY['00000000-0000-4000-8000-000000000001'::uuid, '00000000-0000-4000-8000-000000000002'::uuid]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'plot_id')::uuid (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.plot_id

seagis_environment.demo_grid_mapping.plot_is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Must be true for demo bridge
Valid values / checks	CHECK (plot_is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	caller / SQL default → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.plot_is_synthetic

seagis_environment.demo_grid_mapping.source_grid_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Regional cell and layer reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → grid_id (VERIFIED)

Attribute	Definition
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.source_grid_id

seagis_environment.demo_grid_mapping.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key); FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.dataset_key

seagis_environment.demo_grid_mapping.grid_is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Business / technical meaning	Must be false for real grid in demo bridge
Valid values / checks	CHECK (NOT grid_is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	caller / SQL default → false (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.grid_is_synthetic

seagis_environment.demo_grid_mapping.example_latitude

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	Fictional example latitude, not a survey
Valid values / checks	CHECK (example_latitude::double precision >= '-90'::integer::double precision AND example_latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'latitude')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.example_latitude

seagis_environment.demo_grid_mapping.example_longitude

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Fictional example longitude, not a survey
Valid values / checks	CHECK (example_longitude::double precision >= '-180'::integer::double precision AND example_longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'longitude')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.example_longitude

seagis_environment.demo_grid_mapping.coordinate_status

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	example_not_surveyed in demo bridge
Valid values / checks	CHECK (coordinate_status = 'example_not_surveyed'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → m->>'coordinate_status' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.coordinate_status

seagis_environment.demo_grid_mapping.distance_m

Attribute	Definition
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Plot-to-cell distance in metres
Valid values / checks	CHECK (distance_m::double precision >= 0::double precision AND distance_m::double precision <= 30000::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'distance_m')::float8 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.distance_m

seagis_environment.demo_grid_mapping.sampling_method

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Grid selection method for plot
Valid values / checks	CHECK (sampling_method = ANY (ARRAY['nearest_cell'::text, 'provider_returned_coordinate'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → m->>'sampling_method' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.sampling_method

seagis_environment.demo_grid_mapping.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z

Attribute	Definition
Origin / transformation	caller / SQL default → now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.created_at

Constraints, indexes and triggers

constraints

```
CHECK (coordinate_status = 'example_not_surveyed'::text);
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
CHECK (distance_m::double precision >= 0::double precision AND distance_m::double precision <=
30000::double precision);
CHECK (example_latitude::double precision >= '-90'::integer::double precision AND
example_latitude::double precision <= 90::double precision);
CHECK (example_longitude::double precision >= '-180'::integer::double precision AND
example_longitude::double precision <= 180::double precision);
CHECK (NOT grid_is_synthetic);
PRIMARY KEY (plot_id, source_grid_id);
CHECK (plot_id = ANY (ARRAY['00000000-0000-4000-8000-000000000001'::uuid,
'00000000-0000-4000-8000-000000000002'::uuid]));
FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (plot_is_synthetic);
CHECK (sampling_method = ANY (ARRAY['nearest_cell'::text, 'provider_returned_coordinate'::text]));
FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id,
dataset_key, is_synthetic);
```

indexes

```
CREATE UNIQUE INDEX demo_grid_mapping_pkey ON seagis_environment.demo_grid_mapping USING btree
(plot_id, source_grid_id);
```

triggers

```
CREATE TRIGGER immutable_demo_mapping BEFORE DELETE OR UPDATE ON
seagis_environment.demo_grid_mapping FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.22 seagis_environment.policy

Regional fetch, licence, scheduler cadence, staleness and retention policy.

Attribute	Definition
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

seagis_environment.policy.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.dataset_key

seagis_environment.policy.enabled

Attribute	Definition
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Whether regional policy enables fetching
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	caller / SQL default → true (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.enabled

seagis_environment.policy.purpose

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Purpose noncommercial_educational_prototype
Valid values / checks	CHECK (purpose = 'noncommercial_educational_prototype'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → ' noncommercial_educational_prototype' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.purpose

seagis_environment.policy.licence_url

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Source licence reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → licence_url (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.licence_url

seagis_environment.policy.terms_review

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Provider terms review identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 'cmems-licence-20261022-sections-2.2-2.4-2.6' ELSE ' open-meteo-terms-20261022-noncommercial-cc-by-4.0' END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.terms_review

seagis_environment.policy.attribution

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Attribution text to preserve
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → attribution (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.attribution

seagis_environment.policy.update_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Nominal per-provider scheduler interval in seconds
Valid values / checks	CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds); CHECK (stale_seconds >= update_seconds); CHECK (update_seconds >= 21600)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 86400 ELSE 21600 END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.update_seconds

seagis_environment.policy.release_offset_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Schedule phase offset in seconds
Valid values / checks	CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN dataset_key='cmems_nutrients_daily' THEN 45000 WHEN source_key='copernicus' THEN 30600 ELSE 16200 END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.release_offset_seconds

seagis_environment.policy.stale_seconds

Attribute	Definition
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Maximum vintage age under policy in seconds
Valid values / checks	CHECK (stale_seconds >= update_seconds)
Units / time	s; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 172800 ELSE 43200 END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.stale_seconds

seagis_environment.policy.retention_mode

Attribute	Definition
Type / nullable	text / false
Default / generated	'metadata_only'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	metadata_only retention policy in A2
Valid values / checks	CHECK (retention_mode = 'metadata_only'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'metadata_only'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.retention_mode

seagis_environment.policy.reviewed_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Policy review instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z

Attribute	Definition
Origin / transformation	caller / SQL default → now() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.reviewed_at

Constraints, indexes and triggers

constraints

```
CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds);
CHECK (stale_seconds >= update_seconds);
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
PRIMARY KEY (dataset_key);
CHECK (purpose = 'noncommercial_educational_prototype'::text);
CHECK (retention_mode = 'metadata_only'::text);
CHECK (update_seconds >= 21600);
```

indexes

```
CREATE UNIQUE INDEX policy_pkey ON seagis_environment.policy USING btree (dataset_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.23 seagis_environment.schedule_state

Mutable lease, due time, backoff and last-success state.

Attribute	Definition
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

seagis_environment.schedule_state.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE

Attribute	Definition
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:56 → dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.dataset_key

seagis_environment.schedule_state.lease_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Active fetch lease identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → claim gen_random_uuid(); cleared after persist/failure (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.lease_id

seagis_environment.schedule_state.lease_until

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Lease expiry instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → claim clock_timestamp()+interval 20 minutes; cleared after persist/failure (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.lease_until

seagis_environment.schedule_state.next_attempt_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	'-infinity'::timestamp with time zone

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Earliest next fetch; may initially be -infinity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → '-infinity'::timestamp with time zone (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.next_attempt_at

seagis_environment.schedule_state.last_success_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Last successful fetch/persist instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → persist t=clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_success_at

seagis_environment.schedule_state.last_attempt_at

Attribute	Definition
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Last attempt start instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	owning routine / caller → claim clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_attempt_at

seagis_environment.schedule_state.last_error_code

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Last attempt failure code
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → finish_failure sanitized code; cleared on success (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_error_code

seagis_environment.schedule_state.failures

Attribute	Definition
Type / nullable	integer / false
Default / generated	0
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Failure count used for backoff
Valid values / checks	CHECK (failures >= 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	caller / SQL default → 0 (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.failures

seagis_environment.schedule_state.last_forecast_run_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id)
Business / technical meaning	Last successful vintage reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	owning routine / caller → persist run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_forecast_run_id

Constraints, indexes and triggers

constraints

```
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
CHECK (failures >= 0);
FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id);
PRIMARY KEY (dataset_key);
```

indexes

```
CREATE UNIQUE INDEX schedule_state_pkey ON seagis_environment.schedule_state USING btree
(dataset_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.24 seagis_features.feature_input

Foreign-key input lineage and cutoff guards for feature snapshots.

Attribute	Definition
Schema / domain	seagis_features
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	A0 fixture/test inserts; production builder NOT FOUND
Reader	snapshot/input guards; production ML consumer NOT FOUND
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

seagis_features.feature_input.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.id

seagis_features.feature_input.snapshot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id)
Business / technical meaning	Parent snapshot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied snapshot_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.snapshot_id

seagis_features.feature_input.observation_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Business / technical meaning	Local sensor reading reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied observation_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.observation_id

seagis_features.feature_input.forecast_value_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id)
Business / technical meaning	Single regional model value reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied forecast_value_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.forecast_value_id

Constraints, indexes and triggers

constraints

```
CHECK (num_nonnulls(observation_id, forecast_value_id) = 1);
FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
PRIMARY KEY (id);
FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id);
```

indexes

```
CREATE UNIQUE INDEX feature_input_forecast ON seagis_features.feature_input USING btree
(snapshot_id, forecast_value_id) WHERE (forecast_value_id IS NOT NULL);
CREATE UNIQUE INDEX feature_input_observation ON seagis_features.feature_input USING btree
(snapshot_id, observation_id) WHERE (observation_id IS NOT NULL);
CREATE UNIQUE INDEX feature_input_pkey ON seagis_features.feature_input USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_features.feature_input FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_feature_input BEFORE INSERT ON seagis_features.feature_input FOR EACH ROW
EXECUTE FUNCTION seagis_features.guard_input();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.25 seagis_features.feature_snapshot

Draft/sealed feature container with cutoff and hashes; no production builder found.

Attribute	Definition
Schema / domain	seagis_features
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, as_of, feature_version, input_manifest_sha256)
Rows	UNVERIFIED
Writer	A0 fixture/test inserts; production builder NOT FOUND
Reader	snapshot/input guards; production ML consumer NOT FOUND
Authority	Records of this domain; registry/config or original source remains traceable

Attribute	Definition
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

seagis_features.feature_snapshot.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.id

seagis_features.feature_snapshot.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied plot_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.plot_id

seagis_features.feature_snapshot.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)

Attribute	Definition
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied is_synthetic subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.is_synthetic

seagis_features.feature_snapshot.as_of

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Information cutoff for feature snapshot
Valid values / checks	CHECK (generated_at >= as_of)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied as_of subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.as_of

seagis_features.feature_snapshot.generated_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Feature snapshot generation instant
Valid values / checks	CHECK (generated_at >= as_of)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied generated_at subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.generated_at

seagis_features.feature_snapshot.feature_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Feature definition or builder version
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied feature_version subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.feature_version

seagis_features.feature_snapshot.feature_schema_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Feature JSON structure version
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied feature_schema_version subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.feature_schema_version

seagis_features.feature_snapshot.features

Attribute	Definition
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	NOT APPLICABLE
Business / technical meaning	Feature JSON object; container does not prove calculation correctness
Valid values / checks	CHECK (jsonb_typeof(features) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	A0 fixtures/tests; production builder NOT FOUND → caller-supplied features subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.features

seagis_features.feature_snapshot.missing_context

Attribute	Definition
Type / nullable	text[] / false
Default / generated	'{}'::text[]
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	List of unavailable context
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	[]
Origin / transformation	caller / SQL default → '{}'::text[] (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.missing_context

seagis_features.feature_snapshot.status

Attribute	Definition
Type / nullable	text / false
Default / generated	'draft'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	draft before sealing, sealed after manifest verification
Valid values / checks	CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL)); CHECK (status = ANY (ARRAY['draft'::text, 'sealed'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'draft'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.status

seagis_features.feature_snapshot.input_manifest_sha256

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Digest of ordered lineage-input manifest
Valid values / checks	CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL)); CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	seagis_features.manifest_hash(snapshot) → SHA256 sorted input references at draft -> sealed (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.input_manifest_sha256

seagis_features.feature_snapshot.features_sha256

Attribute	Definition
Type / nullable	text / true
Default / generated	seagis.jsonb_sha256(features)
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	PostgreSQL JSONB feature digest
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	DB generated → seagis.jsonb_sha256(features) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.features_sha256

Constraints, indexes and triggers

constraints

```
CHECK (generated_at >= as_of);
CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL));
CHECK (jsonb_typeof(features) = 'object'::text);
CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text);
PRIMARY KEY (id);
UNIQUE (plot_id, as_of, feature_version, input_manifest_sha256);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (status = ANY (ARRAY['draft'::text, 'sealed'::text]));
```

indexes

```
CREATE UNIQUE INDEX feature_snapshot_pkey ON seagis_features.feature_snapshot USING btree (id);
```

```
CREATE UNIQUE INDEX feature_snapshot_plot_id_as_of_feature_version_input_manife_key ON
seagis_features.feature_snapshot USING btree (plot_id, as_of, feature_version,
input_manifest_sha256);
```

triggers

```
CREATE TRIGGER guard_snapshot BEFORE INSERT OR UPDATE ON seagis_features.feature_snapshot FOR EACH
ROW EXECUTE FUNCTION seagis_features.guard_snapshot();
CREATE TRIGGER no_snapshot_delete BEFORE DELETE ON seagis_features.feature_snapshot FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.26 seagis_quality.alignment_snapshot

Immutable hourly QC/alignment document; always training_authorized=false.

Attribute	Definition
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, mode, start_at, end_at, policy_version, source_sha256)
Rows	UNVERIFIED
Writer	seagis_quality.save_snapshot
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

seagis_quality.alignment_snapshot.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.id

seagis_quality.alignment_snapshot.plot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND

Attribute	Definition
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (plot_id) REFERENCES seagis.plot(id)
Business / technical meaning	Cultivation plot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A3 build -> save_snapshot → document.plot_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.plot_id

seagis_quality.alignment_snapshot.mode

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	demo_review or operational_review according to plot provenance
Valid values / checks	CHECK (mode = ANY (ARRAY['demo_review'::text, 'operational_review'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A3 build -> save_snapshot → document.mode (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.mode

seagis_quality.alignment_snapshot.start_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Inclusive alignment-window start
Valid values / checks	CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A3 build -> save_snapshot → document.start_at (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.start_at

seagis_quality.alignment_snapshot.end_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Exclusive alignment-window end
Valid values / checks	CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval)
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A3 build -> save_snapshot → document.end_at (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.end_at

seagis_quality.alignment_snapshot.policy_version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version)
Business / technical meaning	QC policy version reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A3 build -> save_snapshot → registered policy.version (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.policy_version

seagis_quality.alignment_snapshot.source_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Source-bundle digest for source-change detection
Valid values / checks	CHECK (source_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A3 build -> save_snapshot → document.source_sha256 checked against fresh source_bundle (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.source_sha256

seagis_quality.alignment_snapshot.input_manifest_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Digest of ordered lineage-input manifest
Valid values / checks	CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A3 build -> save_snapshot → SHA256 sorted distinct at|source_ref lines (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.input_manifest_sha256

seagis_quality.alignment_snapshot.body

Attribute	Definition
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Complete alignment document, including rows and QC report
Valid values / checks	CHECK (jsonb_typeof(body) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	A3 build -> save_snapshot → document (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.body

seagis_quality.alignment_snapshot.body_sha256

Attribute	Definition
Type / nullable	text / true
Default / generated	seagis.jsonb_sha256(body)
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Generated digest of JSONB body
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	DB generated → seagis.jsonb_sha256(body) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.body_sha256

seagis_quality.alignment_snapshot.created_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row creation instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A3 build -> save_snapshot → SQL default clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.created_at

Constraints, indexes and triggers

constraints

```
CHECK (jsonb_typeof(body) = 'object'::text);
CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval);
CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (mode = ANY (ARRAY['demo_review'::text, 'operational_review'::text]));
PRIMARY KEY (id);
FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);
UNIQUE (plot_id, mode, start_at, end_at, policy_version, source_sha256);
FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version);
CHECK (source_sha256 ~ '^[0-9a-f]{64}$'::text);
```

indexes

```
CREATE UNIQUE INDEX alignment_snapshot_pkey ON seagis_quality.alignment_snapshot USING btree (id);
CREATE UNIQUE INDEX alignment_snapshot_plot_id_mode_start_at_end_at_policy_vers_key ON
seagis_quality.alignment_snapshot USING btree (plot_id, mode, start_at, end_at, policy_version,
```

source_sha256);

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.alignment_snapshot FOR EACH ROW
 EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.27 seagis_quality.policy

Immutable registry of QC policy versions and builder code hashes.

Attribute	Definition
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (version)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/quality/store.py::register_policy
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

seagis_quality.policy.version

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Policy version identity
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → register_policy policy[version] (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.version

seagis_quality.policy.definition

Attribute	Definition
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE

Attribute	Definition
Business / technical meaning	Registered QC policy document
Valid values / checks	CHECK (jsonb_typeof(definition) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	owning routine / caller → register_policy JSON policy argument (config/a3_quality.json) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.definition

seagis_quality.policy.policy_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Policy-content digest
Valid values / checks	CHECK (policy_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → register_policy policy_hash from digest(policy) (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.policy_sha256

seagis_quality.policy.code_sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Alignment-builder source digest
Valid values / checks	CHECK (code_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	owning routine / caller → register_policy builder_hash from code_hash() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.code_sha256

seagis_quality.policy.registered_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Immutable policy registration instant
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	caller / SQL default → clock_timestamp() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.registered_at

Constraints, indexes and triggers

constraints

```
CHECK (code_sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (jsonb_typeof(definition) = 'object'::text);
PRIMARY KEY (version);
CHECK (policy_sha256 ~ '^[0-9a-f]{64}$'::text);
```

indexes

```
CREATE UNIQUE INDEX policy_pkey ON seagis_quality.policy USING btree (version);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.policy FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.28 seagis_quality.snapshot_input

Per-hour measurement lineage of an alignment snapshot.

Attribute	Definition
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (snapshot_id, aligned_at, source_ref)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_quality.save_snapshot
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

seagis_quality.snapshot_input.snapshot_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id)
Business / technical meaning	Parent snapshot reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A3 document.rows[].cells[].lineage → saved snapshot UUID (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.snapshot_id

seagis_quality.snapshot_input.aligned_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	UTC hour of alignment row
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	A3 document.rows[].cells[].lineage → row.at::timestamptz (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.aligned_at

seagis_quality.snapshot_input.source_ref

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Typed reference observation:uuid or forecast_value:uuid
Valid values / checks	CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE

Attribute	Definition
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	A3 document.rows[].cells[].lineage → lineage ref (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.source_ref

seagis_quality.snapshot_input.observation_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Business / technical meaning	Local sensor reading reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1); CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A3 document.rows[].cells[].lineage → UUID if ref prefix observation else NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.observation_id

seagis_quality.snapshot_input.forecast_value_id

Attribute	Definition
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id)
Business / technical meaning	Single regional model value reference
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1); CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	A3 document.rows[].cells[].lineage → UUID if ref prefix forecast_value else NULL (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.forecast_value_id

Constraints, indexes and triggers

constraints

```
CHECK (num_nonnulls(observation_id, forecast_value_id) = 1);
CHECK (source_ref =
CASE
   WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id
   ELSE 'forecast_value:'::text || forecast_value_id
END);
FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
PRIMARY KEY (snapshot_id, aligned_at, source_ref);
FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id);
```

indexes

```
CREATE UNIQUE INDEX snapshot_input_pkey ON seagis_quality.snapshot_input USING btree (snapshot_id,
aligned_at, source_ref);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.snapshot_input FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

4.29 seagis_raw.raw_payload

Immutable raw manifest: digest, hash basis, licence and retention mode.

Attribute	Definition
Schema / domain	seagis_raw
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic); UNIQUE (ingestion_run_id, sha256)
Rows	UNVERIFIED
Writer	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Reader	A3 source_bundle; provenance guards
Authority	Records of this domain; registry/config or original source remains traceable
Lifecycle	See triggers below and Chapter 9; no verified TTL
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

seagis_raw.raw_payload.id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Row identity under the table primary key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001

Attribute	Definition
Origin / transformation	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.id

seagis_raw.raw_payload.ingestion_run_id

Attribute	Definition
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Business / technical meaning	Source ingestion reference
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → run_id (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.ingestion_run_id

seagis_raw.raw_payload.dataset_key

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Business / technical meaning	Registered dataset key
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → s.dataset_key (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.dataset_key

seagis_raw.raw_payload.is_synthetic

Attribute	Definition
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)

Attribute	Definition
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Business / technical meaning	Synthetic provenance flag, separate from QC
Valid values / checks	CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → s.is_synthetic (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.is_synthetic

seagis_raw.raw_payload.sha256

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Raw digest under hash_basis, not necessarily network bytes
Valid values / checks	CHECK (sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN seagis.jsonb_sha256(packet) ELSE wire_sha256 END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.sha256

seagis_raw.raw_payload.hash_basis

Attribute	Definition
Type / nullable	text / false
Default / generated	'source_bytes'::text
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Byte or serialization convention used for hashing
Valid values / checks	CHECK (hash_basis = ANY (ARRAY['source_bytes'::text, 'synthetic_json_canonical_v1'::text, ' jsonb_utf8_v1'::text, 'toolbox_subset_json_v1'::text])); CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	caller / SQL default → 'source_bytes'::text (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.hash_basis

seagis_raw.raw_payload.received_at

Attribute	Definition
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Raw receipt instant at worker
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; timestamptz instant, PostgreSQL microsecond precision; display follows session. A1 API requires UTC Z. next_attempt_at supports -infinity.
Example	2026-10-21T12:00:00Z
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → received (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.received_at

seagis_raw.raw_payload.storage_mode

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	inline, object or metadata_only determines content location
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL); CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text]))); CHECK (storage_mode = ANY (ARRAY['inline'::text, 'object'::text, 'metadata_only'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN 'inline' ELSE 'metadata_only' END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.storage_mode

seagis_raw.raw_payload.payload

Attribute	Definition
Type / nullable	jsonb / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Synthetic raw JSON content when inline
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL)

Attribute	Definition
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN packet ELSE NULL END (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.payload

seagis_raw.raw_payload.object_uri

Attribute	Definition
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Private object reference in object mode
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.object_uri

seagis_raw.raw_payload.byte_size

Attribute	Definition
Type / nullable	bigint / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Byte count reported at capture
Valid values / checks	CHECK (byte_size >= 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → wire_bytes (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.byte_size

seagis_raw.raw_payload.format

Attribute	Definition
Type / nullable	text / false

Attribute	Definition
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Raw representation media type or format
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → 'application/json' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.format

seagis_raw.raw_payload.licence_snapshot

Attribute	Definition
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Licence statement at capture
Valid values / checks	No local CHECK; cross-table constraints/triggers still apply
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	See enum/CHECK below; arbitrary example is not presumed valid
Origin / transformation	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → 'project-owned; A1 synthetic development / real retention gated' (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.licence_snapshot

seagis_raw.raw_payload.provider_metadata

Attribute	Definition
Type / nullable	jsonb / false
Default / generated	'{}'::jsonb
PK / unique	false / false (single-column uniqueness; composites below)
FK	NOT APPLICABLE
Business / technical meaning	Sanitized provider and request metadata
Valid values / checks	CHECK (jsonb_typeof(provider_metadata) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Origin / transformation	caller / SQL default → '{}'::jsonb (VERIFIED)
Mutation / consumers	Follows table triggers/mutations, not assumed mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.provider_metadata

Constraints, indexes and triggers

constraints

```
CHECK (byte_size >= 0);
CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode
= 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~
'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS
NULL);
CHECK (hash_basis = ANY (ARRAY['source_bytes'::text, 'synthetic_json_canonical_v1'::text,
'jsonb_utf8_v1'::text, 'toolbox_subset_json_v1'::text]));
UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic);
FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id,
dataset_key, is_synthetic);
UNIQUE (ingestion_run_id, sha256);
CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY
(ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])));
PRIMARY KEY (id);
CHECK (jsonb_typeof(provider_metadata) = 'object'::text);
CHECK (sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (storage_mode = ANY (ARRAY['inline'::text, 'object'::text, 'metadata_only'::text]));
```

indexes

```
CREATE UNIQUE INDEX raw_payload_id_ingestion_run_id_dataset_key_is_synthetic_key ON
seagis_raw.raw_payload USING btree (id, ingestion_run_id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX raw_payload_ingestion_run_id_sha256_key ON seagis_raw.raw_payload USING btree
(ingestion_run_id, sha256);
CREATE UNIQUE INDEX raw_payload_pkey ON seagis_raw.raw_payload USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_raw.raw_payload FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_raw BEFORE INSERT ON seagis_raw.raw_payload FOR EACH ROW EXECUTE FUNCTION
seagis.guard_raw();
```

RLS at snapshot: True; FORCE RLS: False. Current membership/grants UNVERIFIED.

5 Entity Relationships

All relationships below are physical foreign keys from the historical catalog and migrations. A parent has 0..N children unless the child join key is unique, limiting it to 0..1; a child references exactly one parent when all columns are NOT NULL, or 0..1 when nullable under MATCH SIMPLE. Composite foreign keys bind dataset/synthetic flags to prevent lineage mixing. A foreign key does not require a parent to have children.

FK1 seagis.source → seagis.dataset

Attribute	Definition
Constraint	dataset_source_key_fkey
Join	(source_key) → (source_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Dataset identity, version, cadence, licence and ingestion gates.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

FOREIGN KEY (source_key) REFERENCES seagis.source(source_key);

FK2 seagis.dataset → seagis.dataset_parameter

Attribute	Definition
Constraint	dataset_parameter_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Source-variable to parameter contract and conversion rules.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK3 seagis.parameter → seagis.dataset_parameter

Attribute	Definition
Constraint	dataset_parameter_parameter_code_unit_fkey
Join	(parameter_code,unit) → (parameter_code,unit)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Source-variable to parameter contract and conversion rules.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit);

FK4 seagis.dataset → seagis.forecast_run

Attribute	Definition
Constraint	forecast_run_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Regional model capture/vintage; unknown publication time remains null.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK5 seagis_raw.raw_payload → seagis.forecast_run

Attribute	Definition
Constraint	forecast_run_raw_payload_id_ingestion_run_id_dataset_key_i_fkey
Join	(raw_payload_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Regional model capture/vintage; unknown publication time remains null.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES
seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);

FK6 seagis.dataset_parameter → seagis.forecast_value

Attribute	Definition
Constraint	forecast_value_dataset_key_parameter_code_native_unit_unit_fkey
Join	(dataset_key,parameter_code,native_unit,unit,conversion_code) → (dataset_key,parameter_code,native_unit,unit,conversion_code)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Parameter value per run, grid and valid time; not duplicated per plot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES
seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);

FK7 seagis.forecast_run → seagis.forecast_value

Attribute	Definition
Constraint	forecast_value_forecast_run_id_dataset_key_is_synthetic_fkey
Join	(forecast_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Parameter value per run, grid and valid time; not duplicated per plot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id,
dataset_key, is_synthetic);

FK8 seagis.source_grid → seagis.forecast_value

Attribute	Definition
Constraint	forecast_value_source_grid_id_dataset_key_is_synthetic_fkey
Join	(source_grid_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Parameter value per run, grid and valid time; not duplicated per plot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id,
dataset_key, is_synthetic);

FK9 seagis.dataset → seagis.ingestion_run

Attribute	Definition
Constraint	ingestion_run_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit of a logical ingestion and its completion state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK10 seagis.dataset_parameter → seagis.observation

Attribute	Definition
Constraint	observation_dataset_key_parameter_code_native_unit_unit_co_fkey
Join	(dataset_key,parameter_code,native_unit,unit,conversion_code) → (dataset_key,parameter_code,native_unit,unit,conversion_code)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Local sensor water-temperature readings, retaining native values, QC and raw lineage.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);

FK11 seagis_raw.raw_payload → seagis.observation

Attribute	Definition
Constraint	observation_raw_payload_id_ingestion_run_id_dataset_key_is_fkey
Join	(raw_payload_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Local sensor water-temperature readings, retaining native values, QC and raw lineage.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);

FK12 seagis.sensor → seagis.observation

Attribute	Definition
Constraint	observation_sensor_id_plot_id_dataset_key_is_synthetic_fkey
Join	(sensor_id,plot_id,dataset_key,is_synthetic) → (id,plot_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Local sensor water-temperature readings, retaining native values, QC and raw lineage.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic);

FK13 seagis.plot → seagis.plot_grid_mapping

Attribute	Definition
Constraint	plot_grid_mapping_plot_id_is_synthetic_fkey

Attribute	Definition
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Operational plot-to-grid mapping with effective and recorded times.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK14 seagis.source_grid → seagis.plot_grid_mapping

Attribute	Definition
Constraint	plot_grid_mapping_source_grid_id_fkey
Join	(source_grid_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Operational plot-to-grid mapping with effective and recorded times.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id);

FK15 seagis.forecast_value → seagis.quality_event

Attribute	Definition
Constraint	quality_event_forecast_value_id_dataset_key_is_synthetic_fkey
Join	(forecast_value_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Append-only QC log for ingestion or one measurement.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic);

FK16 seagis.ingestion_run → seagis.quality_event

Attribute	Definition
Constraint	quality_event_ingestion_run_id_dataset_key_is_synthetic_fkey
Join	(ingestion_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1

Attribute	Definition
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Append-only QC log for ingestion or one measurement.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);

FK17 seagis.observation → seagis.quality_event

Attribute	Definition
Constraint	quality_event_observation_id_ingestion_run_id_dataset_key__fkey
Join	(observation_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Append-only QC log for ingestion or one measurement.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic);

FK18 seagis.dataset_parameter → seagis.sensor

Attribute	Definition
Constraint	sensor_dataset_key_parameter_code_fkey
Join	(dataset_key,parameter_code) → (dataset_key,parameter_code)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Temperature sensor deployment identity, plot, depth and active state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code);

FK19 seagis.dataset → seagis.sensor

Attribute	Definition
Constraint	sensor_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false

Attribute	Definition
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Temperature sensor deployment identity, plot, depth and active state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK20 seagis.plot → seagis.sensor

Attribute	Definition
Constraint	sensor_plot_id_is_synthetic_fkey
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Temperature sensor deployment identity, plot, depth and active state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK21 seagis.dataset → seagis.source_grid

Attribute	Definition
Constraint	source_grid_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Regional cell and depth-layer identity; distinct from cultivation plots.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK22 seagis.sensor → seagis_api.credential

Attribute	Definition
Constraint	credential_sensor_id_fkey
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	API actor identity using token hashes, expiry and revocation.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK23 seagis.ingestion_run → seagis_api.packet_receipt

Attribute	Definition
Constraint	packet_receipt_ingestion_run_id_fkey
Join	(ingestion_run_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Sensor-packet idempotency key and semantic digest.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id);

FK24 seagis.observation → seagis_api.packet_receipt

Attribute	Definition
Constraint	packet_receipt_observation_id_fkey
Join	(observation_id) → (id)
Cardinality	1:0..1; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Sensor-packet idempotency key and semantic digest.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK25 seagis.sensor → seagis_api.packet_receipt

Attribute	Definition
Constraint	packet_receipt_sensor_id_fkey
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Sensor-packet idempotency key and semantic digest.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK26 seagis_api.credential → seagis_api.reader_plot

Attribute	Definition
Constraint	reader_plot_credential_id_fkey
Join	(credential_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Plot allowlist for a reader credential.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);

FK27 seagis.plot → seagis_api.reader_plot

Attribute	Definition
Constraint	reader_plot_plot_id_fkey
Join	(plot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Plot allowlist for a reader credential.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);

FK28 seagis_api.credential → seagis_api.request_event

Attribute	Definition
Constraint	request_event_credential_id_fkey
Join	(credential_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Authenticated API response audit and rate-limit basis.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);

FK29 seagis.sensor → seagis_api.request_event

Attribute	Definition
Constraint	request_event_sensor_id_fkey

Attribute	Definition
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Authenticated API response audit and rate-limit basis.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK30 seagis_environment.policy → seagis_environment.attempt

Attribute	Definition
Constraint	attempt_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit of a fetch attempt, including duplicate and abandoned outcomes.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK31 seagis.forecast_run → seagis_environment.attempt

Attribute	Definition
Constraint	attempt_forecast_run_id_fkey
Join	(forecast_run_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit of a fetch attempt, including duplicate and abandoned outcomes.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id);

FK32 seagis_environment.policy → seagis_environment.demo_grid_mapping

Attribute	Definition
Constraint	demo_grid_mapping_dataset_key_fkey
Join	(dataset_key) → (dataset_key)

Attribute	Definition
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge from two fictional plots to real grids for preview, not training authorization.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK33 seagis.plot → seagis_environment.demo_grid_mapping

Attribute	Definition
Constraint	demo_grid_mapping_plot_id_plot_is_synthetic_fkey
Join	(plot_id,plot_is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge from two fictional plots to real grids for preview, not training authorization.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK34 seagis.source_grid → seagis_environment.demo_grid_mapping

Attribute	Definition
Constraint	demo_grid_mapping_source_grid_id_dataset_key_grid_is_synth_fkey
Join	(source_grid_id,dataset_key,grid_is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge from two fictional plots to real grids for preview, not training authorization.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id,
dataset_key, is_synthetic);

FK35 seagis.dataset → seagis_environment.policy

Attribute	Definition
Constraint	policy_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..1; child → parent 1
Optional	false

Attribute	Definition
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Regional fetch, licence, scheduler cadence, staleness and retention policy.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK36 seagis_environment.policy → seagis_environment.schedule_state

Attribute	Definition
Constraint	schedule_state_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..1; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Mutable lease, due time, backoff and last-success state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK37 seagis.forecast_run → seagis_environment.schedule_state

Attribute	Definition
Constraint	schedule_state_last_forecast_run_id_fkey
Join	(last_forecast_run_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Mutable lease, due time, backoff and last-success state.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id);

FK38 seagis.forecast_value → seagis_features.feature_input

Attribute	Definition
Constraint	feature_input_forecast_value_id_fkey
Join	(forecast_value_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Foreign-key input lineage and cutoff guards for feature snapshots.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);

FK39 seagis.observation → seagis_features.feature_input

Attribute	Definition
Constraint	feature_input_observation_id_fkey
Join	(observation_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Foreign-key input lineage and cutoff guards for feature snapshots.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK40 seagis_features.feature_snapshot → seagis_features.feature_input

Attribute	Definition
Constraint	feature_input_snapshot_id_fkey
Join	(snapshot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Foreign-key input lineage and cutoff guards for feature snapshots.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id);

FK41 seagis.plot → seagis_features.feature_snapshot

Attribute	Definition
Constraint	feature_snapshot_plot_id_is_synthetic_fkey
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Draft/sealed feature container with cutoff and hashes; no production builder found.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK42 seagis.plot → seagis_quality.alignment_snapshot

Attribute	Definition
Constraint	alignment_snapshot_plot_id_fkey
Join	(plot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Immutable hourly QC/alignment document; always training_authorized=false.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);

FK43 seagis_quality.policy → seagis_quality.alignment_snapshot

Attribute	Definition
Constraint	alignment_snapshot_policy_version_fkey
Join	(policy_version) → (version)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Immutable hourly QC/alignment document; always training_authorized=false.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version);

FK44 seagis.forecast_value → seagis_quality.snapshot_input

Attribute	Definition
Constraint	snapshot_input_forecast_value_id_fkey
Join	(forecast_value_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Per-hour measurement lineage of an alignment snapshot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);

FK45 seagis.observation → seagis_quality.snapshot_input

Attribute	Definition
Constraint	snapshot_input_observation_id_fkey

Attribute	Definition
Join	(observation_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Per-hour measurement lineage of an alignment snapshot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK46 seagis_quality.alignment_snapshot → seagis_quality.snapshot_input

Attribute	Definition
Constraint	snapshot_input_snapshot_id_fkey
Join	(snapshot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Per-hour measurement lineage of an alignment snapshot.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id);

FK47 seagis.ingestion_run → seagis_raw.raw_payload

Attribute	Definition
Constraint	raw_payload_ingestion_run_id_dataset_key_is_synthetic_fkey
Join	(ingestion_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Immutable raw manifest: digest, hash basis, licence and retention mode.
Integrity risk	FK rejects orphans; nullable composites use MATCH SIMPLE. QC/time semantics need additional guards.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);

Additional logical relationships: two forecast_value aliases in current_speed join on forecast_run_id, source_grid_id and valid_at; parameter_code distinguishes u/v. This is a view join, not a new FK. source_ref uses typed prefixes, a CHECK and two alternative FKs; no invented FK targets a polymorphic string. No self-referencing FK was found. reader_plot, dataset_parameter, feature_input and snapshot_input act as association/lineage entities, with composite keys/uniqueness recorded in the dictionary. AppPOC Farmer→CultivationPlot is an in-memory relationship and is excluded from the 47 PostgreSQL FKs.

Full and per-schema ERDs are in architecture/ERD.md.

The complete ERD contains every physical-table column; PK/FK/UK retain physical names. FK1–FK47 are explained in Chapter 5. Node IDs use underscores for Mermaid syntax only; labels retain schema.table. current_speed is documented separately because it has no physical FKs.



seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.forecast_run**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

→ **seagis.dataset**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.parameter

parameter_code : text [PK]
unit : text
description : text

No outgoing physical FK

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.plot_grid_mapping

id : uuid [PK]
plot_id : uuid [FK]
source_grid_id : uuid [FK]
is_synthetic : boolean [FK]
sampling_method : text
distance_m : finite_float
representativeness : text
effective_from : timestamptz
effective_until : timestamptz
recorded_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.quality_event

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
event_key : text
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
event_code : text
severity : text
details : jsonb
created_at : timestamptz

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.ingestion_run**
1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

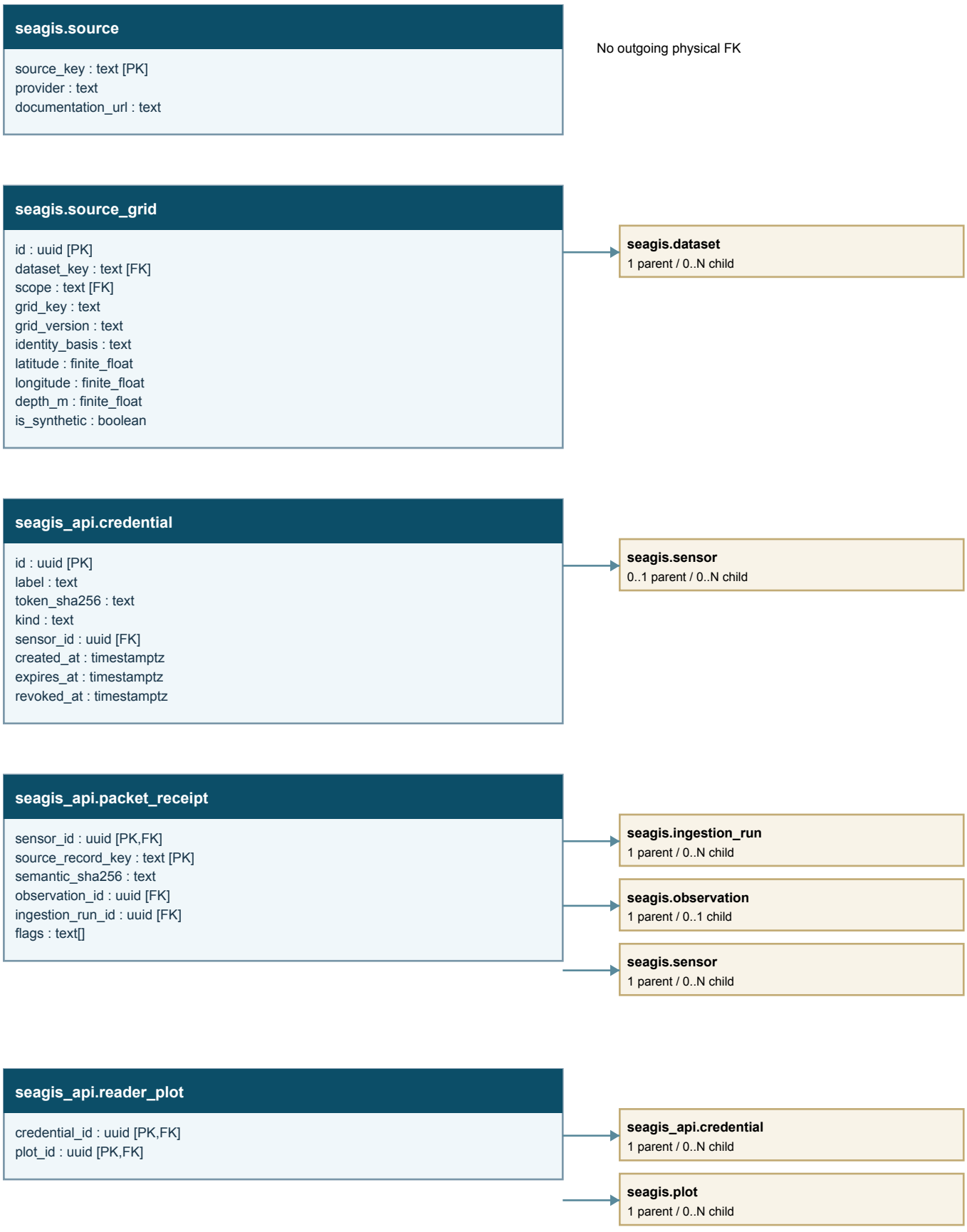
seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child


seagis.source
source_key : text [PK]
provider : text
documentation_url : text
No outgoing physical FK
seagis.source_grid
id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean
seagis.dataset
1 parent / 0..N child
seagis_api.credential
id : uuid [PK]
label : text
token_sha256 : text
kind : text
sensor_id : uuid [FK]
created_at : timestamptz
expires_at : timestamptz
revoked_at : timestamptz
seagis.sensor
0..1 parent / 0..N child
seagis_api.packet_receipt
sensor_id : uuid [PK,FK]
source_record_key : text [PK]
semantic_sha256 : text
observation_id : uuid [FK]
ingestion_run_id : uuid [FK]
flags : text[]
seagis.ingestion_run
1 parent / 0..N child
seagis.observation
1 parent / 0..1 child
seagis.sensor
1 parent / 0..N child
seagis_api.reader_plot
credential_id : uuid [PK,FK]
plot_id : uuid [PK,FK]
seagis_api.credential
1 parent / 0..N child
seagis.plot
1 parent / 0..N child

seagis_api.request_event

id : uuid [PK]
credential_id : uuid [FK]
sensor_id : uuid [FK]
http_status : integer
code : text
created_at : timestamptz

→ **seagis_api.credential**
1 parent / 0..N child

→ **seagis.sensor**
0..1 parent / 0..N child

seagis_api.temperature_policy

id : boolean [PK]
version : text
min_degc : float8
max_degc : float8
max_step_degc : float8
future_tolerance_seconds : integer
stale_after_seconds : integer
requests_per_minute : integer

No outgoing physical FK

seagis_environment.attempt

id : uuid [PK]
dataset_key : text [FK]
started_at : timestamptz
completed_at : timestamptz
status : text
error_code : text
forecast_run_id : uuid [FK]

→ **seagis_environment.policy**
1 parent / 0..N child

→ **seagis.forecast_run**
0..1 parent / 0..N child

seagis_environment.demo_grid_mapping

plot_id : uuid [PK,FK]
plot_is_synthetic : boolean [FK]
source_grid_id : uuid [PK,FK]
dataset_key : text [FK]
grid_is_synthetic : boolean [FK]
example_latitude : finite_float
example_longitude : finite_float
coordinate_status : text
distance_m : finite_float
sampling_method : text
created_at : timestamptz

→ **seagis_environment.policy**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis_environment.policy

dataset_key : text [PK,FK]
enabled : boolean
purpose : text
licence_url : text
terms_review : text
attribution : text
update_seconds : integer
release_offset_seconds : integer
stale_seconds : integer
retention_mode : text
reviewed_at : timestamptz

→ **seagis.dataset**
1 parent / 0..1 child

seagis_environment.schedule_state

dataset_key : text [PK,FK]
lease_id : uuid
lease_until : timestamptz
next_attempt_at : timestamptz
last_success_at : timestamptz
last_attempt_at : timestamptz
last_error_code : text
failures : integer
last_forecast_run_id : uuid [FK]

→ **seagis_environment.policy**
1 parent / 0..1 child

→ **seagis.forecast_run**
0..1 parent / 0..N child

seagis_features.feature_input

id : uuid [PK]
snapshot_id : uuid [FK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

→ **seagis_features.feature_snapshot**
1 parent / 0..N child

seagis_features.feature_snapshot

id : uuid [PK]
plot_id : uuid [FK]
is_synthetic : boolean [FK]
as_of : timestamptz
generated_at : timestamptz
feature_version : text
feature_schema_version : text
features : jsonb
missing_context : text[]
status : text
input_manifest_sha256 : text
features_sha256 : text

→ **seagis.plot**
1 parent / 0..N child

seagis_quality.alignment_snapshot

id : uuid [PK]
plot_id : uuid [FK]
mode : text
start_at : timestamptz
end_at : timestamptz
policy_version : text [FK]
source_sha256 : text
input_manifest_sha256 : text
body : jsonb
body_sha256 : text
created_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis_quality.policy**
1 parent / 0..N child

seagis_quality.policy

version : text [PK]
definition : jsonb
policy_sha256 : text
code_sha256 : text
registered_at : timestamptz

No outgoing physical FK

seagis_quality.snapshot_input

snapshot_id : uuid [PK,FK]
aligned_at : timestamptz [PK]
source_ref : text [PK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

→ **seagis_quality.alignment_snapshot**
1 parent / 0..N child

seagis_raw.raw_payload

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb

→ **seagis.ingestion_run**
1 parent / 0..N child

seagis

seagis.dataset

dataset_key : text [PK]
source_key : text [FK]
provider_product_id : text
provider_dataset_id : text
dataset_version : text
scope : text
native_cadence_seconds : integer
resolution_lat_deg : finite_float
resolution_lon_deg : finite_float
source_timezone : text
licence_url : text
attribution : text
raw_storage_permitted : boolean
real_ingestion_enabled : boolean
metadata_checked_at : timestamptz

→ **seagis.source**
1 parent / 0..N child

seagis.dataset_parameter

dataset_key : text [PK,FK]
parameter_code : text [PK,FK]
source_variable : text
native_unit : text
unit : text [FK]
conversion_code : text
conversion_factor : finite_float
conversion_offset : finite_float
temporal_support : text

seagis.dataset
1 parent / 0..N child

seagis.parameter
1 parent / 0..N child

seagis.forecast_run

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
run_key : text
run_kind : text
provider_run_id : text
model_initialized_at : timestamptz
issued_at : timestamptz
issue_time_basis : text
ingested_at : timestamptz

seagis.dataset
1 parent / 0..N child

seagis_raw.raw_payload
1 parent / 0..N child

seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

seagis.dataset_parameter
1 parent / 0..N child

seagis.forecast_run
1 parent / 0..N child

seagis.source_grid
1 parent / 0..N child

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

→ **seagis.dataset**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.parameter

parameter_code : text [PK]
unit : text
description : text

No outgoing physical FK

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.plot_grid_mapping

id : uuid [PK]
plot_id : uuid [FK]
source_grid_id : uuid [FK]
is_synthetic : boolean [FK]
sampling_method : text
distance_m : finite_float
representativeness : text
effective_from : timestamptz
effective_until : timestamptz
recorded_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.quality_event

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
event_key : text
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
event_code : text
severity : text
details : jsonb
created_at : timestamptz

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.ingestion_run**
1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

seagis.source

source_key : text [PK]
provider : text
documentation_url : text

No outgoing physical FK

seagis.source_grid

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean

seagis.dataset
1 parent / 0..N child

seagis_raw.raw_payload

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb

seagis.ingestion_run
1 parent / 0..N child

seagis_api

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

seagis.dataset
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

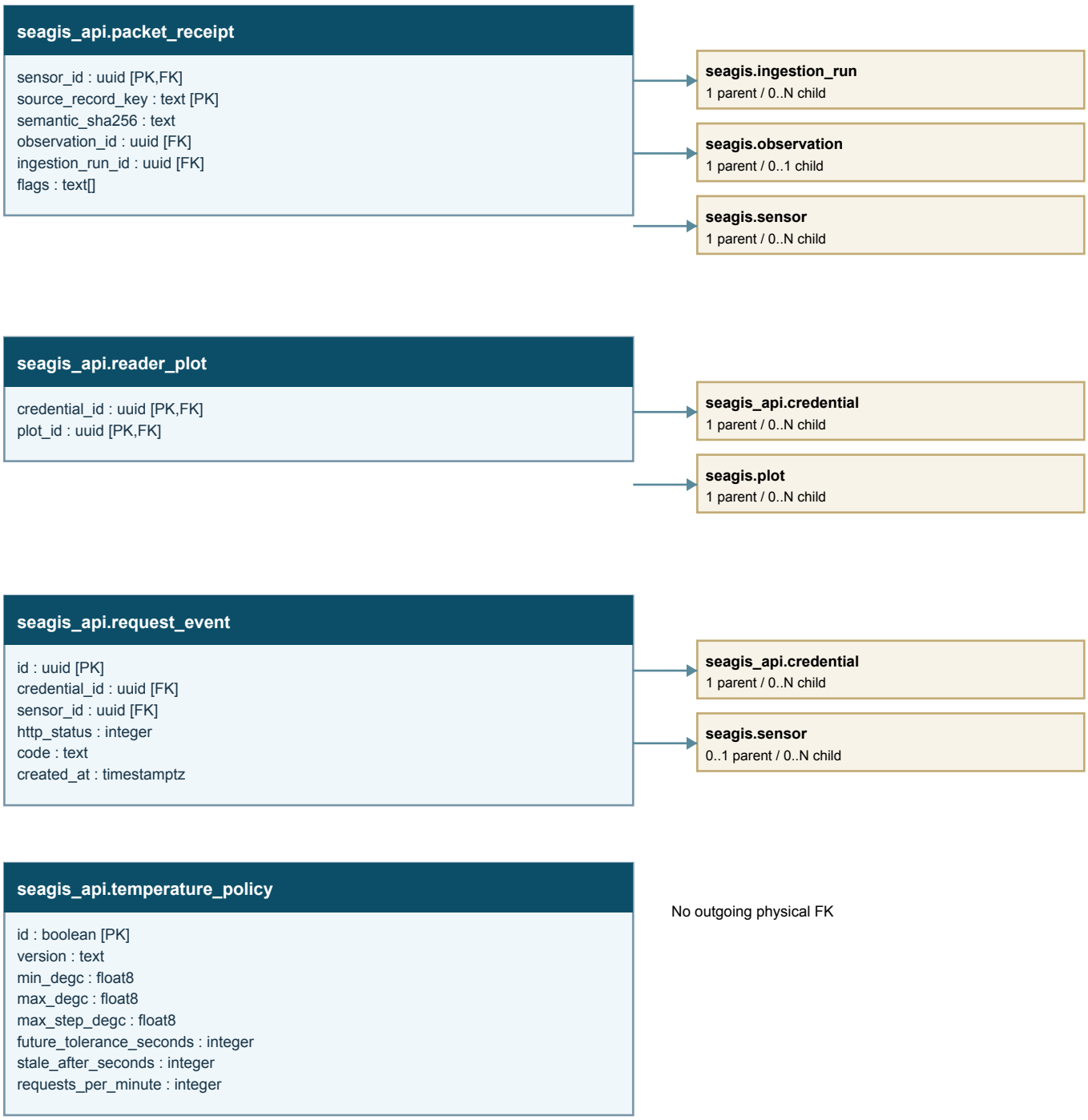
→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

seagis_api.credential

id : uuid [PK]
label : text
token_sha256 : text
kind : text
sensor_id : uuid [FK]
created_at : timestamptz
expires_at : timestamptz
revoked_at : timestamptz

→ **seagis.sensor**
0..1 parent / 0..N child



seagis_environment

seagis.dataset

dataset_key : text [PK]
source_key : text [FK]
provider_product_id : text
provider_dataset_id : text
dataset_version : text
scope : text
native_cadence_seconds : integer
resolution_lat_deg : finite_float
resolution_lon_deg : finite_float
source_timezone : text
licence_url : text
attribution : text
raw_storage_permitted : boolean
real_ingestion_enabled : boolean
metadata_checked_at : timestamptz

→ **seagis.source**
1 parent / 0..N child

seagis.forecast_run

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
run_key : text
run_kind : text
provider_run_id : text
model_initialized_at : timestamptz
issued_at : timestamptz
issue_time_basis : text
ingested_at : timestamptz

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

seagis.plot

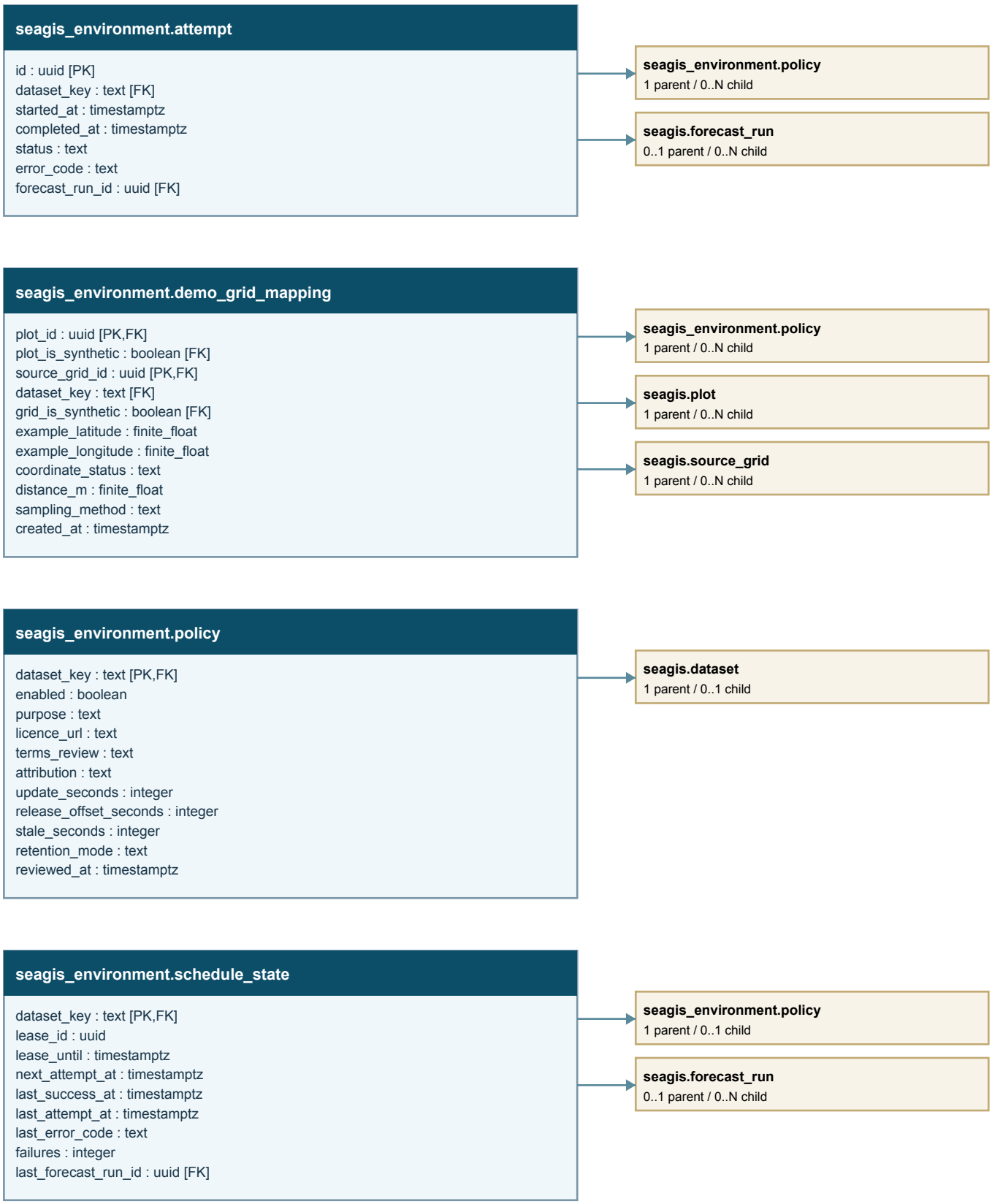
id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.source_grid

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean

→ **seagis.dataset**
1 parent / 0..N child



seagis_features

seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.forecast_run**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

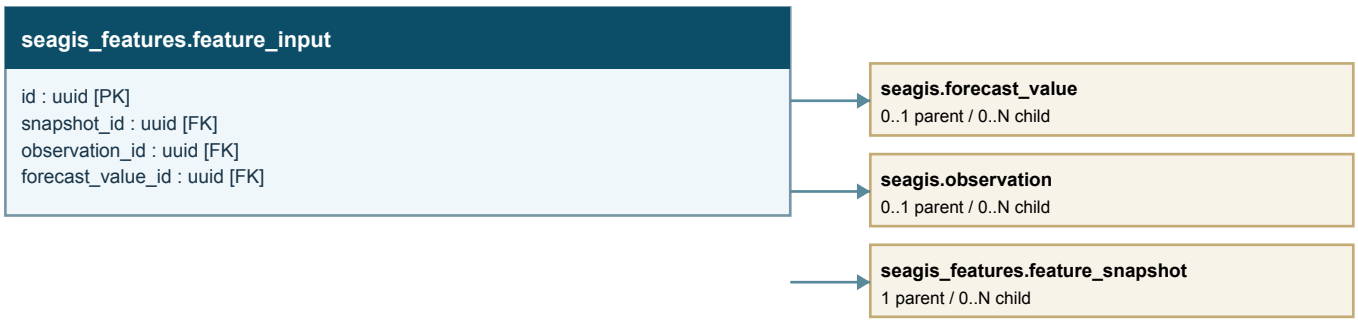
→ **seagis_raw.raw_payload**
1 parent / 0..N child

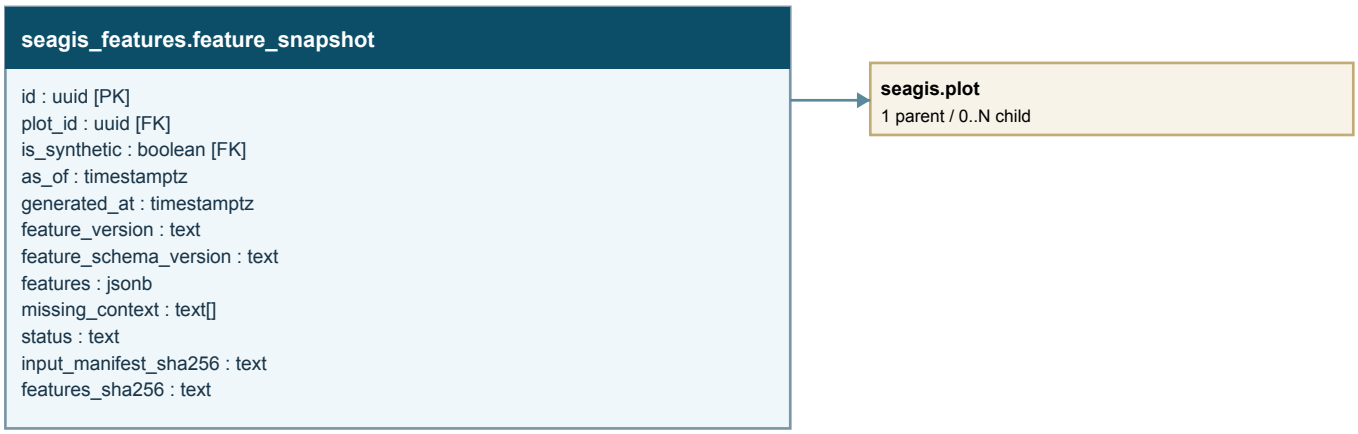
→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

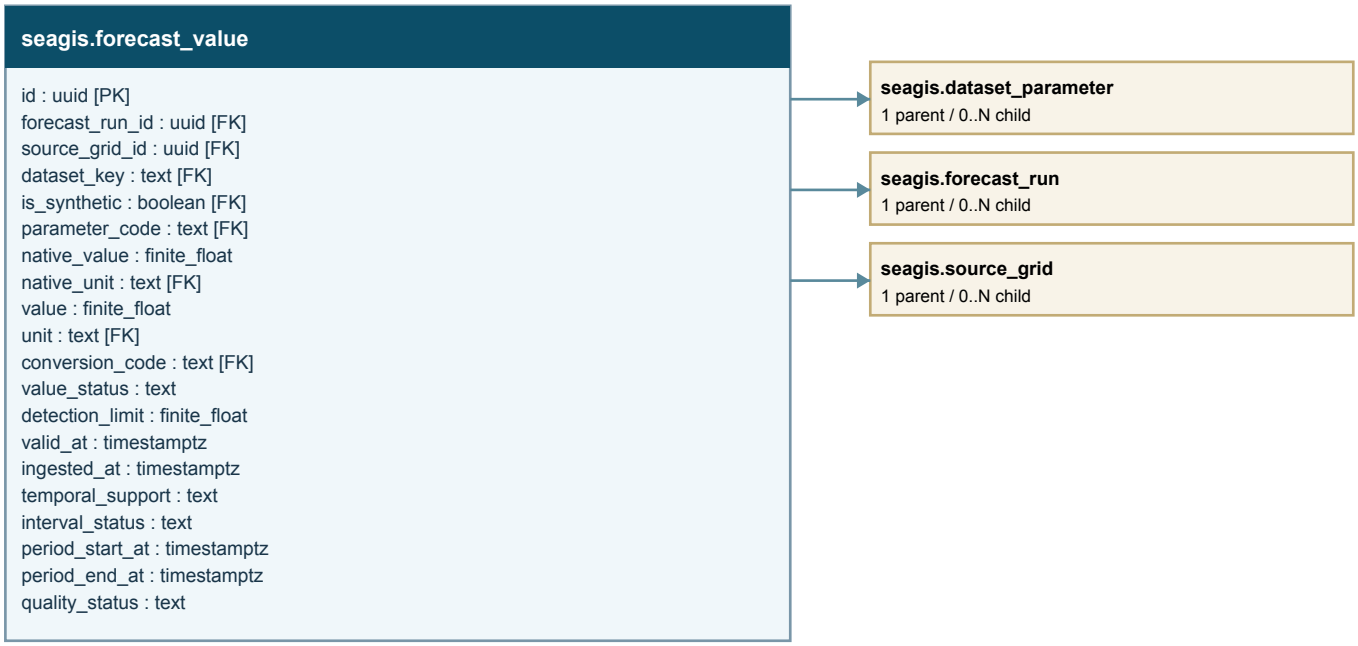
id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK





seagis_quality



seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis_quality.alignment_snapshot

id : uuid [PK]
plot_id : uuid [FK]
mode : text
start_at : timestamptz
end_at : timestamptz
policy_version : text [FK]
source_sha256 : text
input_manifest_sha256 : text
body : jsonb
body_sha256 : text
created_at : timestamptz

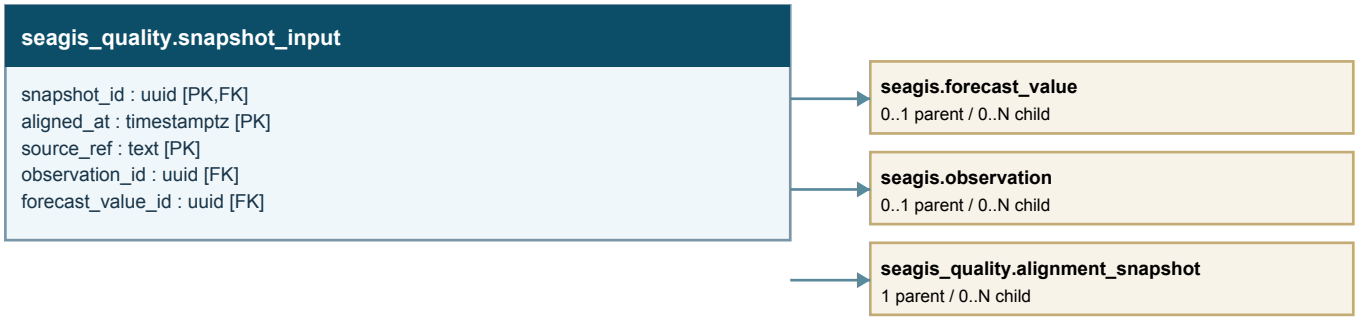
→ **seagis.plot**
1 parent / 0..N child

→ **seagis_quality.policy**
1 parent / 0..N child

seagis_quality.policy

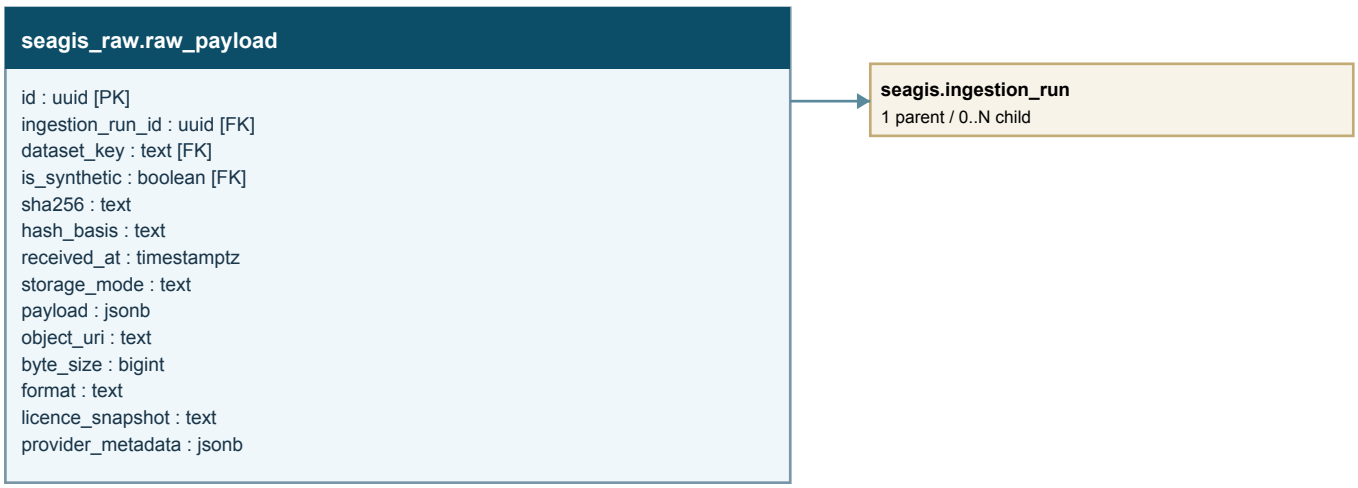
version : text [PK]
definition : jsonb
policy_sha256 : text
code_sha256 : text
registered_at : timestamptz

No outgoing physical FK


seagis_quality.snapshot_input
snapshot_id : uuid [PK,FK]
aligned_at : timestamptz [PK]
source_ref : text [PK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
seagis.forecast_value
0..1 parent / 0..N child
seagis.observation
0..1 parent / 0..N child
seagis_quality.alignment_snapshot
1 parent / 0..N child

seagis_raw


seagis.ingestion_run
id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text
seagis.dataset
1 parent / 0..N child


seagis_raw.raw_payload
id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb
seagis.ingestion_run
1 parent / 0..N child

6 Data Sources and Lineage

A1: simulator/device → JSON bytes → HTTP validation → authentication and sensor identity → SQL validation → semantic digest → raw manifest + observation + quality_event + packet_receipt → history/status. Sensor identity, plot and synthetic provenance are verified against credentials and deployment rather than accepted as arbitrary input. Missing packets do not create zero observations.

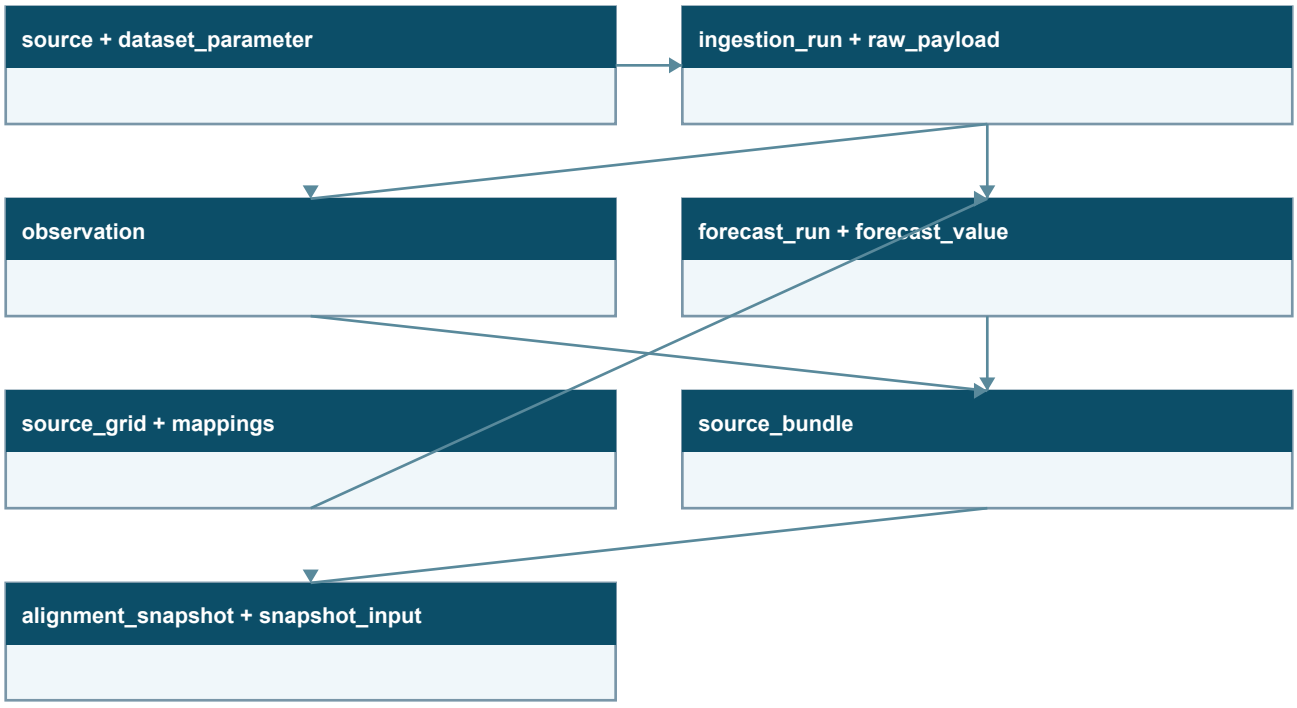
A2: registry/config → lease claim → provider fetch → metadata/unit/cadence/coordinate checks → one Copernicus cell/layer or Open-Meteo response coordinate → deduplication → semantic run_key → metadata_only manifest → forecast_run + source_grid + forecast_value + demo bridge → preview or A3. Open-Meteo source bytes and the canonical Copernicus logical subset have different hash bases. Neither STAC updates nor generationtime_ms become issued_at.

A3: source_bundle joins sensors, parameters, mappings, records and QC events; the maximum window is seven days with a 72-hour lookback and an older sensor sentinel when needed. The builder filters source_time and available_at at each cutoff, selects context, and records spatial metadata, flags and lineage. Persistence rebuilds the source bundle, verifies source/policy/code hashes, and saves a snapshot with hourly inputs. Source changes require a rebuild.

CSV export reads source_bundle in REPEATABLE READ READ ONLY or uses an offline JSON capture. Only two demo plots are supported. Regional CSV selects the latest vintage at capture time, while alignment JSON retains historical cutoffs. These representations are not interchangeable for backtesting. Every exporter output declares training_authorized=false.

Main implementation evidence: Seagis-Model/supabase/functions/temperature/handler.ts:1; Seagis-Model/scripts/environmental/copernicus.py:1; Seagis-Model/scripts/environmental/open_meteo.py:1; Seagis-Model/scripts/quality/alignment.py:1; Seagis-Model/scripts/export_experiment_csv.py:1.

Reproducible diagram: assets/diagrams/lineage.mmd.



Dataset	sensor_temperature_v1
Owner / source	device/simulator
Field	water_temperature
Native unit	degC
Unit	degC
Cadence s	3600

Support	instantaneous
Target	seagis.observation.value

Dataset	cmems_salinity_6h
Owner / source	copernicus
Field	so
Native unit	1e-3
Unit	1e-3
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (salinity)

Dataset	cmems_currents_6h
Owner / source	copernicus
Field	uo
Native unit	m s-1
Unit	m/s
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (current_eastward)

Dataset	cmems_currents_6h
Owner / source	copernicus
Field	vo
Native unit	m s-1
Unit	m/s
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (current_northward)

Dataset	cmems_nutrients_daily
Owner / source	copernicus
Field	no3
Native unit	mmol m-3
Unit	mmol/m3
Cadence s	86400
Support	mean
Target	seagis.forecast_value.value (nitrate)

Dataset	cmems_nutrients_daily
Owner / source	copernicus
Field	po4
Native unit	mmol m-3
Unit	mmol/m3

Cadence s	86400
Support	mean
Target	seagis.forecast_value.value (phosphate)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	precipitation
Native unit	mm
Unit	mm
Cadence s	3600
Support	sum
Target	seagis.forecast_value.value (precipitation)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	wind_speed_10m
Native unit	m/s
Unit	m/s
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (wind_speed_10m)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	wind_direction_10m
Native unit	°
Unit	degree
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (wind_direction_10m)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	cloud_cover
Native unit	%
Unit	percent
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (cloud_cover)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	uv_index
Native unit	

Unit	1
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (uv_index)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	temperature_2m
Native unit	°C
Unit	degC
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (air_temperature_2m)

The 3600-second sensor cadence comes from the pilot seed contract, not proof that every device always transmits on time. Provider reliability was not remeasured; retries/validation do not guarantee an SLA.

Main implementation evidence: Seagis-Model/supabase/seed.sql:1; Seagis-Model/config/a2_sources.json:1.

7 Data Ingestion and Transformation

A1 makes one SQL function call per request. The handler accepts application/json up to 8192 bytes and hashes the bytes with SHA-256 before passing JSON to SQL. SQL requires exactly eight fields, schema_version=1.0.0, degC, a uint32 sequence and UTC Z timestamps with up to six fractional digits. An actor advisory transaction lock protects rate limiting, followed by a sensor lock protecting deduplication and QC. An identical retry returns 200 duplicate; changed content under the same boot/sequence returns 409 packet_conflict. New records return 201 stored. QC may persist a rejected reading; rejected does not automatically mean HTTP rejection.

A2 claims each dataset by locking schedule_state, creating a 20-minute lease and a running attempt. The launchd installer configures a 900-second tick; this is installer configuration, not proof of an active job. Nominal policy fetch intervals are 86400 seconds for Copernicus and 21600 for Open-Meteo; native value cadence remains six-hourly/daily/hourly. Offsets are 30600 seconds for salinity/currents, 45000 for nutrients and 16200 for weather. The worker wrapper has a 900-second timeout; HTTP timeout is 40 seconds, response cap four million bytes, and three retry attempts handle network/timeouts/429/selected 5xx with backoff and jitter. Long Retry-After delays are delegated to scheduler backoff.

Open-Meteo pins gfs_global, UTC, wind_speed_unit=ms and forecast_days=3. Its parser checks units, aligned arrays and timezone before assigning UTC to suffix-free labels. Shared returned cells are deduplicated. Copernicus checks product/dataset/version, STAC cadence/grid/depth/units, loads a fixed nearest cell for the two example plots' centroid, checks catalog revision before/after fetching, and decodes through Toolbox/xarray. Nonfinite/extreme fill values become null missing; daily means are not converted into 24 independent samples.

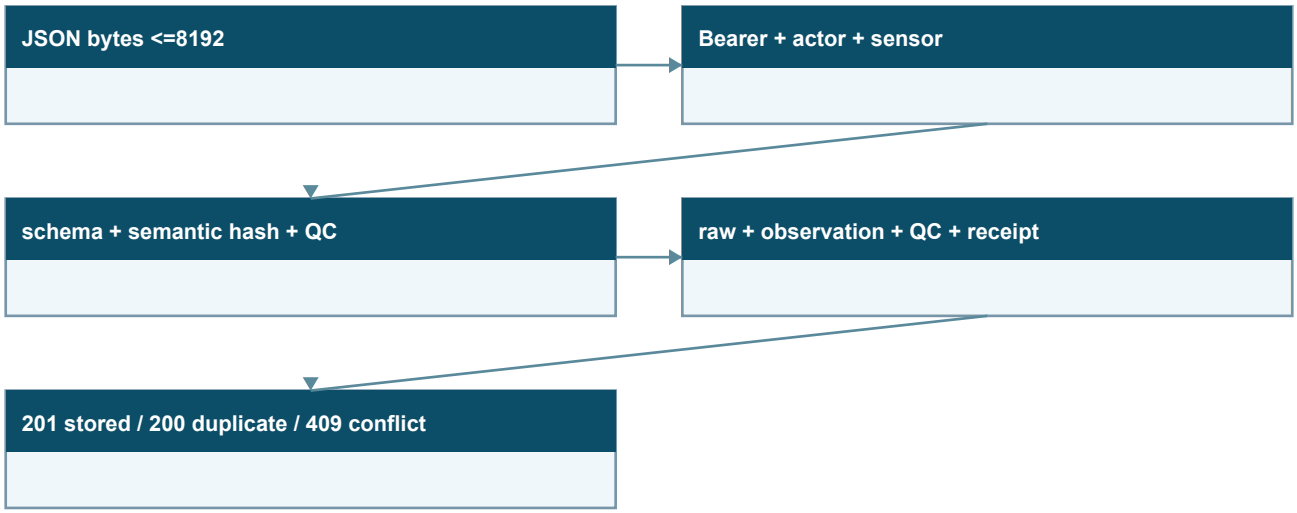
The adapter validates 1–4 grids, exactly two mappings, nonempty values and unique keys; SQL caps batches at two million bytes and 1–5000 values. persist checks lease, policy, dataset version and receipt time before inserting atomically. An existing run_key means duplicate; content changes that alter semantic identity create a new vintage. Grid and demo mapping use ON CONFLICT DO NOTHING with grid identity checks. Statement failure rolls back the entire persistence transaction; claim/attempt state remains and finish_failure is called separately. Lost commit acknowledgement requires status inspection before manual retry.

A3 neither fetches providers nor mutates observations/forecasts. build creates hourly context and detects unit mismatch, missing values, rapid change, flatline candidates, staleness and spatial uncertainty. pilot-quality-2 is an engineering policy. SQL checks lineage, cutoff, selected values, mapping and deterministic output on conflict; it does not rerun all Python QC decisions. This path has no imputation or automatic training.

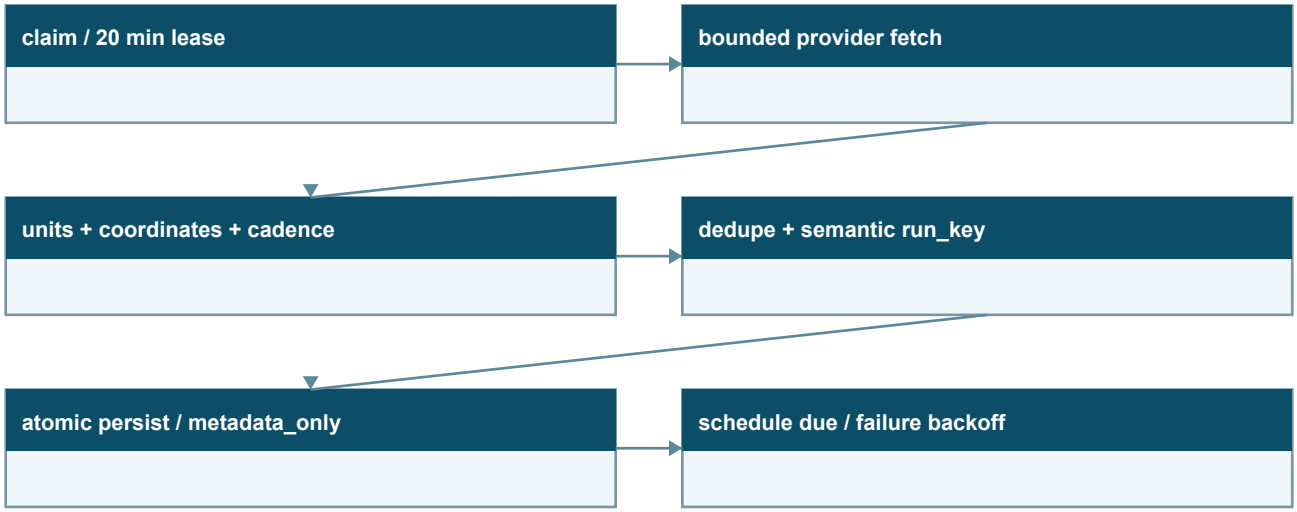
Main implementation evidence: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Main implementation evidence: Seagis-Model/scripts/ingest_environment.py:1;
Seagis-Model/scripts/environmental/common.py:1; Seagis-Model/scripts/install_a2_scheduler.py:1;
Seagis-Model/scripts/run_a2_worker.py:1; Seagis-Model/scripts/curate_environment.py:1.

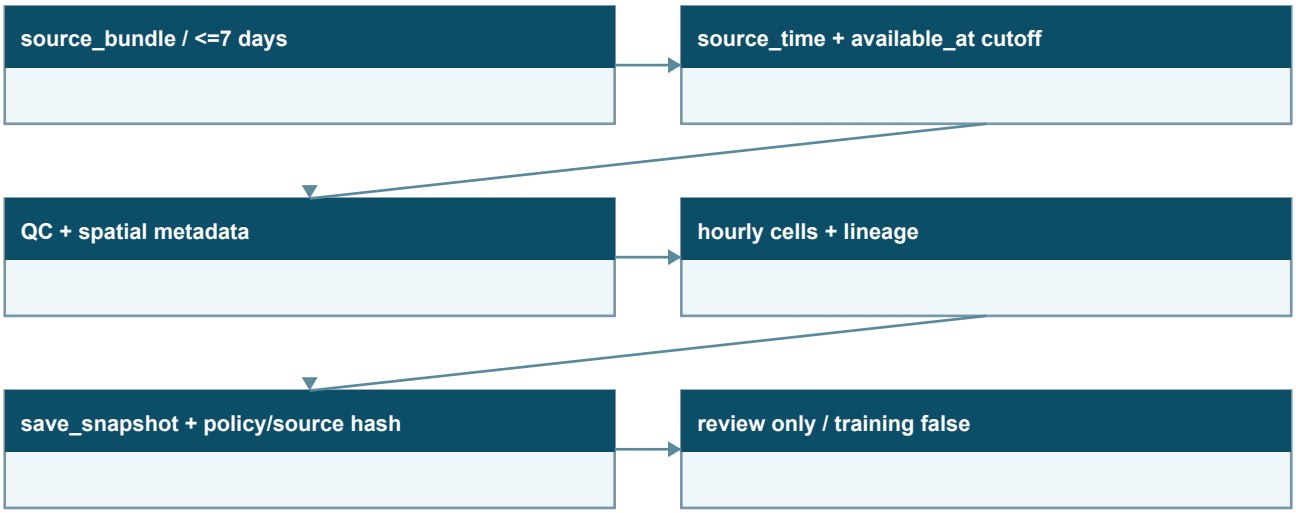
Reproducible diagram: assets/diagrams/a1.mmd.



Reproducible diagram: assets/diagrams/a2.mmd.



Reproducible diagram: assets/diagrams/a3.mmd.



7.1 Source-to-Target Mapping

Every destination column has a mapping row. VERIFIED means an expression/default/join has evidence; an administrative/fixture mapping does not imply a production loader. UNVERIFIED marks unresolved derivations and is counted as a gap. Consult the dictionary for types, meanings and guards; SQL expressions retain original identifiers.

Destination table	seagis.dataset
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	source_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	provider_product_id
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset

Column	provider_dataset_id
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'seagis-water-temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	dataset_version
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	scope
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'local_sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	native_cadence_seconds
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	3600
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	resolution_lat_deg
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	resolution_lon_deg
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	source_timezone
Source / field	caller / SQL default
Transformation	'UTC'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	licence_url
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'project-owned://sensor-data'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	attribution
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'SEAGIS; simulated until hardware commissioning'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	raw_storage_permitted
Source / field	caller / SQL default
Transformation	false
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	real_ingestion_enabled
Source / field	caller / SQL default
Transformation	false
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	metadata_checked_at

Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformation	'2026-10-21T00:00:00Z'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset_parameter
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	parameter_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	source_variable
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	native_unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'identity'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_factor
Source / field	caller / SQL default
Transformation	1
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_offset
Source / field	caller / SQL default
Transformation	0
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	temporal_support
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformation	'instantaneous'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.forecast_run
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168

Transformation	s.dataset_key
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	scope
Source / field	caller / SQL default
Transformation	'regional_model'::text
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	ing
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	raw_payload_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	raw_id
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	run_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	batch->>'run_key'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	run_kind
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	CASE WHEN d.source_key='copernicus' THEN 'analysis_forecast_unspecified' ELSE 'forecast' END
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	provider_run_id
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	model_initialized_at
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	issued_at
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	issue_time_basis
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	'unknown'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	ingested_at
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformation	t

Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_value
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	forecast_run_id
Source / field	A2 adapter batch
Transformation	persist run_id
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	source_grid_id
Source / field	A2 adapter batch
Transformation	grid lookup by dataset/key/synthetic
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	dataset_key
Source / field	A2 adapter batch
Transformation	leased dataset
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	is_synthetic
Source / field	A2 adapter batch
Transformation	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value

Column	parameter_code
Source / field	A2 adapter batch
Transformation	batch.values[].parameter
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	native_value
Source / field	A2 adapter batch
Transformation	batch.values[].value
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	native_unit
Source / field	registered seagis.dataset_parameter par
Transformation	par.native_unit
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	value
Source / field	A2 adapter batch
Transformation	batch.values[].value
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	unit
Source / field	registered seagis.dataset_parameter par
Transformation	par.unit
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	conversion_code
Source / field	registered seagis.dataset_parameter par
Transformation	par.conversion_code
Loader / trigger	seagis_environment.persist; A0 fixtures

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	value_status
Source / field	A2 adapter batch
Transformation	v.value_status
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	detection_limit
Source / field	A2 adapter batch
Transformation	omitted -> SQL NULL
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	valid_at
Source / field	A2 adapter batch
Transformation	v.valid_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	ingested_at
Source / field	A2 adapter batch
Transformation	persist t = clock_timestamp()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	temporal_support
Source / field	registered seagis.dataset_parameter par
Transformation	par.temporal_support
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	interval_status

Source / field	A2 adapter batch
Transformation	v.interval_status
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	period_start_at
Source / field	A2 adapter batch
Transformation	v.period_start_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	period_end_at
Source / field	A2 adapter batch
Transformation	v.period_end_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	quality_status
Source / field	A2 adapter batch
Transformation	default unreviewed
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.ingestion_run
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformation	s.dataset_key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	request_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformation	'a1:'||s.id||':'||key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	adapter_version
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformation	'temperature-1.0.0'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	started_at
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformation	received
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	completed_at
Source / field	owning routine / caller
Transformation	UPDATE at end of A1/A2: clock_timestamp(); NULL while running
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	status
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformation	'running'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191

Transformation	s.is_synthetic
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	error_code
Source / field	owning routine / caller
Transformation	A1/A2 successful rows omit -> NULL; failure codes tracked in environment.attempt
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.observation
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	plot_id
Source / field	A1 packet + registered identity + server
Transformation	registered sensor.plot_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	sensor_id
Source / field	A1 packet + registered identity + server
Transformation	authenticated credential.sensor_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	dataset_key
Source / field	A1 packet + registered identity + server
Transformation	registered sensor.dataset_key
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	ingestion_run_id
Source / field	A1 packet + registered identity + server
Transformation	ingest-generated run_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	raw_payload_id
Source / field	A1 packet + registered identity + server
Transformation	ingest-generated raw_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	is_synthetic
Source / field	A1 packet + registered identity + server
Transformation	registered sensor.is_synthetic
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	source_record_key
Source / field	A1 packet + registered identity + server
Transformation	boot_id || : || sequence
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	parameter_code
Source / field	A1 packet + registered identity + server
Transformation	constant water_temperature
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	native_value
Source / field	A1 packet + registered identity + server
Transformation	packet.water_temperature

Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	native_unit
Source / field	A1 packet + registered identity + server
Transformation	constant degC
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	value
Source / field	A1 packet + registered identity + server
Transformation	packet.water_temperature (identity)
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	unit
Source / field	A1 packet + registered identity + server
Transformation	constant degC
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	conversion_code
Source / field	A1 packet + registered identity + server
Transformation	constant identity
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	value_status
Source / field	A1 packet + registered identity + server
Transformation	packet.value_status
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation

Column	detection_limit
Source / field	A1 packet + registered identity + server
Transformation	omitted -> SQL NULL
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	observed_at
Source / field	A1 packet + registered identity + server
Transformation	packet.observed_at cast timestamptz
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	ingested_at
Source / field	A1 packet + registered identity + server
Transformation	server received = clock_timestamp()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	quality_status
Source / field	A1 packet + registered identity + server
Transformation	pilot policy + flags
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.parameter
Column	parameter_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformation	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.parameter
Column	unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformation	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.parameter
Column	description
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformation	'Submerged local sensor temperature; distinct from air temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.plot
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	plot_code
Source / field	Seagis-Model/supabase/seed.sql:2
Transformation	'plot-a'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	name
Source / field	Seagis-Model/supabase/seed.sql:2
Transformation	'Plot A (synthetic)'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	latitude
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	longitude

Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	is_synthetic
Source / field	Seagis-Model/supabase/seed.sql:2
Transformation	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	location_status
Source / field	Seagis-Model/supabase/seed.sql:2
Transformation	'synthetic'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	created_at
Source / field	caller / SQL default
Transformation	now()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot_grid_mapping
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	plot_id
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied plot_id subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	source_grid_id
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied source_grid_id subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	is_synthetic
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied is_synthetic subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	sampling_method
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied sampling_method subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	distance_m
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied distance_m subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	representativeness
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied representativeness subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	effective_from
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND

Transformation	caller-supplied effective_from subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	effective_until
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformation	caller-supplied effective_until subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	recorded_at
Source / field	caller / SQL default
Transformation	clock_timestamp()
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.quality_event
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	run_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	s.dataset_key
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	s.is_synthetic
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	event_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	flag
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	observation_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	observation_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	forecast_value_id
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	event_code
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	flag
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	severity
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformation	CASE WHEN quality='rejected' THEN 'error' WHEN flag IN ('gap_before','out_of_order') THEN 'info' ELSE 'warning' END

Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	details
Source / field	caller / SQL default
Transformation	'{}'::jsonb
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	created_at
Source / field	caller / SQL default
Transformation	now()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.sensor
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	sensor_code
Source / field	Seagis-Model/supabase/seed.sql:7
Transformation	'simulator-a'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	plot_id
Source / field	Seagis-Model/supabase/seed.sql:7
Transformation	'00000000-0000-4000-8000-000000000001'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor

Column	dataset_key
Source / field	Seagis-Model/supabase/seed.sql:7
Transformation	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	scope
Source / field	caller / SQL default
Transformation	'local_sensor'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	parameter_code
Source / field	caller / SQL default
Transformation	'water_temperature'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	depth_m
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	depth_status
Source / field	Seagis-Model/supabase/seed.sql:7
Transformation	'unknown'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	is_synthetic
Source / field	Seagis-Model/supabase/seed.sql:7
Transformation	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	active
Source / field	caller / SQL default
Transformation	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.source
Column	source_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformation	'sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source
Column	provider
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformation	'SEAGIS local sensor / simulator'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source
Column	documentation_url
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformation	'https://github.com/De-Voeux/Seagis-Model'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source_grid
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	dataset_key

Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	s.dataset_key
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	scope
Source / field	caller / SQL default
Transformation	'regional_model'::text
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	grid_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	g->>'key'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	grid_version
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	d.dataset_version
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	identity_basis
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	g->>'identity_basis'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	latitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	(g->>'latitude')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	longitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	(g->>'longitude')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	depth_m
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	(g->>'depth_m')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformation	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis_api.credential
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	label
Source / field	owning routine / caller
Transformation	provision_a1.create(label) CLI --label
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	token_sha256
Source / field	owning routine / caller

Transformation	sha256((sgi_ + secrets.token_urlsafe(32)).encode()).hexdigest(); original token excluded
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	kind
Source / field	owning routine / caller
Transformation	provision_a1.create(kind), CLI device/reader
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	sensor_id
Source / field	owning routine / caller
Transformation	validated CLI sensor UUID for device, NULL for reader
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	created_at
Source / field	caller / SQL default
Transformation	now()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	expires_at
Source / field	owning routine / caller
Transformation	now()+make_interval(days=>days); CLI 1..365, default 30
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	revoked_at
Source / field	owning routine / caller
Transformation	NULL initially; revoke command UPDATE revoked_at=now()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.packet_receipt
Column	sensor_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	s.id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	source_record_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	key
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	semantic_sha256
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	digest
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	observation_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	observation_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	run_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	flags
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformation	flags

Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.reader_plot
Column	credential_id
Source / field	owning routine / caller
Transformation	provision_a1.create generated credential UUID
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Destination table	seagis_api.reader_plot
Column	plot_id
Source / field	owning routine / caller
Transformation	CLI plots validated with UUID(), FK checked by database
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Destination table	seagis_api.request_event
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	credential_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformation	actor_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	sensor_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformation	sensor
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event

Column	http_status
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformation	status
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	code
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformation	code
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	created_at
Source / field	caller / SQL default
Transformation	clock_timestamp()
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.temperature_policy
Column	id
Source / field	DB default / explicit UUID
Transformation	true
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	version
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	'pilot-qc-1'
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	min_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	-5
Loader / trigger	scripts/provision_a1.py; migration 0002

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	max_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	60
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	max_step_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	3
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	future_tolerance_seconds
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	300
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	stale_after_seconds
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	7200
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	requests_per_minute
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformation	120
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_environment.attempt
Column	id

Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformation	lease
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformation	k
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	started_at
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformation	clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	completed_at
Source / field	owning routine / caller
Transformation	clock_timestamp() at persist/failure/abandon; initially NULL
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	status
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformation	'running'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	error_code
Source / field	owning routine / caller
Transformation	finish_failure sanitized code or lease_expired on abandoned; otherwise NULL
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	forecast_run_id
Source / field	owning routine / caller
Transformation	persist run_id, also on duplicate; NULL before completion
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.current_speed
Column	eastward_value_id
Source / field	seagis.forecast_value u/v
Transformation	u.id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	northward_value_id
Source / field	seagis.forecast_value u/v
Transformation	v.id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	forecast_run_id
Source / field	seagis.forecast_value u/v
Transformation	u.forecast_run_id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	source_grid_id
Source / field	seagis.forecast_value u/v
Transformation	u.source_grid_id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	valid_at
Source / field	seagis.forecast_value u/v

Transformation	u.valid_at
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	ingested_at
Source / field	seagis.forecast_value u/v
Transformation	u.ingested_at
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	available_at
Source / field	seagis.forecast_value u/v
Transformation	greatest(u.ingested_at,v.ingested_at)
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	speed_m_s
Source / field	seagis.forecast_value u/v
Transformation	CASE WHEN both value_status='present' THEN sqrt(u.value*u.value+v.value*v.value) END
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	derivation
Source / field	seagis.forecast_value u/v
Transformation	'sqrt(uo^2+vo^2); a2-vector-magnitude-v1'
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.demo_grid_mapping
Column	plot_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	(m->>'plot_id')::uuid
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	plot_is_synthetic
Source / field	caller / SQL default
Transformation	true
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	source_grid_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	grid_id
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	s.dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	grid_is_synthetic
Source / field	caller / SQL default
Transformation	false
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	example_latitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	(m->>'latitude')::float8
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	example_longitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	(m->>'longitude')::float8

Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	coordinate_status
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	m->>'coordinate_status'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	distance_m
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	(m->>'distance_m')::float8
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	sampling_method
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformation	m->>'sampling_method'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	created_at
Source / field	caller / SQL default
Transformation	now()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.policy
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy

Column	enabled
Source / field	caller / SQL default
Transformation	true
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	purpose
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	'noncommercial_educational_prototype'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	licence_url
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	licence_url
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	terms_review
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	CASE WHEN source_key='copernicus' THEN 'cmems-licence-20261022-sections-2.2-2.4-2.6' ELSE ' open-meteo-terms-20261022-noncommercial-cc-by-4.0' END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	attribution
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	attribution
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	update_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	CASE WHEN source_key='copernicus' THEN 86400 ELSE 21600 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	release_offset_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	CASE WHEN dataset_key='cmems_nutrients_daily' THEN 45000 WHEN source_key='copernicus' THEN 30600 ELSE 16200 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	stale_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformation	CASE WHEN source_key='copernicus' THEN 172800 ELSE 43200 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	retention_mode
Source / field	caller / SQL default
Transformation	'metadata_only'::text
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	reviewed_at
Source / field	caller / SQL default
Transformation	now()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.schedule_state
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:56
Transformation	dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state

Column	lease_id
Source / field	owning routine / caller
Transformation	claim gen_random_uuid(); cleared after persist/failure
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	lease_until
Source / field	owning routine / caller
Transformation	claim clock_timestamp()+interval 20 minutes; cleared after persist/failure
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	next_attempt_at
Source / field	caller / SQL default
Transformation	'-infinity'::timestamp with time zone
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_success_at
Source / field	owning routine / caller
Transformation	persist t=clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_attempt_at
Source / field	owning routine / caller
Transformation	claim clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_error_code
Source / field	owning routine / caller
Transformation	finish_failure sanitized code; cleared on success
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	failures
Source / field	caller / SQL default
Transformation	0
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_forecast_run_id
Source / field	owning routine / caller
Transformation	persist run_id
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_features.feature_input
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	snapshot_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied snapshot_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	observation_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied observation_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	forecast_value_id

Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied forecast_value_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_snapshot
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	plot_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied plot_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	is_synthetic
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied is_synthetic subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	as_of
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied as_of subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	generated_at
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied generated_at subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	feature_version
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied feature_version subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	feature_schema_version
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied feature_schema_version subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	features
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformation	caller-supplied features subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	missing_context
Source / field	caller / SQL default
Transformation	'{}'::text[]
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	status
Source / field	caller / SQL default
Transformation	'draft'::text
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	input_manifest_sha256
Source / field	seagis_features.manifest_hash(snapshot)

Transformation	SHA256 sorted input references at draft -> sealed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	features_sha256
Source / field	DB generated
Transformation	seagis.jsonb_sha256(features)
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_quality.alignment_snapshot
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	plot_id
Source / field	A3 build -> save_snapshot
Transformation	document.plot_id
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	mode
Source / field	A3 build -> save_snapshot
Transformation	document.mode
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	start_at
Source / field	A3 build -> save_snapshot
Transformation	document.start_at
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	end_at
Source / field	A3 build -> save_snapshot
Transformation	document.end_at
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	policy_version
Source / field	A3 build -> save_snapshot
Transformation	registered policy.version
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	source_sha256
Source / field	A3 build -> save_snapshot
Transformation	document.source_sha256 checked against fresh source_bundle
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	input_manifest_sha256
Source / field	A3 build -> save_snapshot
Transformation	SHA256 sorted distinct at|source_ref lines
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	body
Source / field	A3 build -> save_snapshot
Transformation	document
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	body_sha256
Source / field	DB generated
Transformation	seagis.jsonb_sha256(body)

Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	created_at
Source / field	A3 build -> save_snapshot
Transformation	SQL default clock_timestamp()
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.policy
Column	version
Source / field	owning routine / caller
Transformation	register_policy policy[version]
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	definition
Source / field	owning routine / caller
Transformation	register_policy JSON policy argument (config/a3_quality.json)
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	policy_sha256
Source / field	owning routine / caller
Transformation	register_policy policy_hash from digest(policy)
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	code_sha256
Source / field	owning routine / caller
Transformation	register_policy builder_hash from code_hash()
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy

Column	registered_at
Source / field	caller / SQL default
Transformation	clock_timestamp()
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.snapshot_input
Column	snapshot_id
Source / field	A3 document.rows[].cells[].lineage
Transformation	saved snapshot UUID
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	aligned_at
Source / field	A3 document.rows[].cells[].lineage
Transformation	row.at::timestamptz
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	source_ref
Source / field	A3 document.rows[].cells[].lineage
Transformation	lineage ref
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	observation_id
Source / field	A3 document.rows[].cells[].lineage
Transformation	UUID if ref prefix observation else NULL
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	forecast_value_id
Source / field	A3 document.rows[].cells[].lineage
Transformation	UUID if ref prefix forecast_value else NULL
Loader / trigger	seagis_quality.save_snapshot

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_raw.raw_payload
Column	id
Source / field	DB default / explicit UUID
Transformation	gen_random_uuid()
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	run_id
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	s.dataset_key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	s.is_synthetic
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	sha256
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	CASE WHEN s.is_synthetic THEN seagis.jsonb_sha256(packet) ELSE wire_sha256 END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	hash_basis

Source / field	caller / SQL default
Transformation	'source_bytes'::text
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	received_at
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	received
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	storage_mode
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	CASE WHEN s.is_synthetic THEN 'inline' ELSE 'metadata_only' END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	payload
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	CASE WHEN s.is_synthetic THEN packet ELSE NULL END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	object_uri
Source / field	active loader omits column / nullable schema
Transformation	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	byte_size
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	wire_bytes
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	format
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	'application/json'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	licence_snapshot
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformation	'project-owned; A1 synthetic development / real retention gated'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	provider_metadata
Source / field	caller / SQL default
Transformation	'{}'::jsonb
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

8 Data Contracts

Four contracts are explicit JSON Schemas; seven additional contracts are implementation-derived. This classification does not make them all public endpoints. No general version negotiation was found; mismatched constant versions/schemas are rejected. Breaking changes need a new version and consumer review, not assumed automatic compatibility. JSON errors below come from handler/SQL, not an invented standard error format.

ID	C01
Contract	A0 source envelope
Producer	fixture/provider logical extract
Consumer	validate_source
Interface	JSON / Draft 2020-12
Class	explicit
Database	seagis.ingestion_run + seagis_raw.raw_payload
Evidence	Seagis-Model/scripts/validate_contracts.py:1

ID	C02
Contract	A0 normalized envelope
Producer	fixture normalization
Consumer	validate_normalized + SQL guards
Interface	JSON / Draft 2020-12
Class	explicit
Database	observation / forecast_run / forecast_value / source_grid
Evidence	Seagis-Model/scripts/validate_contracts.py:1

ID	C03
Contract	A1 device packet
Producer	device/simulator
Consumer	temperature handler + seagis_api.ingest
Interface	JSON / HTTP POST
Class	explicit
Database	observation + raw_payload + packet_receipt + quality_event
Evidence	Seagis-Model/supabase/functions/temperature/handler.ts:1

ID	C04
Contract	A3 alignment document
Producer	scripts/quality/alignment.py::build
Consumer	seagis_quality.save_snapshot
Interface	JSON / SQL jsonb
Class	explicit
Database	seagis_quality.alignment_snapshot + snapshot_input
Evidence	Seagis-Model/scripts/curate_environment.py:1

ID	C05
Contract	A1 history/status/replies
Producer	seagis_api.history/status/reply

Consumer	authorized reader
Interface	JSON / HTTP GET
Class	implementation-derived
Database	observation + sensor + packet_receipt
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

ID	C06
Contract	A2 adapter batch/lease
Producer	Adapter.envelope + claim
Consumer	seagis_environment.persist
Interface	JSON / SQL jsonb
Class	implementation-derived
Database	raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
Evidence	Seagis-Model/scripts/environmental/common.py:1

ID	C07
Contract	A3 source_bundle
Producer	seagis_quality.source_bundle
Consumer	build / CSV exporter
Interface	JSON / SQL jsonb
Class	implementation-derived
Database	plot + sensor + dataset_parameter + mappings + observation + forecast_value
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1

ID	C08
Contract	CSV experiment-csv-2
Producer	export_experiment_csv.py::export
Consumer	offline experiments / developer
Interface	CSV + JSON manifest
Class	implementation-derived
Database	derived files; no database writes
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

ID	C09
Contract	iOS Assessment domain
Producer	MockAssessmentRepository
Consumer	LoadDemoAssessment / ViewModel
Interface	Swift structs
Class	implementation-derived / mock
Database	NOT APPLICABLE: no backend mapping
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

ID	C10
Contract	Open-Meteo response

Producer	Open-Meteo forecast API
Consumer	OpenMeteo.parse/normalize
Interface	JSON hourly arrays
Class	implementation-derived
Database	forecast_value + grid + raw manifest
Evidence	Seagis-Model/scripts/environmental/open_meteo.py:1

ID	C11
Contract	Copernicus subset
Producer	Copernicus Toolbox/xarray + STAC
Consumer	Copernicus.parse/normalize
Interface	xarray + canonical logical JSON subset
Class	implementation-derived
Database	forecast_value + grid + raw manifest
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1

C01 A0 source envelope

Attribute	Definition
Producer → consumer	fixture/provider logical extract → validate_source
Interface / format	JSON / Draft 2020-12
Classification	explicit
Database mapping	seagis.ingestion_run + seagis_raw.raw_payload
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/validate_contracts.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Contract structure version; not a biological model version
Validation / allowed values	{"const": "0.1.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"0.1.0"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	source_type
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Source adapter kind determining the fixture payload

Validation / allowed values	{"enum": ["sensor", "copernicus", "open_meteo"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	dataset_key
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Registered dataset key
Validation / allowed values	{"enum": ["sensor_temperature_v1", "cmems_salinity_6h", "cmems_currents_6h", "cmems_nutrients_daily", "open_meteo_gfs"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor_temperature_v1"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	is_synthetic
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Synthetic provenance flag, separate from QC
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	received_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Raw receipt instant at worker
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	provider_metadata
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND

Meaning	Sanitized provider and request metadata
Validation / allowed values	{"type": "object"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	payload
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Synthetic raw JSON content when inline
Validation / allowed values	{"type": "object"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Complete schema for conditional rules and polymorphic structures (required above applies within each parent):


```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "schema_version": {
      "const": "0.1.0"
    },
    "source_type": {
      "enum": [
        "sensor",
        "copernicus",
        "open_meteo"
      ]
    },
    "dataset_key": {
      "enum": [
        "sensor_temperature_v1",
        "cmems_salinity_6h",
        "cmems_currents_6h",
        "cmems_nutrients_daily",
        "open_meteo_gfs"
      ]
    },
    "is_synthetic": {
      "type": "boolean"
    },
    "received_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    },
    "provider_metadata": {
      "type": "object"
    },
    "payload": {
      "type": "object"
    }
  },
  "required": [
    "schema_version",
    "source_type",
    "dataset_key",
    "is_synthetic",
    "received_at",
    "provider_metadata",
    "payload"
  ],
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/source.schema.json",
```

```
"title": "SEAGIS source envelopes v0.1.0",
"allOf": [
 {
  "if": {
   "properties": {
    "source_type": {
     "const": "sensor"
    }
   }
  },
  "then": {
   "properties": {
    "payload": {
     "type": "object",
     "additionalProperties": false,
     "properties": {
      "sensor_id": {
       "type": "string",
       "format": "uuid"
      },
      "boot_id": {
       "type": "string",
       "minLength": 1
      },
      "sequence": {
       "type": "integer",
       "minimum": 0
      },
      "observed_at": {
       "type": "string",
       "format": "date-time",
       "pattern": "Z$"
      },
      "water_temperature": {
       "type": [
        "number",
        "null"
       ]
      },
      "unit": {
       "const": "degC"
      },
      "value_status": {
       "enum": [
        "present",
        "missing",
        "below_detection_limit",
        "not_available"
       ]
      },
      "detection_limit": {
       "type": [
        "number",
        "null"
       ]
      }
     },
     "required": [
      "sensor_id",
      "boot_id",
      "sequence",
      "observed_at",
      "water_temperature",
      "unit",
      "value_status",
      "detection_limit"
     ]
    },
    "dataset_key": {
     "enum": [
      "sensor_temperature_v1"
     ]
    }
   }
  }
 },
 {
  "if": {
   "properties": {
    "source_type": {
     "const": "copernicus"
    }
   }
```

```
    },
    "then": {
      "properties": {
        "payload": {
          "type": "object",
          "additionalProperties": false,
          "properties": {
            "representation": {
              "const": "netcdf-logical-extract"
            },
            "coordinates": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "time": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "string",
                    "format": "date-time",
                    "pattern": "Z$"
                  }
                },
                "depth": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": 0
                  }
                },
                "latitude": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": -90,
                    "maximum": 90
                  }
                },
                "longitude": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": -180,
                    "maximum": 180
                  }
                }
              },
              "required": [
                "time",
                "depth",
                "latitude",
                "longitude"
              ]
            },
            "variables": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "so": {
                  "type": "object",
                  "additionalProperties": false,
                  "properties": {
                    "unit": {
                      "const": "1e-3"
                    },
                    "fill_value": {
                      "type": "number"
                    },
                    "values": {
                      "type": "array",
                      "minItems": 1,
                      "items": {
                        "type": [
                          "number",
                          "null"
                        ]
                      }
                    }
                  },
                  "required": [
```

```
      "unit",
      "fill_value",
      "values"
    ]
  },
  "uo": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "m s-1"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
      "values"
    ]
  },
  "vo": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "m s-1"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
      "values"
    ]
  },
  "no3": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "mmol m-3"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
```

```
              "values"
             ]
            },
            "po4": {
             "type": "object",
             "additionalProperties": false,
             "properties": {
              "unit": {
               "const": "mmol m-3"
              },
              "fill_value": {
               "type": "number"
              },
              "values": {
               "type": "array",
               "minItems": 1,
               "items": {
                "type": [
                 "number",
                 "null"
                ]
               }
              }
             },
             "required": [
              "unit",
              "fill_value",
              "values"
             ]
            }
           },
           "minProperties": 1
          }
         },
         "required": [
          "representation",
          "coordinates",
          "variables"
         ]
        },
        "dataset_key": {
         "enum": [
          "cmems_salinity_6h",
          "cmems_currents_6h",
          "cmems_nutrients_daily"
         ]
        }
       }
      }
     },
     {
      "if": {
       "properties": {
        "source_type": {
         "const": "open_meteo"
        }
       }
      },
      "then": {
       "properties": {
        "payload": {
         "type": "object",
         "additionalProperties": false,
         "properties": {
          "latitude": {
           "type": "number"
          },
          "longitude": {
           "type": "number"
          },
          "generationtime_ms": {
           "type": "number"
          },
          "utc_offset_seconds": {
           "const": 0
          },
          "timezone": {
           "enum": [
            "GMT",
            "UTC"
           ]
          },
          "timezone_abbreviation": {
```

```
      "enum": [
        "GMT",
        "UTC"
      ]
    },
    "hourly_units": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "time": {
          "const": "iso8601"
        },
        "precipitation": {
          "const": "mm"
        },
        "wind_speed_10m": {
          "const": "m/s"
        },
        "wind_direction_10m": {
          "const": "°"
        },
        "cloud_cover": {
          "const": "%"
        },
        "uv_index": {
          "const": ""
        },
        "temperature_2m": {
          "const": "°C"
        }
      },
      "required": [
        "time",
        "precipitation",
        "wind_speed_10m",
        "wind_direction_10m",
        "cloud_cover",
        "uv_index",
        "temperature_2m"
      ]
    },
    "hourly": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "time": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": "string",
            "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}$"
          }
        },
        "precipitation": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
        "wind_speed_10m": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
        "wind_direction_10m": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
```

```
            "cloud_cover": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            },
            "uv_index": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            },
            "temperature_2m": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            }
          },
          "required": [
            "time",
            "precipitation",
            "wind_speed_10m",
            "wind_direction_10m",
            "cloud_cover",
            "uv_index",
            "temperature_2m"
          ]
        }
      },
      "required": [
        "latitude",
        "longitude",
        "generationtime_ms",
        "utc_offset_seconds",
        "timezone",
        "timezone_abbreviation",
        "hourly_units",
        "hourly"
      ]
    },
    "dataset_key": {
      "enum": [
        "open_meteo_gfs"
      ]
    }
  }
 }
 ]
}
```


C02 A0 normalized envelope

Attribute	Definition
Producer → consumer	fixture normalization → validate_normalized + SQL guards
Interface / format	JSON / Draft 2020-12
Classification	explicit
Database mapping	observation / forecast_run / forecast_value / source_grid
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/validate_contracts.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Contract structure version; not a biological model version
Validation / allowed values	{"const": "0.1.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"0.1.0"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	dataset_key
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Registered dataset key
Validation / allowed values	{"enum": ["sensor_temperature_v1", "cmems_salinity_6h", "cmems_currents_6h", "cmems_nutrients_daily", "open_meteo_gfs"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor_temperature_v1"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "request_key", "adapter_version", "started_at", "completed_at", "status", "is_synthetic"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"

DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.request_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Logical ingestion retry key per dataset
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.adapter_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Ingestion adapter implementation version
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.started_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Ingestion or attempt start instant
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.completed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Completion instant; null while running
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"

DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"const": "succeeded"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"succeeded"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.is_synthetic
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Synthetic provenance flag, separate from QC
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "sha256", "received_at", "format", "licence_snapshot"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.sha256
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Raw digest under hash_basis, not necessarily network bytes
Validation / allowed values	{"type": "string", "pattern": "^[0-9a-f]{64}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.received_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Raw receipt instant at worker
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.format
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Raw representation media type or format
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.licence_snapshot
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Licence statement at capture
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "run_key", "run_kind", "provider_run_id", " model_initialized_at", "issued_at", "issue_time_basis", "ingested_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.run_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Semantic capture identity per dataset
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.run_kind
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	forecast, analysis or analysis_forecast_unspecified
Validation / allowed values	{"enum": ["forecast", "analysis"]}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"forecast"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.provider_run_id
Type / ref	["string", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Provider run ID only when actually available
Validation / allowed values	{"type": ["string", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.model_initialized_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.model_initialized_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issued_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issued_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issue_time_basis
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Known/unknown publication-time basis
Validation / allowed values	{"enum": ["provider_verified", "unknown"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"provider_verified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Instant data became available to the system
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "grid_key", "grid_version", "identity_basis", " latitude", "longitude", "depth_m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.grid_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Cell key based on coordinates/depth or fixture
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.grid_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Grid identity version

Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.identity_basis
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Identity basis: native cell, response coordinate or synthetic
Validation / allowed values	{"enum": ["provider_cell", "provider_returned_coordinate", "synthetic"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"provider_cell"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.latitude
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	WGS84 latitude in degrees
Validation / allowed values	{"type": "number", "minimum": -90, "maximum": 90}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.longitude
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	WGS84 longitude in degrees
Validation / allowed values	{"type": "number", "minimum": -180, "maximum": 180}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.depth_m
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Depth positive downward in metres

Validation / allowed values	{"type": "number", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Cross-field-validated observation/forecast_value list
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "parameter_code", "native_value", "native_unit", "value", "unit", "conversion_code", "value_status", "detection_limit", "ingested_at", "quality_status", "kind", "plot_id", "sensor_id", "source_record_key", "observed_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.parameter_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Environmental parameter identity
Validation / allowed values	{"enum": ["water_temperature", "salinity", "current_eastward", "current_northward", "nitrate", "phosphate", " precipitation", "wind_speed_10m", "wind_direction_10m", "cloud_cover", "uv_index", "air_temperature_2m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"water_temperature"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.native_value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Physical value in source units
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.native_unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Original unit before normalization
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.value
Type / ref	["number", "null"]
Required	True
Nullable	True

Default	NOT FOUND
Meaning	Normalized value; null follows value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Normalized value unit from registry
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.conversion_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Unit-conversion method identity
Validation / allowed values	{"enum": ["identity", "unit_notation"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"identity"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Numeric availability, separate from quality
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.detection_limit
Type / ref	["number", "null"]
Required	True
Nullable	True

Default	NOT FOUND
Meaning	Detection limit in normalized units when applicable
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Instant data became available to the system
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.quality_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	QC state unreviewed, accepted, suspect or rejected; not a diagnosis
Validation / allowed values	{"enum": ["unreviewed", "accepted", "suspect", "rejected"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"unreviewed"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.kind
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Actor kind device or reader
Validation / allowed values	{"const": "observation"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"observation"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.plot_id
Type / ref	"string"
Required	True
Nullable	False

Default	NOT FOUND
Meaning	Cultivation plot reference
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.sensor_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Sensor deployment reference
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.source_record_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	boot_id:sequence for A1 packets
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.observed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Sensor measurement instant, distinct from receipt
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1
Type / ref	"object"
Required	False
Nullable	False

Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "parameter_code", "native_value", "native_unit", " value", "unit", "conversion_code", "value_status", "detection_limit", "ingested_at", "quality_status", "kind", " valid_at", "temporal_support", "interval_status", "period_start_at", "period_end_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Row identity under the table primary key
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.parameter_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Environmental parameter identity
Validation / allowed values	{"enum": ["water_temperature", "salinity", "current_eastward", "current_northward", "nitrate", "phosphate", " precipitation", "wind_speed_10m", "wind_direction_10m", "cloud_cover", "uv_index", "air_temperature_2m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"water_temperature"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.native_value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Physical value in source units
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.native_unit
Type / ref	"string"

Required	True
Nullable	False
Default	NOT FOUND
Meaning	Original unit before normalization
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Normalized value; null follows value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Normalized value unit from registry
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.conversion_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Unit-conversion method identity
Validation / allowed values	{"enum": ["identity", "unit_notation"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"identity"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.value_status
Type / ref	"const/enum"

Required	True
Nullable	False
Default	NOT FOUND
Meaning	Numeric availability, separate from quality
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.detection_limit
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Detection limit in normalized units when applicable
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Instant data became available to the system
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.quality_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	QC state unreviewed, accepted, suspect or rejected; not a diagnosis
Validation / allowed values	{"enum": ["unreviewed", "accepted", "suspect", "rejected"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"unreviewed"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.kind
Type / ref	"const/enum"

Required	True
Nullable	False
Default	NOT FOUND
Meaning	Actor kind device or reader
Validation / allowed values	{"const": "forecast_value"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"forecast_value"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.valid_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Representative time of model value
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.temporal_support
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Temporal support: instantaneous, mean or sum
Validation / allowed values	{"enum": ["instantaneous", "mean", "sum"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"instantaneous"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.interval_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	known, unknown or not_applicable interval status
Validation / allowed values	{"enum": ["not_applicable", "known", "unknown"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"not_applicable"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_start_at/anyOf/0
Type / ref	"string"

Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_start_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_end_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_end_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Complete schema for conditional rules and polymorphic structures (required above applies within each parent):

{

```
 "type": "object",
 "additionalProperties": false,
 "properties": {
  "schema_version": {
   "const": "0.1.0"
  },
  "dataset_key": {
   "enum": [
    "sensor_temperature_v1",
    "cmems_salinity_6h",
    "cmems_currents_6h",
    "cmems_nutrients_daily",
    "open_meteo_gfs"
   ]
  },
  "ingestion_run": {
   "type": "object",
   "additionalProperties": false,
   "properties": {
    "id": {
     "type": "string",
     "format": "uuid"
    },
    "request_key": {
     "type": "string",
     "minLength": 1
    },
    "adapter_version": {
     "type": "string",
     "minLength": 1
    },
    "started_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "completed_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "status": {
     "const": "succeeded"
    },
    "is_synthetic": {
     "type": "boolean"
    }
   },
   "required": [
    "id",
    "request_key",
    "adapter_version",
    "started_at",
    "completed_at",
    "status",
    "is_synthetic"
   ]
  },
  "raw_payload": {
   "type": "object",
   "additionalProperties": false,
   "properties": {
    "id": {
     "type": "string",
     "format": "uuid"
    },
    "sha256": {
     "type": "string",
     "pattern": "^[0-9a-f]{64}$"
    },
    "received_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "format": {
     "type": "string",
     "minLength": 1
    },
    "licence_snapshot": {
     "type": "string",
     "minLength": 1
    }
```

```
    },
    "required": [
      "id",
      "sha256",
      "received_at",
      "format",
      "licence_snapshot"
    ]
  },
  "forecast_run": {
    "anyOf": [
      {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "run_key": {
            "type": "string",
            "minLength": 1
          },
          "run_kind": {
            "enum": [
              "forecast",
              "analysis"
            ]
          },
          "provider_run_id": {
            "type": [
              "string",
              "null"
            ]
          },
          "model_initialized_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "issued_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "issue_time_basis": {
            "enum": [
              "provider_verified",
              "unknown"
            ]
          },
          "ingested_at": {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          }
        },
        "required": [
          "id",
          "run_key",
          "run_kind",
          "provider_run_id",
          "model_initialized_at",
          "issued_at",
          "issue_time_basis",
          "ingested_at"
        ]
      },
      {
```

```
        "type": "null"
      }
    ]
  },
  "source_grid": {
    "anyOf": [
      {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "grid_key": {
            "type": "string",
            "minLength": 1
          },
          "grid_version": {
            "type": "string",
            "minLength": 1
          },
          "identity_basis": {
            "enum": [
              "provider_cell",
              "provider_returned_coordinate",
              "synthetic"
            ]
          },
          "latitude": {
            "type": "number",
            "minimum": -90,
            "maximum": 90
          },
          "longitude": {
            "type": "number",
            "minimum": -180,
            "maximum": 180
          },
          "depth_m": {
            "type": "number",
            "minimum": 0
          }
        },
        "required": [
          "id",
          "grid_key",
          "grid_version",
          "identity_basis",
          "latitude",
          "longitude",
          "depth_m"
        ]
      },
      {
        "type": "null"
      }
    ]
  },
  "measurements": {
    "type": "array",
    "minItems": 1,
    "items": {
      "oneOf": [
        {
          "type": "object",
          "additionalProperties": false,
          "properties": {
            "id": {
              "type": "string",
              "format": "uuid"
            },
            "parameter_code": {
              "enum": [
                "water_temperature",
                "salinity",
                "current_eastward",
                "current_northward",
                "nitrate",
                "phosphate",
                "precipitation",
                "wind_speed_10m",
                "wind_direction_10m",
```

```
        "cloud_cover",
        "uv_index",
        "air_temperature_2m"
      ]
    },
    "native_value": {
      "type": [
        "number",
        "null"
      ]
    },
    "native_unit": {
      "type": "string"
    },
    "value": {
      "type": [
        "number",
        "null"
      ]
    },
    "unit": {
      "type": "string",
      "minLength": 1
    },
    "conversion_code": {
      "enum": [
        "identity",
        "unit_notation"
      ]
    },
    "value_status": {
      "enum": [
        "present",
        "missing",
        "below_detection_limit",
        "not_available"
      ]
    },
    "detection_limit": {
      "type": [
        "number",
        "null"
      ]
    },
    "ingested_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    },
    "quality_status": {
      "enum": [
        "unreviewed",
        "accepted",
        "suspect",
        "rejected"
      ]
    },
    "kind": {
      "const": "observation"
    },
    "plot_id": {
      "type": "string",
      "format": "uuid"
    },
    "sensor_id": {
      "type": "string",
      "format": "uuid"
    },
    "source_record_key": {
      "type": "string",
      "minLength": 1
    },
    "observed_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    }
  },
  "required": [
    "id",
    "parameter_code",
    "native_value",
    "native_unit",
```

```
      "value",
      "unit",
      "conversion_code",
      "value_status",
      "detection_limit",
      "ingested_at",
      "quality_status",
      "kind",
      "plot_id",
      "sensor_id",
      "source_record_key",
      "observed_at"
    ]
  },
  {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "id": {
        "type": "string",
        "format": "uuid"
      },
      "parameter_code": {
        "enum": [
          "water_temperature",
          "salinity",
          "current_eastward",
          "current_northward",
          "nitrate",
          "phosphate",
          "precipitation",
          "wind_speed_10m",
          "wind_direction_10m",
          "cloud_cover",
          "uv_index",
          "air_temperature_2m"
        ]
      },
      "native_value": {
        "type": [
          "number",
          "null"
        ]
      },
      "native_unit": {
        "type": "string"
      },
      "value": {
        "type": [
          "number",
          "null"
        ]
      },
      "unit": {
        "type": "string",
        "minLength": 1
      },
      "conversion_code": {
        "enum": [
          "identity",
          "unit_notation"
        ]
      },
      "value_status": {
        "enum": [
          "present",
          "missing",
          "below_detection_limit",
          "not_available"
        ]
      },
      "detection_limit": {
        "type": [
          "number",
          "null"
        ]
      },
      "ingested_at": {
        "type": "string",
        "format": "date-time",
        "pattern": "Z$"
      },
      "quality_status": {
```

```
            "enum": [
              "unreviewed",
              "accepted",
              "suspect",
              "rejected"
            ]
          },
          "kind": {
            "const": "forecast_value"
          },
          "valid_at": {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          "temporal_support": {
            "enum": [
              "instantaneous",
              "mean",
              "sum"
            ]
          },
          "interval_status": {
            "enum": [
              "not_applicable",
              "known",
              "unknown"
            ]
          },
          "period_start_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "period_end_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          }
        },
        "required": [
          "id",
          "parameter_code",
          "native_value",
          "native_unit",
          "value",
          "unit",
          "conversion_code",
          "value_status",
          "detection_limit",
          "ingested_at",
          "quality_status",
          "kind",
          "valid_at",
          "temporal_support",
          "interval_status",
          "period_start_at",
          "period_end_at"
        ]
      }
    ]
  }
 }
},
"required": [
  "schema_version",
  "dataset_key",
  "ingestion_run",
  "raw_payload",
```

```
    "forecast_run",
    "source_grid",
    "measurements"
  ],
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/normalized.schema.json",
  "title": "SEAGIS normalized envelope v0.1.0",
  "allOf": [
    {
      "if": {
        "properties": {
          "dataset_key": {
            "const": "sensor_temperature_v1"
          }
        }
      },
      "then": {
        "properties": {
          "forecast_run": {
            "type": "null"
          },
          "source_grid": {
            "type": "null"
          },
          "measurements": {
            "type": "array",
            "minItems": 1,
            "items": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "id": {
                  "type": "string",
                  "format": "uuid"
                },
                "parameter_code": {
                  "enum": [
                    "water_temperature",
                    "salinity",
                    "current_eastward",
                    "current_northward",
                    "nitrate",
                    "phosphate",
                    "precipitation",
                    "wind_speed_10m",
                    "wind_direction_10m",
                    "cloud_cover",
                    "uv_index",
                    "air_temperature_2m"
                  ]
                },
                "native_value": {
                  "type": [
                    "number",
                    "null"
                  ]
                },
                "native_unit": {
                  "type": "string"
                },
                "value": {
                  "type": [
                    "number",
                    "null"
                  ]
                },
                "unit": {
                  "type": "string",
                  "minLength": 1
                },
                "conversion_code": {
                  "enum": [
                    "identity",
                    "unit_notation"
                  ]
                },
                "value_status": {
                  "enum": [
                    "present",
                    "missing",
                    "below_detection_limit",
                    "not_available"
                  ]
```

```
            },
            "detection_limit": {
              "type": [
                "number",
                "null"
              ]
            },
            "ingested_at": {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            "quality_status": {
              "enum": [
                "unreviewed",
                "accepted",
                "suspect",
                "rejected"
              ]
            },
            "kind": {
              "const": "observation"
            },
            "plot_id": {
              "type": "string",
              "format": "uuid"
            },
            "sensor_id": {
              "type": "string",
              "format": "uuid"
            },
            "source_record_key": {
              "type": "string",
              "minLength": 1
            },
            "observed_at": {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            }
          },
          "required": [
            "id",
            "parameter_code",
            "native_value",
            "native_unit",
            "value",
            "unit",
            "conversion_code",
            "value_status",
            "detection_limit",
            "ingested_at",
            "quality_status",
            "kind",
            "plot_id",
            "sensor_id",
            "source_record_key",
            "observed_at"
          ]
        }
      }
    }
  },
  "else": {
    "properties": {
      "forecast_run": {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "run_key": {
            "type": "string",
            "minLength": 1
          },
          "run_kind": {
            "enum": [
              "forecast",
              "analysis"
            ]
          },
```

```
      "provider_run_id": {
        "type": [
          "string",
          "null"
        ]
      },
      "model_initialized_at": {
        "anyOf": [
          {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          {
            "type": "null"
          }
        ]
      },
      "issued_at": {
        "anyOf": [
          {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          {
            "type": "null"
          }
        ]
      },
      "issue_time_basis": {
        "enum": [
          "provider_verified",
          "unknown"
        ]
      },
      "ingested_at": {
        "type": "string",
        "format": "date-time",
        "pattern": "Z$"
      }
    },
    "required": [
      "id",
      "run_key",
      "run_kind",
      "provider_run_id",
      "model_initialized_at",
      "issued_at",
      "issue_time_basis",
      "ingested_at"
    ]
  },
  "source_grid": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "id": {
        "type": "string",
        "format": "uuid"
      },
      "grid_key": {
        "type": "string",
        "minLength": 1
      },
      "grid_version": {
        "type": "string",
        "minLength": 1
      },
      "identity_basis": {
        "enum": [
          "provider_cell",
          "provider_returned_coordinate",
          "synthetic"
        ]
      },
      "latitude": {
        "type": "number",
        "minimum": -90,
        "maximum": 90
      },
      "longitude": {
        "type": "number",
```

```
        "minimum": -180,
        "maximum": 180
      },
      "depth_m": {
        "type": "number",
        "minimum": 0
      }
    },
    "required": [
      "id",
      "grid_key",
      "grid_version",
      "identity_basis",
      "latitude",
      "longitude",
      "depth_m"
    ]
  },
  "measurements": {
    "type": "array",
    "minItems": 1,
    "items": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "id": {
          "type": "string",
          "format": "uuid"
        },
        "parameter_code": {
          "enum": [
            "water_temperature",
            "salinity",
            "current_eastward",
            "current_northward",
            "nitrate",
            "phosphate",
            "precipitation",
            "wind_speed_10m",
            "wind_direction_10m",
            "cloud_cover",
            "uv_index",
            "air_temperature_2m"
          ]
        },
        "native_value": {
          "type": [
            "number",
            "null"
          ]
        },
        "native_unit": {
          "type": "string"
        },
        "value": {
          "type": [
            "number",
            "null"
          ]
        },
        "unit": {
          "type": "string",
          "minLength": 1
        },
        "conversion_code": {
          "enum": [
            "identity",
            "unit_notation"
          ]
        },
        "value_status": {
          "enum": [
            "present",
            "missing",
            "below_detection_limit",
            "not_available"
          ]
        },
        "detection_limit": {
          "type": [
            "number",
            "null"
          ]
```

```
        },
        "ingested_at": {
          "type": "string",
          "format": "date-time",
          "pattern": "Z$"
        },
        "quality_status": {
          "enum": [
            "unreviewed",
            "accepted",
            "suspect",
            "rejected"
          ]
        },
        "kind": {
          "const": "forecast_value"
        },
        "valid_at": {
          "type": "string",
          "format": "date-time",
          "pattern": "Z$"
        },
        "temporal_support": {
          "enum": [
            "instantaneous",
            "mean",
            "sum"
          ]
        },
        "interval_status": {
          "enum": [
            "not_applicable",
            "known",
            "unknown"
          ]
        },
        "period_start_at": {
          "anyOf": [
            {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            {
              "type": "null"
            }
          ]
        },
        "period_end_at": {
          "anyOf": [
            {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            {
              "type": "null"
            }
          ]
        }
      },
      "required": [
        "id",
        "parameter_code",
        "native_value",
        "native_unit",
        "value",
        "unit",
        "conversion_code",
        "value_status",
        "detection_limit",
        "ingested_at",
        "quality_status",
        "kind",
        "valid_at",
        "temporal_support",
        "interval_status",
        "period_start_at",
        "period_end_at"
      ]
    }
  }
}
```

```
    }
   }
  ]
}
```

C03 A1 device packet

Attribute	Definition
Producer → consumer	device/simulator → temperature handler + seagis_api.ingest
Interface / format	JSON / HTTP POST
Classification	explicit
Database mapping	observation + raw_payload + packet_receipt + quality_event
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/supabase/functions/temperature/handler.ts:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Contract structure version; not a biological model version
Validation / allowed values	{"const": "1.0.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"1.0.0"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	sensor_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Sensor deployment reference
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	boot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Sensor boot-session identity for idempotency
Validation / allowed values	{"type": "string", "pattern": "^[A-Za-z0-9._-]{1,64}$"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	sequence
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	uint32 counter within one boot; key combined with boot_id
Validation / allowed values	{"type": "integer", "minimum": 0, "maximum": 4294967295}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	observed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Sensor measurement instant, distinct from receipt
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,6})?Z$"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	water_temperature
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Water temperature in degC or null missing; distinct from air temperature
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	unit
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Normalized value unit from registry
Validation / allowed values	{"const": "degC"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"degC"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Numeric availability, separate from quality
Validation / allowed values	{"enum": ["present", "missing"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Complete schema for conditional rules and polymorphic structures (required above applies within each parent):


```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/iot-packet/1.0.0",
  "title": "SEAGIS temperature packet 1.0.0",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "schema_version",
    "sensor_id",
    "boot_id",
    "sequence",
    "observed_at",
    "water_temperature",
    "unit",
    "value_status"
  ],
  "properties": {
    "schema_version": {
      "const": "1.0.0"
    },
    "sensor_id": {
      "type": "string",
      "format": "uuid"
    },
    "boot_id": {
      "type": "string",
      "pattern": "^[A-Za-z0-9._-]{1,64}$"
    },
    "sequence": {
      "type": "integer",
      "minimum": 0,
      "maximum": 4294967295
    },
    "observed_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,6})?Z$"
    },
    "water_temperature": {
      "type": [
        "number",
        "null"
      ]
    },
    "unit": {
      "const": "degC"
    },
    "value_status": {
      "enum": [
        "present",
        "missing"
      ]
    }
  },
  "allOf": [
```

```
    {
      "if": {
        "properties": {
          "value_status": {
            "const": "present"
          }
        }
      },
      "then": {
        "properties": {
          "water_temperature": {
            "type": "number"
          }
        }
      },
      "else": {
        "properties": {
          "water_temperature": {
            "type": "null"
          }
        }
      }
    }
  ]
}
```


C04 A3 alignment document

Attribute	Definition
Producer → consumer	scripts/quality/alignment.py::build → seagis_quality.save_snapshot
Interface / format	JSON / SQL jsonb
Classification	explicit
Database mapping	seagis_quality.alignment_snapshot + snapshot_input
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/curate_environment.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Contract structure version; not a biological model version
Validation / allowed values	{"const": "a3-alignment-1"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"a3-alignment-1"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	builder_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	policy_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	QC policy version reference
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	policy_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Policy-content digest
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	code_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Alignment-builder source digest
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Source-bundle digest for source-change detection
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	plot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Cultivation plot reference
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	mode
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	demo_review or operational_review according to plot provenance
Validation / allowed values	{"enum": ["demo_review", "operational_review"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"demo_review"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	use
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Document use restriction quality_alignment_review
Validation / allowed values	{"const": "quality_alignment_review"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"quality_alignment_review"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	training_authorized
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Must be false; review/demo is not an approved training dataset
Validation / allowed values	{"const": false}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	false
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_registry
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Parameter-contract copy checked against the SQL bundle
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_registry[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Parameter-contract copy checked against the SQL bundle
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "native_unit", "unit", "conversion_code", " native_cadence_seconds", "attribution", "licence_url"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	sensor_deployments
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Deployment-identity copy for consistency checks
Validation / allowed values	{"type": "array"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	sensor_deployments[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Deployment-identity copy for consistency checks
Validation / allowed values	{"type": "object"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	start_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Inclusive alignment-window start
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	end_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Exclusive alignment-window end
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UTC hourly context rows; not additional independent samples
Validation / allowed values	{"type": "array", "minItems": 1, "maxItems": 168}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UTC hourly context rows; not additional independent samples
Validation / allowed values	{"type": "object", "required": ["at", "cells"], "additionalProperties": false}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UTC hourly cutoff for context selection
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].cells
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Per-dataset/parameter context at the hourly cutoff
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].cells[]
Type / ref	"#/$defs/cell"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Per-dataset/parameter context at the hourly cutoff
Validation / allowed values	{"$ref": "#/$defs/cell"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	QC summary and coverage based on hourly context
Validation / allowed values	{"type": "object", "required": ["plot_id", "start_at", "end_at", "coverage_basis", "drift_assessment", "parameters"]}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.plot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Cultivation plot reference
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.start_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Inclusive alignment-window start
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.end_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Exclusive alignment-window end
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.coverage_basis
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "string"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.drift_assessment
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"const": "not_assessable_without_calibrated_colocated_reference"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"not_assessable_without_calibrated_colocated_reference"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "array"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "expected_hourly_contexts", "usable_contexts", "unique_selected_source_records", "native_cadence_seconds", "coverage_fraction", "value_status_counts", "flag_counts", "input_qc_flag_counts_at_final_cutoff"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].expected_hourly_contexts
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Hourly slots in window, not native sample count

Validation / allowed values	{"type": "integer", "minimum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].usable_contexts
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "integer", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].unique_selected_source_records
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Count of distinct selected source records
Validation / allowed values	{"type": "integer", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].coverage_fraction
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	usable_contexts / expected_hourly_contexts
Validation / allowed values	{"type": "number", "minimum": 0, "maximum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.hash
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified

Validation / allowed values	{"type": "string", "pattern": "^[0-9a-f]{64}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.nullableTime
Type / ref	["string", "null"]
Required	False
Nullable	True
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": ["string", "null"], "format": "date-time"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Meaning	UNVERIFIED: contract structure/field; detailed meaning follows validator and source examples, not fully verified
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "scope", "native_cadence_seconds", "source_id", "value", "value_status", "unit", "source_time", "available_at", "age_seconds", "vintage_age_seconds", "lineage", "spatial", "flags", "blocking_flags", "usable_context"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.dataset_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Registered dataset key
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.parameter_code
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Meaning	Environmental parameter identity
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.scope
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Origin: local_sensor or regional_model
Validation / allowed values	{"enum": ["local_sensor", "regional_model"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"local_sensor"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.native_cadence_seconds
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Native sample cadence, distinct from fetch interval
Validation / allowed values	{"type": "integer", "minimum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.source_id
Type / ref	["string", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Typed reference to selected source record, or null
Validation / allowed values	{"type": ["string", "null"], "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND

Meaning	Normalized value; null follows value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.native_value
Type / ref	["number", "null"]
Required	False
Nullable	True
Default	NOT FOUND
Meaning	Physical value in source units
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Numeric availability, separate from quality
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Normalized value unit from registry
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.source_time
Type / ref	"#/$defs/nullableTime"
Required	True
Nullable	True
Default	NOT FOUND

Meaning	Sensor observed_at or model valid_at
Validation / allowed values	{"$ref": "#/$defs/nullableTime"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.available_at
Type / ref	"#/$defs/nullableTime"
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Conservative input availability time
Validation / allowed values	{"$ref": "#/$defs/nullableTime"}
Units / timestamp	Units/time follow unit and parameter registry; UTC for normalized envelopes.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.age_seconds
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Age of source_time relative to cutoff in seconds
Validation / allowed values	{"type": ["number", "null"], "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.vintage_age_seconds
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Run age relative to cutoff in seconds
Validation / allowed values	{"type": ["number", "null"], "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.lineage
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND

Meaning	Typed references to selected records and supporting QC inputs
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.lineage[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Typed references to selected records and supporting QC inputs
Validation / allowed values	{"type": "string", "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.spatial
Type / ref	["object", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Meaning	Grid mapping, resolution, distance and depth metadata
Validation / allowed values	{"type": ["object", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.flags
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	QC codes from packet acceptance
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.flags[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Meaning	QC codes from packet acceptance
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.blocking_flags
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Reasons context is unusable under the builder
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.blocking_flags[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Reasons context is unusable under the builder
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"See complete fixture/schema; standalone field example unverified"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.usable_context
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Meaning	Builder QC decision for review; not training authorization
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Complete schema for conditional rules and polymorphic structures (required above applies within each parent):


```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/alignment.schema.json",
  "title": "A3 quality and hourly alignment snapshot",
  "type": "object",
  "required": [
    "schema_version",
```

```
    "builder_version",
    "policy_version",
    "policy_sha256",
    "code_sha256",
    "source_sha256",
    "plot_id",
    "mode",
    "use",
    "training_authorized",
    "source_registry",
    "sensor_deployments",
    "start_at",
    "end_at",
    "rows",
    "quality_report"
  ],
  "additionalProperties": false,
  "properties": {
    "schema_version": {
      "const": "a3-alignment-1"
    },
    "builder_version": {
      "type": "string",
      "minLength": 1
    },
    "policy_version": {
      "type": "string",
      "minLength": 1
    },
    "policy_sha256": {
      "$ref": "#/$defs/hash"
    },
    "code_sha256": {
      "$ref": "#/$defs/hash"
    },
    "source_sha256": {
      "$ref": "#/$defs/hash"
    },
    "plot_id": {
      "type": "string",
      "format": "uuid"
    },
    "mode": {
      "enum": [
        "demo_review",
        "operational_review"
      ]
    },
    "use": {
      "const": "quality_alignment_review"
    },
    "training_authorized": {
      "const": false
    },
    "source_registry": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": [
          "dataset_key",
          "parameter_code",
          "native_unit",
          "unit",
          "conversion_code",
          "native_cadence_seconds",
          "attribution",
          "licence_url"
        ]
      }
    },
    "sensor_deployments": {
      "type": "array",
      "items": {
        "type": "object"
      }
    },
    "start_at": {
      "type": "string",
      "format": "date-time"
    },
    "end_at": {
      "type": "string",
```

```
    "format": "date-time"
  },
  "rows": {
    "type": "array",
    "minItems": 1,
    "maxItems": 168,
    "items": {
      "type": "object",
      "required": [
        "at",
        "cells"
      ],
      "additionalProperties": false,
      "properties": {
        "at": {
          "type": "string",
          "format": "date-time"
        },
        "cells": {
          "type": "array",
          "minItems": 1,
          "items": {
            "$ref": "#/$defs/cell"
          }
        }
      }
    }
  },
  "quality_report": {
    "type": "object",
    "required": [
      "plot_id",
      "start_at",
      "end_at",
      "coverage_basis",
      "drift_assessment",
      "parameters"
    ],
    "properties": {
      "plot_id": {
        "type": "string",
        "format": "uuid"
      },
      "start_at": {
        "type": "string",
        "format": "date-time"
      },
      "end_at": {
        "type": "string",
        "format": "date-time"
      },
      "coverage_basis": {
        "type": "string"
      },
      "drift_assessment": {
        "const": "not_assessable_without_calibrated_colocated_reference"
      },
      "parameters": {
        "type": "array",
        "items": {
          "type": "object",
          "required": [
            "dataset_key",
            "parameter_code",
            "expected_hourly_contexts",
            "usable_contexts",
            "unique_selected_source_records",
            "native_cadence_seconds",
            "coverage_fraction",
            "value_status_counts",
            "flag_counts",
            "input_qc_flag_counts_at_final_cutoff"
          ],
          "properties": {
            "expected_hourly_contexts": {
              "type": "integer",
              "minimum": 1
            },
            "usable_contexts": {
              "type": "integer",
              "minimum": 0
            },
            "unique_selected_source_records": {
```

```
            "type": "integer",
            "minimum": 0
          },
          "coverage_fraction": {
            "type": "number",
            "minimum": 0,
            "maximum": 1
          }
        }
      }
    }
  }
 },
 "$defs": {
  "hash": {
   "type": "string",
   "pattern": "^[0-9a-f]{64}$"
  },
  "nullableTime": {
   "type": [
    "string",
    "null"
   ],
   "format": "date-time"
  },
  "cell": {
   "type": "object",
   "required": [
    "dataset_key",
    "parameter_code",
    "scope",
    "native_cadence_seconds",
    "source_id",
    "value",
    "value_status",
    "unit",
    "source_time",
    "available_at",
    "age_seconds",
    "vintage_age_seconds",
    "lineage",
    "spatial",
    "flags",
    "blocking_flags",
    "usable_context"
   ],
   "properties": {
    "dataset_key": {
     "type": "string"
    },
    "parameter_code": {
     "type": "string"
    },
    "scope": {
     "enum": [
      "local_sensor",
      "regional_model"
     ]
    },
    "native_cadence_seconds": {
     "type": "integer",
     "minimum": 1
    },
    "source_id": {
     "type": [
      "string",
      "null"
     ],
     "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"
    },
    "value": {
     "type": [
      "number",
      "null"
     ]
    },
    "native_value": {
     "type": [
      "number",
      "null"
     ]
    },
```

```
      "value_status": {
        "enum": [
          "present",
          "missing",
          "below_detection_limit",
          "not_available"
        ]
      },
      "unit": {
        "type": "string"
      },
      "source_time": {
        "$ref": "#/$defs/nullableTime"
      },
      "available_at": {
        "$ref": "#/$defs/nullableTime"
      },
      "age_seconds": {
        "type": [
          "number",
          "null"
        ],
        "minimum": 0
      },
      "vintage_age_seconds": {
        "type": [
          "number",
          "null"
        ],
        "minimum": 0
      },
      "lineage": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string",
          "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"
        }
      },
      "spatial": {
        "type": [
          "object",
          "null"
        ]
      },
      "flags": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string"
        }
      },
      "blocking_flags": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string"
        }
      },
      "usable_context": {
        "type": "boolean"
      }
    }
  }
 }
}
```


C05 A1 history/status/replies

Attribute	Definition
Producer → consumer	seagis_api.history/status/reply → authorized reader
Interface / format	JSON / HTTP GET
Classification	implementation-derived
Database mapping	observation + sensor + packet_receipt
Compatibility	Constant version where defined; general negotiation NOT FOUND

Attribute	Definition
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

Field / group	Type / required	Meaning / validation / mapping
from, to	string / required	UTC Z; from<to; <=7 days; observation.observed_at [from,to)
limit	integer string / optional; default 200	1..500
cursor	UUID / optional	observation.id inside plot/window; ordering observed_at,id
code, request_id	string, UUID / response	SQL reply / handler error
observations[]	array / history response	id,sensor_id,value,unit,value_status,observed_at,ingested_at,quality_status,flags,ingestion_run_id,is_synthetic
coverage	object	expected_hours,hours_with_packets,hours_with_usable_values,missing_hours,unusable_hours
next_cursor	UUID|null	last returned row if another page exists

C06 A2 adapter batch/lease

Attribute	Definition
Producer → consumer	Adapter.envelope + claim → seagis_environment.persist
Interface / format	JSON / SQL jsonb
Classification	implementation-derived
Database mapping	raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/environmental/common.py:1

Field	Required	Validation / mapping
dataset_key	True	Registered dataset key; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
purpose	True	Purpose noncommercial_educational_prototype; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
terms_review	True	Provider terms review identity; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
dataset_version	True	Pinned dataset version; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
provider_dataset_id	True	Original provider dataset identity; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
run_key	True	Semantic capture identity per dataset; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
raw_sha256	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
raw_byte_size	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
hash_basis	True	Byte or serialization convention used for hashing; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
format	True	Raw representation media type or format; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
started_at	True	Ingestion or attempt start instant; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
received_at	True	Raw receipt instant at worker; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping

Field	Required	Validation / mapping
metadata	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
grids	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
mappings	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
values	True	See persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping

Nested grids contain key, latitude, longitude, depth_m and identity_basis; values contain grid_key, parameter, value, native_unit, unit, conversion_code, value_status, valid_at, temporal_support, interval_status and period bounds. Mappings carry plot identity and example distance. SQL checks envelope field presence, lease, purpose/licence, 64-hex hashes and grid/value/mapping counts; this is not a universal JSON Schema.

C07 A3 source_bundle

Attribute	Definition
Producer → consumer	seagis_quality.source_bundle → build / CSV exporter
Interface / format	JSON / SQL jsonb
Classification	implementation-derived
Database mapping	plot + sensor + dataset_parameter + mappings + observation + forecast_value
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1

Field	Type / required	Mapping
schema_version	constant a3-source-1	internal
plot	object	seagis.plot
start_at,end_at	timestamptz / required	query bounds
sensors	array	seagis.sensor
parameters	array	dataset_parameter + dataset + environment.policy
mappings	array	plot_grid_mapping + demo_grid_mapping + source_grid
records	array	observation + forecast_value + quality_event
source_sha256	64-hex string	seagis.jsonb_sha256(bundle)

C08 CSV experiment-csv-2

Attribute	Definition
Producer → consumer	export_experiment_csv.py::export → offline experiments / developer
Interface / format	CSV + JSON manifest
Classification	implementation-derived
Database mapping	derived files; no database writes
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

Manifest export_version=experiment-csv-2 records capture/export times, bounds, units, unique record counts, CSV counts and per-file checksums. Temperature CSV carries timestamp_utc, timestamp_wib, plot_id, value, raw_value, gap/synthetic/QC

fields, observation_id, available_at_utc, training_authorized=false, parameter_code, unit, source_kind and value_policy. Regional CSV carries flattened source-record fields, cadence, attribution, licence_url, selection_basis and capture_at_utc. Dynamic source fields mean the CSV schema follows bundle records rather than a universal fixed column contract.

C09 iOS Assessment domain

Attribute	Definition
Producer → consumer	MockAssessmentRepository → LoadDemoAssessment / ViewModel
Interface / format	Swift structs
Classification	implementation-derived / mock
Database mapping	NOT APPLICABLE: no backend mapping
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

Swift field	id
Type	String
Required	True
Meaning / mapping	Row identity under the table primary key

Swift field	plotID
Type	String
Required	True
Meaning / mapping	Demo domain field; not a backend DTO

Swift field	title
Type	String
Required	True
Meaning / mapping	Demo domain field; not a backend DTO

Swift field	summary
Type	String
Required	True
Meaning / mapping	Demo domain field; not a backend DTO

Swift field	provenance
Type	AssessmentProvenance.syntheticFixture
Required	True
Meaning / mapping	Demo domain field; not a backend DTO

Swift field	evidenceIDs
Type	[String]
Required	True
Meaning / mapping	Demo domain field; not a backend DTO

C10 Open-Meteo response

Attribute	Definition
Producer → consumer	Open-Meteo forecast API → OpenMeteo.parse/normalize
Interface / format	JSON hourly arrays
Classification	implementation-derived
Database mapping	forecast_value + grid + raw manifest
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/environmental/open_meteo.py:1

Response must be a list matching the two plots, each item carrying latitude/longitude/elevation, timezone UTC/GMT, utc_offset_seconds=0, hourly_units.time=iso8601 and aligned hourly arrays. Variable fields and units come from a2_sources.json. generationtime_ms is not issue time. Null/numeric handling follows common.point. No stable provider response_error format is claimed.

C11 Copernicus subset

Attribute	Definition
Producer → consumer	Copernicus Toolbox/xarray + STAC → Copernicus.parse/normalize
Interface / format	xarray + canonical logical JSON subset
Classification	implementation-derived
Database mapping	forecast_value + grid + raw manifest
Compatibility	Constant version where defined; general negotiation NOT FOUND
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1

STAC requires id, collection/product, cube:dimensions time/elevation/latitude/longitude and cube:variables unit/standardName. The xarray dataset has one latitude/longitude/depth and unique ordered times at native cadence; variables are transposed time,depth,latitude,longitude. The logical subset contains cell,time,variables,native_units and is not a provider wire format. Unknown original metadata is not fabricated.

HTTP routes and responses

Method	GET
Path (temperature base)	/health
Auth	none
Result / errors	200 service=seagis-temperature, version=1.0.0

Method	POST
Path (temperature base)	/v1/ingest
Auth	device Bearer
Result / errors	201 stored; 200 duplicate; 409 packet_conflict; 422 invalid_packet; 403 sensor_forbidden/sensor_inactive; 503 commissioning_disabled

Method	GET
Path (temperature base)	/v1/plots/{plot_id}/temperature
Auth	reader Bearer + reader_plot
Result / errors	200 observations/coverage; 400 invalid_query/invalid_plot_id; 403 plot_forbidden

Method	GET
Path (temperature base)	/v1/status

Auth	reader Bearer
Result / errors	200 checked_at,stale_after_seconds,plots[]

Method	all
Path (temperature base)	known protected paths
Auth	Bearer sgi_ pattern
Result / errors	401 unauthorized; 429 rate_limited

Method	all
Path (temperature base)	HTTP boundary
Auth	handler
Result / errors	404 not_found; 405 method_not_allowed; 415 json_required; 413 payload_too_large; 400 invalid_json/empty_body/unexpected_query; 503 temporarily_unavailable

Deployment base /functions/v1/temperature; the handler also accepts a /temperature prefix. Responses use no-store. Retry-After is 60 for 429 and 5 for 503. Status plots[] contains plot_id,plot_code,is_synthetic,latest_usable_observed_at,last_ingested_at,age_seconds,stale,observations_last_24h,rejected_last_24h. last_ingested_at need not belong to the latest_usable_observed_at record. Null time/age means unavailable.

A1 payloads and errors

Synthetic structurally valid example; persistence requires a registered credential/sensor and active policy.


```
{
  "schema_version": "1.0.0",
  "sensor_id": "00000000-0000-4000-8000-000000000101",
  "boot_id": "synthetic-boot-a",
  "sequence": 1,
  "observed_at": "2026-10-21T12:00:00Z",
  "water_temperature": 29.2,
  "unit": "degC",
  "value_status": "present"
}
```


Invalid example: change unit to kelvin. SQL returns 422 invalid_packet after successful authentication.


```
{
  "code": "invalid_packet",
  "request_id": "00000000-0000-4000-8000-000000000999"
}
```


A stored response has observation_id, ingestion_run_id, observed_at, quality_status, flags, is_synthetic, code and request_id. UUIDs/times below are illustrative, not a live response.


```
{
  "observation_id": "00000000-0000-4000-8000-000000000201",
  "ingestion_run_id": "00000000-0000-4000-8000-000000000301",
  "observed_at": "2026-10-21T12:00:00.000000Z",
  "quality_status": "accepted",
  "flags": [],
  "is_synthetic": true,
  "code": "stored",
  "request_id": "00000000-0000-4000-8000-000000000999"
}
```


Boundary	A0 vs A1 packet
Rule / discrepancy	A0 fixture uses a 0.1.0 envelope and detection_limit; A1 accepts exactly eight 1.0.0 fields and rejects extras.
Classification	VERIFIED
Evidence	Seagis-Model/contracts/iot-packet.schema.json:1

Boundary	A0 run_kind vs A2
Rule / discrepancy	0003 extends run_kind with analysis_forecast_unspecified; do not rely on the old A0 list.

Classification	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1

Boundary	A3 JSON schema vs runtime
Rule / discrepancy	Tests validate the full schema; curate/build does not call a general JSON Schema validator and SQL enforces selected checks.
Classification	VERIFIED
Evidence	Seagis-Model/scripts/curate_environment.py:1

Boundary	API time vs PostgreSQL
Rule / discrepancy	A1 UTC Z; timestamptz session display / microseconds
Classification	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

Boundary	iOS Assessment vs DB
Rule / discrepancy	No verified backend assessment DTO/contract.
Classification	NOT FOUND
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

9 Data Lifecycle

Creation: callers or SQL supply UUIDs as specified in the dictionary; registry natural keys are pinned in seed/config. Raw data, measurements, forecasts, QC and snapshots have mutation guards. Dataset changes are restricted to certain operational gates; semantic changes to dataset, parameter or grid identities require new versions. Location/depth changes require a new sensor deployment; active can be disabled.

Ingestion transitions from running to succeeded/failed/partial under checks and guards. Regional attempts transition from running to succeeded/duplicate/failed/abandoned. An expired lease is marked abandoned on the next claim. Feature snapshots transition draft → sealed; the input manifest must match and a parent row lock prevents races during sealing. A3 snapshots are immutable immediately after insertion. Credentials are revoked using revoked_at or expire through expires_at; neither deletes observations.

A0 initially allowed closing effective_until on mappings. Migration 0005 adds recorded_at and an immutable_alignment_mapping trigger before UPDATE/DELETE. Therefore older instructions to close mappings through UPDATE no longer apply after the full migration chain. New interval resolution requires a design that preserves history without silently bypassing guards.

No age-based cleanup/retention job, general soft delete, tested backup/restore, cross-database replication or iOS-cloud synchronization was found. A2 metadata_only is a raw-content policy, not a TTL for all tables. Foreign keys without explicit ON DELETE/UPDATE use NO ACTION; cascade deletion must not be assumed. Prototype FarmStore deletes farmers and their plots in memory; it does not issue backend SQL DELETE statements.

Migration sequence: 0001 initial, 0002 temperature API, 0003 regional ingestion, 0004 provider_updating, 0005 quality alignment. SQL migrations are forward-only; rollback/down scripts were NOT FOUND. Deployment scripts use administrator access for provisioning and worker-role setup; none was run during this audit. Migration adoption should include backups, empty-schema tests and upgrade tests; these are recommendations, not session test results.

Security: migrations enable RLS on application tables; FORCE RLS=false in the historical snapshot. PUBLIC/anon/authenticated access is restricted; gateway/workers receive selected EXECUTE permissions using SECURITY DEFINER and search_path=pg_catalog. A2/A3 use TLS verify-full with a pinned CA hash. Original tokens are not printed; credentials store hashes. Active administrator privileges, role membership, bucket policies, encryption at rest, actual rotation, production backup and recovery remain UNVERIFIED. Rate-limit logs, attempts, QC events and hashes provide partial auditing; no dead-letter queue or configured external alerting was found on these paths.

Main implementation evidence: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1; Seagis-Model/supabase/migrations/0002_temperature_api.sql:1; Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1; Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1; Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Main implementation evidence: Seagis-Model/scripts/environmental/store.py:1; Seagis-Model/scripts/quality/store.py:1; Seagis-Model/supabase/functions/temperature/connection.ts:1.

10 Application Data Dependencies

Temperature workflow: device packet → handler → seagis_api.ingest → receipt/observation/QC → authorized reader history call with from/to and cursor → data/coverage JSON. History orders by observed_at,id; from is inclusive and to exclusive, with a seven-day maximum window and default limit 200/maximum 500. Coverage uses UTC hourly buckets and may differ from paginated row counts. Status summarizes deployments/last packets; it is not a disease assessment.

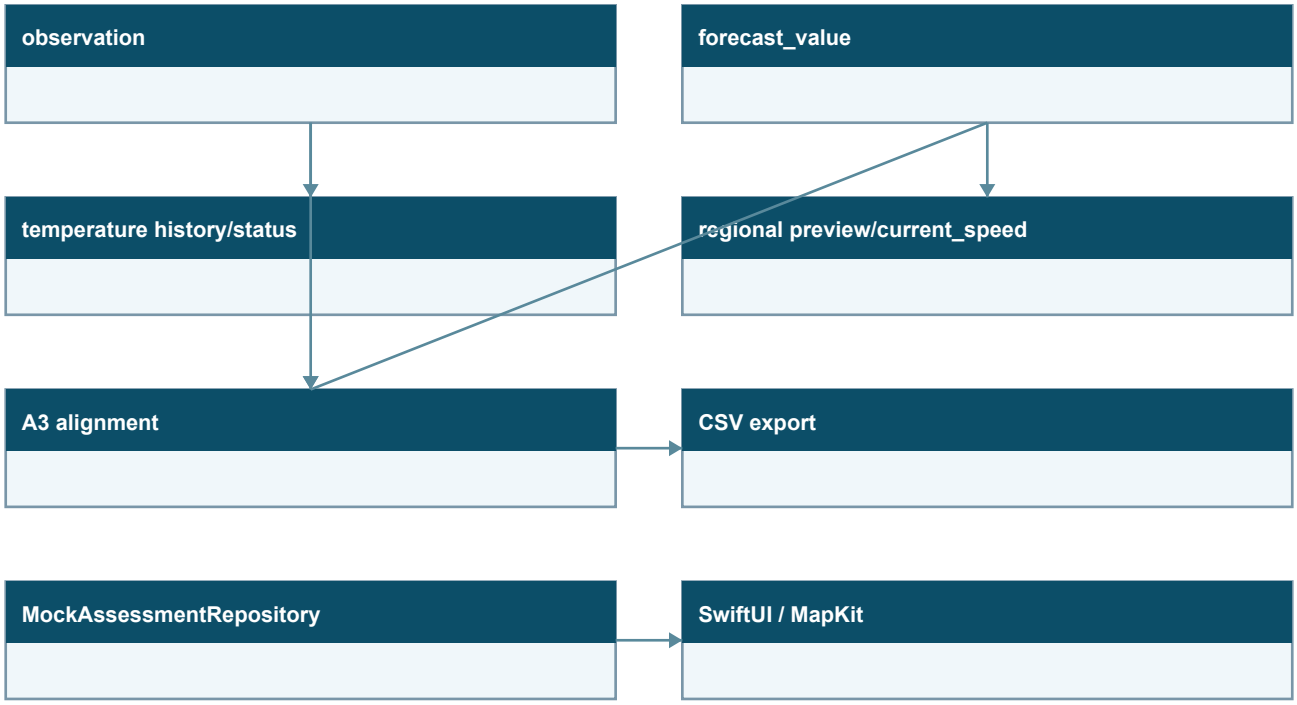
Regional workflow: dispatcher → claim/fetch/persist → preview selects the latest run available at cutoff and not stale under policy, limits valid_at to the next two days and returns at most 1500 rows. current_speed is an inner join of components; incomplete pairs produce no view row, whereas complete pairs containing missing values produce null speed. Demo preview preserves grid identity and representativeness limitations.

The iOS workflow is AppContainer dependency injection → MockAssessmentRepository → LoadDemoAssessment → BootstrapViewModel → SwiftUI. The Assessment domain model only has synthetic_fixture provenance and no verified backend DTO. AppPOC FarmStore keeps Farmer/CultivationPlot arrays, generates UUIDs, trims names and deletes relationships in memory. No verified mapping connects prototype farmer/plot IDs to seagis.plot. Production local caching/persistence and synchronization conflict handling were not found implemented.

Verified analytical consumers are QC building and CSV export; no production feature builder or training consumer was found for seagis_features. Hourly coverage is not counted as independent observations. No trained ML model producing ice-ice probabilities was found on these backend paths.

Main implementation evidence: Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift:1; AppPOC/POCApp/Shared/Data/FarmStore.swift:1; Seagis-Model/supabase/migrations/0002_temperature_api.sql:1; Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1.

Reproducible diagram: assets/diagrams/dependencies.mmd.



Data entity	seagis.dataset
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Dataset identity, version, cadence, licence and ingestion gates.

Data entity	seagis.dataset_parameter
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Source-variable to parameter contract and conversion rules.

Data entity	seagis.forecast_run
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Regional model capture/vintage; unknown publication time remains null.

Data entity	seagis.forecast_value
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Parameter value per run, grid and valid time; not duplicated per plot.

Data entity	seagis.ingestion_run
Producer / writers	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Readers	A3 source_bundle; provenance guards
API / feature	private SQL / demo / offline
Downstream	Audit of a logical ingestion and its completion state.

Data entity	seagis.observation
Producer / writers	seagis_api.ingest; A0 fixtures
Readers	seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
API / feature	temperature history/status
Downstream	Local sensor water-temperature readings, retaining native values, QC and raw lineage.

Data entity	seagis.parameter
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Registry of parameters and normalized units.

Data entity	seagis.plot
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Plot identity and survey status; nullable coordinates are not evidence of a surveyed location.

Data entity	seagis.plot_grid_mapping
Producer / writers	A0 fixture/admin SQL
Readers	A3 source_bundle; feature_input guards
API / feature	private SQL / demo / offline

Downstream	Operational plot-to-grid mapping with effective and recorded times.

Data entity	seagis.quality_event
Producer / writers	seagis_api.ingest; A0 fixtures
Readers	A3 source_bundle/alignment
API / feature	private SQL / demo / offline
Downstream	Append-only QC log for ingestion or one measurement.

Data entity	seagis.sensor
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Temperature sensor deployment identity, plot, depth and active state.

Data entity	seagis.source
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Registry of dataset providers.

Data entity	seagis.source_grid
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Regional cell and depth-layer identity; distinct from cultivation plots.

Data entity	seagis_api.credential
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	API actor identity using token hashes, expiry and revocation.

Data entity	seagis_api.packet_receipt
Producer / writers	seagis_api.ingest/reply
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Sensor-packet idempotency key and semantic digest.

Data entity	seagis_api.reader_plot
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Plot allowlist for a reader credential.

Data entity	seagis_api.request_event
Producer / writers	seagis_api.ingest/reply

Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Authenticated API response audit and rate-limit basis.

Data entity	seagis_api.temperature_policy
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Pilot engineering QC settings and request limit; not disease thresholds.

Data entity	seagis_environment.attempt
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Audit of a fetch attempt, including duplicate and abandoned outcomes.

Data entity	seagis_environment.current_speed
Producer / writers	NOT APPLICABLE: view
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Current-magnitude view joining components from the same run, grid and time.

Data entity	seagis_environment.demo_grid_mapping
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Bridge from two fictional plots to real grids for preview, not training authorization.

Data entity	seagis_environment.policy
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Regional fetch, licence, scheduler cadence, staleness and retention policy.

Data entity	seagis_environment.schedule_state
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Mutable lease, due time, backoff and last-success state.

Data entity	seagis_features.feature_input
Producer / writers	A0 fixture/test inserts; production builder NOT FOUND
Readers	snapshot/input guards; production ML consumer NOT FOUND
API / feature	private SQL / demo / offline
Downstream	Foreign-key input lineage and cutoff guards for feature snapshots.

Data entity	seagis_features.feature_snapshot
Producer / writers	A0 fixture/test inserts; production builder NOT FOUND
Readers	snapshot/input guards; production ML consumer NOT FOUND
API / feature	private SQL / demo / offline
Downstream	Draft/sealed feature container with cutoff and hashes; no production builder found.

Data entity	seagis_quality.alignment_snapshot
Producer / writers	seagis_quality.save_snapshot
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Immutable hourly QC/alignment document; always training_authorized=false.

Data entity	seagis_quality.policy
Producer / writers	scripts/quality/store.py::register_policy
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Immutable registry of QC policy versions and builder code hashes.

Data entity	seagis_quality.snapshot_input
Producer / writers	seagis_quality.save_snapshot
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Per-hour measurement lineage of an alignment snapshot.

Data entity	seagis_raw.raw_payload
Producer / writers	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Readers	A3 source_bundle; provenance guards
API / feature	private SQL / demo / offline
Downstream	Immutable raw manifest: digest, hash basis, licence and retention mode.

11 Data Quality and Architecture Audit

Read-only audit of code and historical metadata; severity expresses potential impact within the prototype scope, not a measured production incident. No application/schema changes were applied.

F01 iOS assessment is not connected to backend

Attribute	Finding
Severity / category	High / Contract / integration
Status	VERIFIED
Description / impact	The mock returns synthetic_fixture; no assessment API adapter was found. New consumers must not treat the domain model as a wire contract.
Affected	Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift, Seagis/Seagis/Models/Shared/Assessment.swift
Evidence	Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift:1; Seagis/Seagis/Models/Shared/Assessment.swift:1
Recommendation	Agree an executable contract and provenance/error mapping before integration.

F02 Mapping immutability after 0005 changes the old closing procedure

Attribute	Finding
Severity / category	Medium / Lifecycle
Status	VERIFIED
Description / impact	A0 guard_mapping permits interval closure; the 0005 trigger rejects all UPDATE/DELETE. A0 architecture instructions for closing mappings can fail on the complete schema.
Affected	supabase/migrations/0001_initial_schema.sql, supabase/migrations/0005_quality_alignment.sql, docs/architecture.md
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1; Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1; Seagis-Model/docs/architecture.md:1
Recommendation	Define append-only supersession and test interval overlap/cutoff.

F03 History grows without a discovered retention job

Attribute	Finding
Severity / category	Medium / Governance / retention
Status	INFERRED
Description / impact	request_event, observations, forecasts and snapshots are append-only; scheduled TTL/cleanup and recovery tests were not found in the checkouts. Capacity risk is inferred, not benchmarked.
Affected	supabase/migrations/0001_initial_schema.sql, supabase/migrations/0002_temperature_api.sql, supabase/migrations/0005_quality_alignment.sql
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1; Seagis-Model/supabase/migrations/0002_temperature_api.sql:1; Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1
Recommendation	Set domain retention, archival, backups and restore drills before production scale.

F04 A3 SQL checks lineage but does not recompute all QC

Attribute	Finding
Severity / category	Medium / Validation boundary
Status	VERIFIED
Description / impact	save_snapshot verifies policy/code hashes and source values, but usable_context/flags/blocking_flags decisions are not recomputed through Python inside the DB. The worker role remains a trust boundary.
Affected	supabase/migrations/0005_quality_alignment.sql, scripts/quality/alignment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1; Seagis-Model/scripts/quality/alignment.py:1
Recommendation	Document worker trust; add deterministic QC verification and payload-tampering tests where required.

F05 Metadata-only limits raw reproducibility

Attribute	Finding
Severity / category	Medium / Data provenance
Status	VERIFIED
Description / impact	A2 retains no raw content. A Toolbox subset hash is not an original NetCDF/Zarr byte hash; refetching a provider aggregate does not guarantee the same vintage.
Affected	scripts/environmental/copernicus.py, supabase/migrations/0003_environment_ingestion.sql
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1; Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1
Recommendation	Preserve hash_basis; if licence and cost allow, propose immutable versioned subset/object retention.

F06 Capture CSV is not a historical backtest

Attribute	Finding
Severity / category	High / Analytics semantics
Status	VERIFIED
Description / impact	The exporter selects latest_available_at_capture; A3 uses a per-hour cutoff. Treating the CSV as historical availability risks leakage; code explicitly sets training_authorized=false.
Affected	scripts/export_experiment_csv.py, scripts/quality/alignment.py
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1; Seagis-Model/scripts/quality/alignment.py:1
Recommendation	Use all-vintage source bundles and explicit cutoffs when designing backtests; retain the training gate until validation.

F07 A0 documentation predates A1–A3 implementation

Attribute	Finding
Severity / category	Low / Documentation drift
Status	VERIFIED
Description / impact	docs/architecture.md says ingestion/scheduler/QC are not implemented, while subsequent migrations and workers exist. Statements require phase/date context.
Affected	docs/architecture.md, scripts/ingest_environment.py, scripts/curate_environment.py
Evidence	Seagis-Model/docs/architecture.md:1; Seagis-Model/scripts/ingest_environment.py:1; Seagis-Model/scripts/curate_environment.py:1
Recommendation	Use a phase-status index and source manifest for documentation updates.

F08 Catalog snapshot is five days old

Attribute	Finding
Severity / category	Medium / Runtime verification
Status	UNVERIFIED
Description / impact	The 22 October snapshot is not a 27 October runtime audit. Grants, row counts, active schedules and current cloud migrations were not reverified.
Affected	runtime catalog
Evidence	schema-metadata-20261022.json inspected_at
Recommendation	Run a read-only catalog diff and grants review before production decisions.

Priority: P1 agree integration/contracts and prevent leakage (F01/F06); P2 address trust, retention, mapping lifecycle and raw reproducibility (F02–F05); P3 refresh documentation status/catalog (F07/F08). Index/performance evaluation needs EXPLAIN and actual workloads: no missing-index or slow-query claim is established merely from column names.

12 Glossary

Term	Definition
Origin	Data origin: sensor, regional model or derived value
Ingestion	Receipt and persistence of one capture
Vintage	Preserved model capture version
Native cadence	Original sample interval, distinct from scheduler tick
Availability cutoff	Boundary of information available at decision time
Value status	Numeric availability, not disease state
QC	Data quality checks; not diagnosis
Idempotency	Identical retry does not add a measurement
Lineage	Reference trail from output to source
Manifest	Input/file list with identities and digests
RLS	Row Level Security; effect depends on roles and grants
FK	Foreign key: referential integrity between rows
NULL	Unavailable value; distinct from numeric zero
Metadata-only	Provenance/digest retention without raw content
Sealed snapshot	Snapshot that cannot be mutated after sealing
Ice-ice	Cultivation disease context; this documentation does not validate its prediction

13 Appendices

13.1 Evidence base and audit boundaries

Repository	Commit	Path
Seagis-Model	291b6c5fdc7c990660ea98b1dcf676f6c0b58148	/Users/cindyram/Documents/Apple/Seagis-Model
Seagis	109619ba0cb43035ce102e187158fbce9a7a42c1	/Users/cindyram/Documents/Apple/Seagis
AppPOC	c68d72ae90457a9e8def40b57dfae9e86607aff8	/Users/cindyram/Documents/Apple/AppPOC

Implementation files were read through the documentation generator and targeted searches; the manifest records per-file hashes. No external source substitutes for implementation. No migrations, ingestion workers, database-writing tests or deployment were run. Pure/offline contract tests are reported separately in validation.json. Environment files, credentials and table contents were not copied.

13.2 Measured coverage

Component	Application databases
Discovered	1
ID documented	1
EN documented	1
Evidence / gaps	Code + historical catalog; current runtime unresolved

Component	Application schemas
Discovered	6
ID documented	6
EN documented	6
Evidence / gaps	historical catalog + migrations

Component	Physical tables
Discovered	28
ID documented	28
EN documented	28
Evidence / gaps	row counts UNVERIFIED

Component	Views
Discovered	1
ID documented	1
EN documented	1
Evidence / gaps	current_speed

Component	Columns (tables + view)
Discovered	268
ID documented	268
EN documented	268
Evidence / gaps	259 table + 9 view fields; some meanings remain limited

Component	Physical FK
Discovered	47
ID documented	47
EN documented	47
Evidence / gaps	historical catalog + SQL

Component	Column-origin mappings
Discovered	268
ID documented	268
EN documented	268
Evidence / gaps	268 VERIFIED; 0 UNVERIFIED

Component	Pipeline paths
Discovered	4
ID documented	4
EN documented	4
Evidence / gaps	A1, A2, A3, CSV export

Component	Boundary contracts
Discovered	11
ID documented	11
EN documented	11
Evidence / gaps	4 explicit; 7 implementation-derived

Component	JSON-schema field/structure entries
Discovered	147
ID documented	147
EN documented	147
Evidence / gaps	Not all nested semantics resolved; full schemas included

Component	Object dependency rows
Discovered	29
ID documented	29
EN documented	29
Evidence / gaps	writers/readers from code; deployment unverified

Component	Runtime rows/grants/backups
Discovered	UNVERIFIED
ID documented	UNVERIFIED
EN documented	UNVERIFIED
Evidence / gaps	No live access in this audit

Coverage of 29/29 snapshot objects and 268/268 columns is not a claim of 100% production-architecture coverage. Internal schemas, sources outside the three checkouts, runtime deployment and ambiguous nested fields have no known complete denominator.

13.2.1 Observed local file datasets

CSV metadata below was inspected without copying row values. Counts belong to local capture files, not database tables. Regional CSV retains native cadence and latest_available_at_capture; plot temperatures are synthetic. source_bundles/alignment/manifest JSON are exporter/curator outputs under C04/C07/C08; worker.log is operational logging and the CA certificate is trust material, not an environmental dataset.

File	Rows observed	Columns / contract
Seagis-Model/data/exports/forecast-20261023T015005Z/features/air_temperature_2m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/cloud_cover.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/current_eastward.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/current_northward.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/nitrate.csv	2	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/phosphate.csv	2	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/precipitation.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at

File	Rows observed	Columns / contract
Seagis-Model/data/exports/forecast-20261023T015005Z/features/salinity.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/uv_index.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/water_temperature_plot-a.csv	72	timestamp_utc, timestamp_wib, plot_id, is_gap, is_synthetic, value_status, quality_status, qc_flags, strict_source_qc_pass, observation_id, available_at_utc, training_authorized, value, raw_value, parameter_code, unit, source_kind, value_policy
Seagis-Model/data/exports/forecast-20261023T015005Z/features/water_temperature_plot-b.csv	72	timestamp_utc, timestamp_wib, plot_id, is_gap, is_synthetic, value_status, quality_status, qc_flags, strict_source_qc_pass, observation_id, available_at_utc, training_authorized, value, raw_value, parameter_code, unit, source_kind, value_policy
Seagis-Model/data/exports/forecast-20261023T015005Z/features/wind_direction_10m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/wind_speed_10m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at

13.3 Open questions

- Have cloud schemas and grants changed since 22 October? - How are immutable mapping supersession and overlap resolution defined? - Who owns backup/retention operations and how is restore verified? - Which assessment contract will backend/iOS approve? - When will real plots/locations/sensor depths be validated and the training gate evaluated?

13.4 Illustrative read-only SQL

The following query was not executed against the cloud in this audit.

```
BEGIN READ ONLY;
SELECT schemaname, tablename FROM pg_catalog.pg_tables
WHERE schemaname IN ('seagis','seagis_raw','seagis_api',
 'seagis_environment','seagis_quality','seagis_features')
ORDER BY schemaname, tablename;
ROLLBACK;
```

13.5 SQL function registry

Function	Arguments	Evidence
seagis.valid_value	v double precision, native double precision, status text, lim double precision	Seagis-Model/supabase/migrations/0001_initial_schema.sql:168
seagis.valid_support	t timestamptz, support text, status text, a timestamptz, b timestamptz	Seagis-Model/supabase/migrations/0001_initial_schema.sql:176
seagis.jsonb_sha256	value jsonb	Seagis-Model/supabase/migrations/0001_initial_schema.sql:296
seagis.reject_mutation		Seagis-Model/supabase/migrations/0001_initial_schema.sql:330
seagis.guard_ingestion		Seagis-Model/supabase/migrations/0001_initial_schema.sql:334
seagis.guard_dataset_identity		Seagis-Model/supabase/migrations/0001_initial_schema.sql:347
seagis.guard_sensor_deployment		Seagis-Model/supabase/migrations/0001_initial_schema.sql:357
seagis.guard_raw		Seagis-Model/supabase/migrations/0001_initial_schema.sql:366
seagis.guard_measurement		Seagis-Model/supabase/migrations/0001_initial_schema.sql:379
seagis.guard_run		Seagis-Model/supabase/migrations/0001_initial_schema.sql:406
seagis.guard_mapping		Seagis-Model/supabase/migrations/0001_initial_schema.sql:415
seagis.guard_quality_event		Seagis-Model/supabase/migrations/0001_initial_schema.sql:424
seagis_features.guard_input		Seagis-Model/supabase/migrations/0001_initial_schema.sql:434
seagis_features.manifest_hash	snapshot uuid	Seagis-Model/supabase/migrations/0001_initial_schema.sql:458
seagis_features.guard_snapshot		Seagis-Model/supabase/migrations/0001_initial_schema.sql:465
seagis_api.utc	t timestamptz	Seagis-Model/supabase/migrations/0002_temperature_api.sql:88
seagis_api.actor	token text, actor_kind text	Seagis-Model/supabase/migrations/0002_temperature_api.sql:90
seagis_api.reply	actor_id uuid, status integer, code text, body jsonb DEFAULT '{}', sensor uuid DEFAULT NULL	Seagis-Model/supabase/migrations/0002_temperature_api.sql:96
seagis_api.rate_ok	actor_id uuid	Seagis-Model/supabase/migrations/0002_temperature_api.sql:105
seagis_api.ingest	token text, packet jsonb, wire_sha256 text, wire_bytes integer	Seagis-Model/supabase/migrations/0002_temperature_api.sql:113
seagis_api.history	token text, target_plot uuid, spec jsonb	Seagis-Model/supabase/migrations/0002_temperature_api.sql:222
seagis_api.status	token text	Seagis-Model/supabase/migrations/0002_temperature_api.sql:274
seagis_environment.claim	k text, force_fetch boolean DEFAULT false	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:99
seagis_environment.finish_failure	lease uuid, code text	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:117
seagis_environment.persist	lease uuid, batch jsonb	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:131
seagis_environment.status		Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:212
seagis_environment.preview	plot uuid, as_of timestamptz DEFAULT now(	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:227
seagis_environment.finish_failure	lease uuid, code text	Seagis-Model/supabase/migrations/0004_provider_update_status.sql:2
seagis_quality.source_bundle	p_plot uuid,p_start timestamptz,p_end timestamptz,p_demo boolean DEFAULT false	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:47
seagis_quality.save_snapshot	document jsonb	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:121
seagis_quality.snapshot	identity uuid	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:232

13.6 Source register

Path	Lines	SHA-256
Seagis-Model/README.md	71	4db1d01136db09e662c2bb005264cd7ff657c5fc6b2a2851b2ef9ccb5010354a
Seagis-Model/config/a2_sources.json	15	a53addfbf7a0b2733bdcedaabf6c85dad2c428d3afd05eeffbdf79af459393b2
Seagis-Model/config/a3_quality.json	33	0f4e803b19b129edd3d49be6466bcaf0e22eafb293001290c5893e23614bae5e
Seagis-Model/contracts/alignment.schema.json	81	7cf0f75151aaf3b1e73da9091ad225abaeee979aba5a1a4aa48d3bd5fff91f66
Seagis-Model/contracts/iot-packet.schema.json	19	1a03ae7fad5b1d3a4f328b879cd68e432544ebeed819f77fa97437aa627581c4
Seagis-Model/contracts/normalized.schema.json	918	e3d1605ce38afb2c77cee8ca3a669f3dfbb32ebce927155c65429b74d1b13309
Seagis-Model/contracts/source.schema.json	527	818d451814b97aba7d40d75576d60e5c0a72aea4e110cee900cff795fc7aab08
Seagis-Model/docs/a0_review.md	73	3de9ffcf90fe3abfece563347e5447dfb7bacb56dbc03c265c0d10120025966e
Seagis-Model/docs/a1/iot_handover.md	131	7fbd035f3e1b843ca05b69958f7a442515c51e4df5ca33b36bb4c1a6f0914333
Seagis-Model/docs/a1/operations.md	111	01d9de4516a22f3683b8065a5fd0b3a02941e027bba38e81307b87e821c0c36e
Seagis-Model/docs/a1/review.md	53	b2d95bdf10bff99cf560b93d0a86c2227525f882412d04b1b031d7e3cbdf705c
Seagis-Model/docs/a1/validation_20261022.json	112	1ad7b37395480199d2b6b2fa55cab7455862212901b8474c0e4bbc05577dae99
Seagis-Model/docs/a2/architecture.md	74	530a0c4d72f8d9b05b53b0c09dee3d1c30064a6446a8dba7220fc9bd019f5655
Seagis-Model/docs/a2/ingestion_report_20261022.json	432	894bb8e96fa756341075c760c2841db21ff3d95e098aa44540e7557a29c34ddd
Seagis-Model/docs/a2/licenses.md	27	c1551d495c52a5054f7974d2d7397d1e9ea3f841619b77ed4d504b4570d1743a
Seagis-Model/docs/a2/operations.md	69	d0a4e9894254025cc7514a9fe808dd25bd929c0053c389f8ea05ff4208c2acfa
Seagis-Model/docs/a2/review.md	52	ea22ff34a20cbf1691ecfcbe1b3269aeeb46d5d4d6bb5f450b8f9571667442a2
Seagis-Model/docs/a2/source_evidence_20261022.json	55	a80fd90bae4da2694c049feed7b34bde43ff0ace51d5ddbb66389536b797b0e5
Seagis-Model/docs/a3/architecture.md	59	9e209b5b3840be8dbdeb0948f2faa9c1c479aaa22bcdec70e231c6c0c94dbb24
Seagis-Model/docs/a3/operations.md	99	a5f17669b2400757e76bd6a3fd27caf29a6c6e6966a2bb3bb21d5fc5a7db676e
Seagis-Model/docs/a3/quality_report_20261022.json	1002	2e67a3b630917bda4d20fe4ef031d393d3ba61292ea90de15f4ce4a8a8500223
Seagis-Model/docs/a3/review.md	63	a0d533104fbfbedad18ab435a0a2702364fecd8392e3b7b9983410afbbd5eb33
Seagis-Model/docs/architecture.md	85	b5a7ff5659e4aad20a726218f1333ae814b60963ae5ae272071621b7c9d7f95f
Seagis-Model/docs/data_contracts.md	103	836f6ed8879a23efbd67c24fe199246d065c4aae9b73c662b9a8a47ce15a293d
Seagis-Model/docs/data_dictionary.md	115	175a21dc9d633b89ad0a605c1747296cefa287a9da134b78cac1f0a2d0f4d8c0
Seagis-Model/docs/decisions/ADR-001-data-storage.md	50	bc2fcffc9170b6034ae9707d56b4382fa54ea7a1dfda6a83acb273e4b55d49d3
Seagis-Model/docs/experiment_csv.md	75	67c912601b6c228f0140eea4726cfed34481cf3e6eaa9fe850dc33f4ad8f4b7a

Path	Lines	SHA-256
Seagis-Model/docs/metadata/source_catalogue_20261021.json	79	8ef86e2ca4db05315efb5b4cececc23951c55fa63295ffa31460cef4b96f0eee
Seagis-Model/docs/source_registry.md	54	6310bce3d80e1e7fdea1d2855fd9446febf133902a007af56304254c1a950569
Seagis-Model/fixtures/a1/hourly_synthetic.json	1970	8d833be53dbfe0c2fd5c590830821374f027a5090f8804d2085f686855add22c
Seagis-Model/fixtures/a3/synthetic_bundle.json	670	522bd51062e91a1419bf85f0b62e8b206e5e5208df17f3cf5bd81f9606c39d89
Seagis-Model/fixtures/normalized/copernicus_currents.json	79	c5fb39ed7cccb2dbba94135d3dda19464bf8287b6e082d224f006ab0461c4913
Seagis-Model/fixtures/normalized/copernicus_nutrients.json	79	a30d3dae6c700faa491662773bcda03efe08d0c93d8d136f68c84264f115fa85
Seagis-Model/fixtures/normalized/copernicus_salinity.json	60	9c262f0b2922d3f100a8d35ad326b53fc850e688e2d0d9742c4aeb0aa4e4104a
Seagis-Model/fixtures/normalized/open_meteo.json	155	8b4d8cdd2732ca0b88bcd6db6f920e12b4b6746187bb080aa135dbaadf4e94d3
Seagis-Model/fixtures/normalized/sensor.json	42	7ab0f72245e724d6ad36c5c65abcbdd24775f7708b3846d4c31a4b3b3c224d8f
Seagis-Model/fixtures/source/copernicus_currents.json	48	ee2a87156030ab83902eee430ac9ef0cbf9dc508cf2b48a5981a9780ceded27d
Seagis-Model/fixtures/source/copernicus_nutrients.json	48	5f1424796bf98e90ce129f477d063cbd286a4302f7f53e79869157ba4e342675
Seagis-Model/fixtures/source/copernicus_salinity.json	41	ba795d27a545d2830c2b05d2ffe3a8e9c4fd7c8d72874edc3493195f94436046
Seagis-Model/fixtures/source/open_meteo.json	62	fe3c4e57479ee18e2e295b23a6d4240136f89f6a3dc011950056f4f5fa5ea2f2
Seagis-Model/fixtures/source/sensor.json	20	8c18a6aff2a6b72f77bcce58e8d1a8a4a7cd1d378e2d2d70abea0150a3a9b844
Seagis-Model/pyproject.toml	19	bc2c936d59d74591a7c3714c70201d9583e5723c7a386b0e33d11954b91b4c91
Seagis-Model/scripts/check_a0.py	140	27a2ccc2ca016c93bf0436da5884f6ec32598847856fdcc36f5533a9499f728d
Seagis-Model/scripts/check_a1.py	10	9223b11860bc099ef330c604d51fb6e53afdeec7651d110ab1dcba32ff27b792
Seagis-Model/scripts/check_a1_http.py	161	19e104cc1cdcd406ef60b1c83a95ef5f9d68460c68dda5b7acc73772bcd773c2
Seagis-Model/scripts/check_a1_live.py	109	6c5992eb844030de9dd96f255c273b46d1aed6bbb3e9fc43d13f0c8c066dbe6d
Seagis-Model/scripts/curate_environment.py	63	a16b2cce80ef1bc84d514e616c06609a4ac6ecc3f1a3d40bad1ab5f137018c33
Seagis-Model/scripts/deploy_a1.py	130	38bb8c515829ddfb81d2698d27d2a197a4ec49f5d82ef670ccd8327fe8f09c0f
Seagis-Model/scripts/deploy_a2.py	36	70b9e672dc5ab270984998320c329fe7ba24b0f6700989e8033c8ac461bc0ab2
Seagis-Model/scripts/deploy_a3.py	56	24bd90b43ea4ec355745f335a51520cc693ce249f9c8289e016a335eb876c9fe
Seagis-Model/scripts/environmental/__init__.py	1	1f689d46a84d43a484eae691eb87eb1c5a18fcde06b197d7a02f80441e5126d1
Seagis-Model/scripts/environmental/common.py	158	83237385df19e57171552ab6c42551fcd531d156500780f84e838a964e0adae8
Seagis-Model/scripts/environmental/copernicus.py	133	a55dc9b11c14f9b094e72ffb832844b6cc4f530963270f0f6d98b23801e1ef04

Path	Lines	SHA-256
Seagis-Model/scripts/environmental/open_meteo.py	65	684ab244e25e6eafac81a2f910462fa326444bfdab26a49c06ed450085eae361
Seagis-Model/scripts/environmental/store.py	68	6e81e722210f5d39f15595e0039f16185ef31d18521c8a206f5de5424ea76708
Seagis-Model/scripts/export_experiment_csv.py	245	7a3eea3739e9aa8938a32b403599d911515b3d4fefe86fdb065ef12c7072c4fa
Seagis-Model/scripts/ingest_environment.py	76	10a83e77f4f42ed58095385aedb95aa92e620dd458bc673341de074086952ebb
Seagis-Model/scripts/install_a2_scheduler.py	52	1d3bfd452f5b1a6adf850b7d5169d698967d8691785bec1aa37006995ef36fb0
Seagis-Model/scripts/provision_a1.py	111	224bc7a7ce5dcce83e290fd6261c8ddd6349a86d07e4041ac51d0741405503bc
Seagis-Model/scripts/quality/__init__.py	1	c5b07be1b5223a81fc5ea6a9cf99f794c7a96360a552d23eb3c4300783592ebf
Seagis-Model/scripts/quality/alignment.py	387	dde4eb5b79edfc3fff1e0ed62befdeea58b162541d9e53962f5dc0f508384ca6
Seagis-Model/scripts/quality/store.py	66	204f411108496fe63d242a8fdc6c338f7e17f52def9cc909affef4e010115802
Seagis-Model/scripts/rotate_worker_credentials.py	105	2a8f5cd0606861abb48ffa145bd4a4f2629f244272cfd931f1d1da31edd6aa73
Seagis-Model/scripts/run_a2_worker.py	31	00fb29d0cb12ef882e729d965bbf788aefdb2f47a2597c5d446cd02c1016e3cb
Seagis-Model/scripts/simulate_temperature.py	216	1c452139a18e64d91d4292e0d1370fb010988a58a310a2632c253d312240f4fe
Seagis-Model/scripts/supabase_admin.py	81	ff4f5e9b5d2849a435fb3b8508b11c4e20b255840c4a89c67e10a04b382fa21d
Seagis-Model/scripts/validate_contracts.py	263	bd0c9589beb03ae16c5c74141c739d73c980599eda46fb078578df07cfad2a0b
Seagis-Model/supabase/config.toml	420	5e2f8fab3c2bcd9dd70e290b6a739257df5d8e2b5729084bb4314174fe9fcceb
Seagis-Model/supabase/functions/temperature/connection.ts	27	9eb48afaba8117353276eabfdfc76a37b24c3d1eaac576049c233fc564b27f1a
Seagis-Model/supabase/functions/temperature/deno.json	15	a16b21ff8a72b6d280187e532a40636be001322c6c86fda1df87485160b8feec
Seagis-Model/supabase/functions/temperature/handler.test.ts	88	7f231639b4160931ce21e2f32bdf8f420ba83a6bb5ab1c219c46f865bf7067e7
Seagis-Model/supabase/functions/temperature/handler.ts	113	e8391e427cab3979b1076a4aba695da63072a770923d6f7716ef6d4776d08bd7
Seagis-Model/supabase/functions/temperature/index.ts	31	1e74cd3a0eadd83a8145b352603fd9ef3c116057c7663d7183327c5c09fa1b44
Seagis-Model/supabase/migrations/0001_initial_schema.sql	544	f83e16be7d676cd1552266f2648444edbe12c01aac6b6ac94d4c6d32161c9b42
Seagis-Model/supabase/migrations/0002_temperature_api.sql	297	c1547f70b3bf29ea5a54171ff547177713bbd0f4d9bac8e33d47d1c940fd992f
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql	261	d49a5897bca967b33e3ab79745bb4f1bb9e76babc318544bfc7c024c8993d1f5
Seagis-Model/supabase/migrations/0004_provider_update_status.sql	15	bf92788ed13dd0ffd60a8f21b25f84627976f5e9b46e27f98b00f4062a4cbe0a
Seagis-Model/supabase/migrations/0005_quality_alignment.sql	249	afc5d564ed54adf9cf353865f0859f414469016b02823707f3d872453608da65
Seagis-Model/supabase/seed.sql	10	4809f15911388f46d0b94fcc09d413e341f7aadf4eb67fb88dadf6e6f0748f79

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Seagis-Model/tests/bootstrap_roles.sql	8	4fe6ea493501b15e67aa693fce29a08f24096f1c95d5f6f52af79e3943338f08
Seagis-Model/tests/test_a1.py	268	49ffb98dbf091d5343be35d7f0ef532e27beccdef5af9b2a343bc30b1e8e758b
Seagis-Model/tests/test_a1_tools.py	85	e9a79d56845040e695774e8205a2aa5ef9ea85bc3dd6448f07479ffc5e0a0bb9
Seagis-Model/tests/test_a3.py	418	607f4ec2cf2bb737a93d4412294da99b49d5bfea04ff1ae886d677f23a2f1df1
Seagis-Model/tests/test_contracts.py	108	af73d14cd7b5d49e91555b1040a9520d2ca2eb2ccc7def15f40e59ab31296ea9
Seagis-Model/tests/test_database.py	489	19be7574738537be1358ed6b36ca1cf586be5affd9e9b846f92032e755d56e44
Seagis/AGENTS.md	28	9aeaecda1a0926efa09176c2adbb543b06c4c7041d7fed671cb7a43177fee1e2
Seagis/README.md	124	c40fef74d33539a20f283c742175e2f1fd4a747e4d909a9aa6cc152ef29682ba
Seagis/SPEC_REFERENCE.md	25	dc29fa4fcf7879059b83c255fd604ab0a6d8eb7376deb2a0114ca544db329c8a
Seagis/Seagis/App/AppContainer.swift	20	30f78023407d3e3f85dd4f1f3de7be9e8c0db823843ac2435f3134ebc0f85421
Seagis/Seagis/App/SeagisApp.swift	10	d6f0f8373869116965c49bbf404a64a1138cc939fd2a6703cdf4ee71ab99797a
Seagis/Seagis/Models/FarmMap/FarmPlotDisplayStatus.swift	25	30863de6a7400f137297a0668d175b3c5814f7db7720ad3699325b56ee07a43e
Seagis/Seagis/Models/PlotCapture/CaptureMapArea.swift	20	c58772b18a16045b75b00f60e86a56a2f034e3da4ba0cb80d4b3c47b030f169f
Seagis/Seagis/Models/PlotDetail/ConditionDay.swift	16	10924ce38ee6f243f009d260ca0e099bc7cdd05f12b07c02ffce73009f78fd9e
Seagis/Seagis/Models/PlotDetail/ConditionPoint.swift	23	916c2b770f0bf4311257d690d03b3fe2091fa8e9e54f7b41428bae9ade438249
Seagis/Seagis/Models/PlotDetail/ParameterColorScale.swift	64	182ab73e68f99e7b1af35d363bffd616a8b810401e54a4df75c43f45148ddcb8
Seagis/Seagis/Models/PlotDetail/PlotDetailStatus.swift	30	1701f9fca3172013ffdc396ee32f5b55cc10db9c12a1c7e8976b0602bffbf50e
Seagis/Seagis/Models/PlotDetail/WaterParameter.swift	55	1b1c583b79a385e9b1327ba5466dc0c1fb7eea2e2ae3d00e3f58d7f6a2256137
Seagis/Seagis/Models/PlotDetail/WaterParameterInfo.swift	32	10782cf6ba9b090cc5e8acd5c0614f2b76813b650dd44462a1f49207e7b744f8
Seagis/Seagis/Models/Shared/Assessment.swift	25	349be21ee4ddaf9dec672875837b2876893658b53c5343ac9a67dc4dda07a70e
Seagis/Seagis/Models/Shared/ConditionWindow.swift	37	22019f46b70b9ef012f97407e7df3dba338baa6dcd4f79419a2ecba3e2478a54
Seagis/Seagis/Models/Shared/MetricSummaryItem.swift	56	9705a185e0d121695da2018450df2f49662e68834a7c9d752b5968e5dee46d2d
Seagis/Seagis/Repositories/Shared/AssessmentRepository.swift	17	37c4c063a83327523bb7d3ab186cddf6c780d8c651e498f8fe6165425e52df2f
Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift	21	41618019f73ba0774b04dd43fb2b1643f9b11e17ba7caaff41811907bee4effc
Seagis/Seagis/Services/Intelligence/InsightGenerator.swift	6	5cde5f99c69be2fd73e14f8984928dbb930a732e308429542cb22d673bdafcdc
Seagis/Seagis/Services/Intelligence/TemplateInsightGenerator.swift	13	945cfd6d3ef1e12ed8d3eef12d58e8a7157618a969d2c7485a0ad9eeae2eae17

Path	Lines	SHA-256
Seagis/Seagis/UseCases/Bootstrap/LoadDemoAssessment.swift	13	657ae092c4e13b79772fd05be04606b860cc3897574068535738ac3f6652b9f9
Seagis/Seagis/Utilities/PlotDetail/ConditionDayFormatter.swift	26	7ad2a51263fee6b90eaf933a6ebf0457a776214011e47f93d76c57ab64e44f1f
Seagis/Seagis/Utilities/PlotDetail/DailyConditionSummaryFormatter.swift	29	98775dad31d7afa55b4bb62b076b0ddcd5e11a167daeaad56fb15f254c37bd52
Seagis/Seagis/ViewModels/Bootstrap/BootstrapViewModel.swift	33	bc8f0d8ee67ae582bddfefe68043892be053d75e8083226d19423f871998d142
Seagis/Seagis/Views/Components/BootstrapView.swift	75	55d25cf1d0f7a6a8786a918c46ed9b989b11b649c19eee9425fbb757ba747b75
Seagis/Seagis/Views/Components/ComponentGalleryPreview.swift	92	c3a459bc889713e0201bbe669012e5244f5c6c41a719a15419bd56e9ce566a07
Seagis/Seagis/Views/Components/FarmMap/FarmMapSheetHeader.swift	165	4ed2fca1dbaaf83fcf9a41ff4e2c41fe5c72b38b487e54774a145f61a0c7ff7d
Seagis/Seagis/Views/Components/FarmMap/FarmerRow.swift	95	4e21e4e2b7620a06a04132d07af57f3c55e523b1892c02dd995b667526cdbb92
Seagis/Seagis/Views/Components/FarmMap/PlotMapLabels.swift	102	5c87d86f283db0fc7e590dbc69da2f0d346b2923d36b86c52ec09efc90c77fbb
Seagis/Seagis/Views/Components/FarmMap/PlotRow.swift	60	6c1dec4c8355c3f7df0787275d46d54723b93115fc7d70c955b6b9d3ae47388d
Seagis/Seagis/Views/Components/PlotCapture/ARSceneView.swift	56	34785ca13eecd86ab33b063f8e4dcf6c2035409b4b09c0e898270dacf9532a8e
Seagis/Seagis/Views/Components/PlotCapture/CaptureChrome.swift	223	bb7a4f5b14b98aef0fdb7a02985255be9d494dae2cea391c5840d1c291a5e60f
Seagis/Seagis/Views/Components/PlotCapture/CaptureControls.swift	141	d7cb9b0d0c169567c1e21788cd04bea0da420f5e24a4e533c58e8a1a68c44761
Seagis/Seagis/Views/Components/PlotCapture/CaptureMapView.swift	364	de0facdb408e2cda3cbe8701783863accd8417558486de75780324cdfa304d54
Seagis/Seagis/Views/Components/PlotCapture/CaptureStepsOverlay.swift	80	e643bea1cbd3373a5a22cac83cf7b93846f4a1332a001c0007093d643824fb84
Seagis/Seagis/Views/Components/PlotCapture/HatchedOutline.swift	61	4fdacc0fd4b5ff87139787b8020605536ea141107e294923bca595a71ac67bc0
Seagis/Seagis/Views/Components/PlotCapture/OutlinePreview.swift	110	98724e172d88169bcf13fb11453d5cdbea88c1a67f8ed2e64faee8da72184195
Seagis/Seagis/Views/Components/PlotDetail/ConditionWidget.swift	91	732c4cb26477995528a12f41f4728ffa77d18a7ab3206abc2e28c561d8bfe31d
Seagis/Seagis/Views/Components/PlotDetail/CurrentWidget.swift	85	48d87c6ff150c26d8e27bafad217c6114a1d616fe9bfac6d035c1450bd7c4122
Seagis/Seagis/Views/Components/PlotDetail/GaugeWidget.swift	48	7655e21e12000acbc45586fd5e7a349c4f132be1fe51663e743be0e14ec5e302
Seagis/Seagis/Views/Components/PlotDetail/HourlyConditionChart.swift	96	c0ebef731591ac7c84c634950e0ca0352ce52a1454b2cecbb8d45f3fa49b1cf0
Seagis/Seagis/Views/Components/PlotDetail/ParameterDayCarousel.swift	73	525e536ecda2a4dfddc2d7940a27bef59f18f6f798b97dad78ad35a507ab5c21
Seagis/Seagis/Views/Components/PlotDetail/PlotDetailPreviewData.swift	29	d2d0b94eee2d321b75d4f0ebc338032172a423e54bce95272f1783920e4b441e

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Seagis/Seagis/Views/Components/PlotDetail/PlotOutlineShape.swift	30	d9f42846399d4c218fa4efabf55943c4bd5e46e31ff520dcf910d9daaa3b9f52
Seagis/Seagis/Views/Components/PlotDetail/PlotStatusCard.swift	88	4b9e3437150e8013e5154079f647c97e24d3d6344f3fe6b60cb259f2fc51b4ff
Seagis/Seagis/Views/Components/PlotDetail/TemperatureWidget.swift	95	401025ba3aa721025753419da186b8fcbda7ee3bcd23e91c45d12e4a218b6935
Seagis/Seagis/Views/Components/Shared/AdaptiveGlassModifier.swift	43	070be23bb434650845804f3227749464f01283aec5e22c9083f68b2d9b0122c2
Seagis/Seagis/Views/Components/Shared/CircleIconButton.swift	125	9f8458edcff549fd16c0df7d9fccdca5f28990e7de4fe71cb2a252cd2069d276
Seagis/Seagis/Views/Components/Shared/StatusBadge.swift	115	0f7de149618bfe6c82e7ed946f70de7a8d09c18269da97307f917d8cb60a9348
Seagis/Seagis/Views/Components/Shared/StatusBanner.swift	93	7eab25a99e8c8c006c060ebd8e6ac33ea262cc6ae67f1e5bc135691dff7360f1
Seagis/SeagisTests/Services/Intelligence/TemplateInsightGeneratorTests.swift	15	1a23b61f6a886ec61810f3e60725321fec4b4f29dba2ed93355f4bd839488e99
Seagis/SeagisTests/UseCases/Bootstrap/LoadDemoAssessmentTests.swift	17	f12da530e126fbd6e0eabcb87b1706b5de3b9bd1b26f252d769f680ef5259ae2
Seagis/SeagisTests/ViewModels/Bootstrap/BootstrapViewModelTests.swift	47	29c0ee996c8aba71f50e15936b01affa343ed2c4abc989851b7c1dee389c4368
Seagis/docs/architecture.md	67	9a3a73e6763f42c40d2609e8ff35f9e1680894340a9fc560ab5855662bd1b1a2
Seagis/docs/decisions/ADR-001-engine-placement.md	27	0292e6f7fbc53ea13e987c619e862f4a58d3429a4bdf4c4ba27621d3480b7b0f
Seagis/docs/development-workflow.md	27	b6202666a9d68357b8d07f8aef9762410319be76d3eecb1d8fe8addbd28e4e73
Seagis/docs/integration-boundaries.md	29	a214c3e3b6432fcbc27afed03009e95d854cbf3b03dee4c013f8dddab5242a61
Seagis/docs/ios2-scaffold.md	85	377a42d94241f06d4b5adb8588c746a6992c4cdc1e6024026c28941c4c47e34b
Seagis/docs/ios3-reusable-components.md	1347	c866cb9afc1d1c7a30501c015a90f4410a4f6e2f5262cbf0f48c63d0384dbab4
Seagis/docs/iot-dataml-architecture-mapping.md	149	200129d604a399f5da6e0f15928c2bc5f7583484b48f2adcb0e992c24d0ab088
Seagis/docs/open-decisions.md	17	3a1c1673c8337da046d48935f57b97c0347c7fc01501eab9b02b1cf85196fcd1
AppPOC/AGENTS.md	9	39ac4b865dfefe8101538df5c7703e9dc039e7da7ca2c0b7ca873ec0470c8919
AppPOC/Documentation/MODEL_CARD.md	29	86ce30190d9d267bb6757d57e857f6120411178ab0ef96ca063a40427aa07e5d
AppPOC/Documentation/SOURCE_REQUIREMENTS.md	150	3c130872b1e857653e8ff47af3f9d82e571e7ad0e1655860f69d7d263d3896ed
AppPOC/Documentation/UI_REDESIGN.md	34	52013450c9b0e04a263a452cfba15f8553272dc07a353356a57734c991342b4d
AppPOC/Documentation/VERIFICATION.md	25	5237b4ff2b43b40fc98cf68647c485e79787d75dbc8e9c884f43e3c75494f494
AppPOC/POCApp/App/POCApp.swift	9	8e14c4b8dde5c3b11de73ed22dddf1861cf2d69bd382e8dc9b51e065ad388ab2

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AppPOC/POCApp/Features/FarmMap/ViewModels/FarmMapViewModel.swift	151	e6eb210fc07dfec6f38df729129707343905498d8c30e23485eefc0636e11881
AppPOC/POCApp/Features/FarmMap/Views/FarmMapSheetHeader.swift	53	296dc9226891ad521b2df0c18e8d1190d57782be1d800d442a91f5ee9a6c60be
AppPOC/POCApp/Features/FarmMap/Views/FarmMapView.swift	127	06f85b64f137a63b64cd839fc39fead1b1f5824e1258cb9582f00cba08e9b134
AppPOC/POCApp/Features/FarmMap/Views/FarmerListView.swift	22	ee5b06f9d4aa1d1e3ed1eab2aecd0925f5c01480abdf0f692a49e0e38e25e77d
AppPOC/POCApp/Features/FarmMap/Views/FarmerRow.swift	29	2bb17c91bd9ee33de4929f1070a7d544415068fe75035d9c919598f044537285
AppPOC/POCApp/Features/FarmMap/Views/PlotListView.swift	47	47f3db3934b6aded271d696420487ae58b58a081512e2fceee9a746f698ec578
AppPOC/POCApp/Features/FarmMap/Views/PlotMapLabels.swift	46	9c00ca470f4921b0f440573f9f54b26b3b518f0654adca72db7f5b991aef7389
AppPOC/POCApp/Features/PlotCapture/Models/ARSite.swift	110	c68a4dcff8c56b8094e70deb68ee43821040e8b8383caab967d97a74c340a57b
AppPOC/POCApp/Features/PlotCapture/Models/CaptureArea.swift	44	f6709a92b1898e1fddb15e31c73224830af42342444bb95fe81f60c16b7514e0
AppPOC/POCApp/Features/PlotCapture/ViewModels/ARCaptureViewModel.swift	445	673b9744f7e416d9bea503f09caf54d848d49468a59d24a8000ec94ac301362b
AppPOC/POCApp/Features/PlotCapture/ViewModels/CaptureNudge.swift	20	b3987dab6afbc02080a2450f0297ec5c028f9c3cf2b6c35688b1a3d09f3f8625
AppPOC/POCApp/Features/PlotCapture/ViewModels/GPSCaptureViewModel.swift	122	885bef577472ce779e1e296f98659ce30b9be3d086d13d29bb8686608e47a2fa
AppPOC/POCApp/Features/PlotCapture/Views/ARCaptureView.swift	71	3dd05c13eea600040c88d516c1224af4533fdf6bbff8b4ec003315f3ab19eaf9
AppPOC/POCApp/Features/PlotCapture/Views/ARSceneView.swift	19	16c02fd37f704d0840992a1a34f3beb27b03c37a5a77d732f2558a45edf2c667
AppPOC/POCApp/Features/PlotCapture/Views/CaptureMapView.swift	201	c60881ad2bd06fbd2e603a07bfdc1e783d5a13baf86e74a02a88b38d2b190ecb
AppPOC/POCApp/Features/PlotCapture/Views/Components/CaptureChrome.swift	56	5d311360e4bcbd2d5750b02d513fe5e9c3129398c3e9fb705f7a4f6f80cccb77
AppPOC/POCApp/Features/PlotCapture/Views/Components/CaptureStepsOverlay.swift	45	ed7dfe27275a27e141a7783eb8cf4130922c673f1de10a78e7dbda9000528438
AppPOC/POCApp/Features/PlotCapture/Views/Components/GlassControls.swift	82	59cb3693cf0192152574e8baba54d830909890dfb3a5cf44ea04499c8c483af7
AppPOC/POCApp/Features/PlotCapture/Views/GPSCaptureView.swift	125	94cc2cff9e9c98e7ef305be4002e2260e97fe9fa784c38e7d3e4cfb07f089da8
AppPOC/POCApp/Features/PlotCapture/Views/PlotCaptureFlow.swift	49	2e36716eba568cb10ce43f290824045c4da81731835581c1c030cb124c91aa91
AppPOC/POCApp/Features/PlotCapture/Views/SavePlotSheet.swift	191	f3c02cba58ab38c054d254de7bdd2a10d972a1941489cebe5f3511f168eecdbc

Path	Lines	SHA-256
AppPOC/POCApp/Features/PlotDetail/Models/DailySummary.swift	44	893581a30491604589955e6902042ff015b465c235cdb73c984ffd9f87af441b
AppPOC/POCApp/Features/PlotDetail/Models/ParameterColorScale.swift	60	8ef76246dead359265e5c94ce1105a757c271c8183b4632ae785c1baed4e1848
AppPOC/POCApp/Features/PlotDetail/Models/PlotConditions.swift	53	680e31bb9aab5efad8d867b0e02255adad2247979b225837377fa6b34d26df1f
AppPOC/POCApp/Features/PlotDetail/Models/PlotStatus.swift	40	8f70f34713993f4392eb319f854bbe7817c6872ead25547bf4e6f5f1d06174ec
AppPOC/POCApp/Features/PlotDetail/Models/WaterParameter.swift	61	78a6dca94962d9e271c77fc2ddf3ef6692eb9987e796b46e8bd43a4207735564
AppPOC/POCApp/Features/PlotDetail/Models/WaterParameterInfo.swift	34	3b4582d97e9394fc28c73491127d5cbbb8eea446e03b8aa7c2ce145e4a452e5b
AppPOC/POCApp/Features/PlotDetail/Services/WaterConditionProvider.swift	52	e4db4bc86a9737e1ae041b1e8a89eb0f5b02d91d7ae7e3602cdf3028d4beb125
AppPOC/POCApp/Features/PlotDetail/ViewModels/PlotDetailViewModel.swift	41	cd36ed23c519d7e076cc32e77061febcd37396a624928dc38373cff1ebdc7ba1
AppPOC/POCApp/Features/PlotDetail/Views/ConditionDayFormat.swift	31	2557c452929fb3da14a8b4a044b526245b3af87bbc1ab81b11cb69456c176ef5
AppPOC/POCApp/Features/PlotDetail/Views/Detail/HourlyConditionChart.swift	115	4258f8d2bc53cf3fb7b9cc2b2ba4f7df73390f8ea920e16a9822cd91afb0a3d3
AppPOC/POCApp/Features/PlotDetail/Views/Detail/ParameterDayCarousel.swift	59	ff39a72b4e7b4a287c311f243836cbac1a0be3742eb860da13e753f6b0cdfd51
AppPOC/POCApp/Features/PlotDetail/Views/Detail/ParameterDetailView.swift	126	b3cbd2b5bd967678b6c863a89d9275d7553ddc80f864e93fd0cedd944e5e3fc2
AppPOC/POCApp/Features/PlotDetail/Views/Detail/PlotOutlineShape.swift	19	e59eb2b7041c030193659355a9f8f65604b54fec564ff6097cd85296fb8a4dbe
AppPOC/POCApp/Features/PlotDetail/Views/PlotDetailView.swift	57	8d076b7b181e40108086fe24a57d9ff04cb84dce3fdba2d07b167600065980c7
AppPOC/POCApp/Features/PlotDetail/Views/StatusCard.swift	83	4a6fe7b331dcb0accd4ccd1cb784e47ae1243a78e45a0b7a62d0a6409d845b52
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/ConditionWidget.swift	76	7b8865207bb5d336feb7dd3ac4b93e596723863833d563647aeb01a79ebeae88
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/CurrentWidget.swift	74	3334408a2ecc5d412aaf2cb6f297fc0f940c3e58428f5f5b0b38c87e582c65c9
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/GaugeWidget.swift	30	498490bb259c057497df3f47bd7a108447b14111efef2a932f71c8e340d9cf7d
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/TemperatureWidget.swift	100	59c03a89512341b8f2fd2c7c9852a7943fcce45de6c142c5d674d9cfd126633d
AppPOC/POCApp/Shared/Data/FarmStore.swift	56	35e892e5fa3bc5c8e7a6bcc6b50d03e2fa02712366099d48b0416597b1d3b78f
AppPOC/POCApp/Shared/DesignSystem/CircleButton.swift	25	b5032491bff7586d8b555e4e3e1a741592e357304a699b650026cc6b7bff13b2

Path	Lines	SHA-256
AppPOC/POCApp/Shared/DesignSystem/Glass.swift	13	e53326f3fdd43b3043bedfe64dde78791469d0060e8407b99f022646fb4e43d9
AppPOC/POCApp/Shared/DesignSystem/Marine.swift	17	09771fbfbbb29f7c24fb72c2f85c496d43ea3523c6729c3cefbeee8b6f171061
AppPOC/POCApp/Shared/Geo/Geo.swift	102	5a98921b657536f8ad0e25f24e9deda3655415d0fcba80dc185538145f364928
AppPOC/POCApp/Shared/Location/LocationManager.swift	77	e907be2920f002826f63db9988e95b67521888a9f75e304c088137226fe116b1
AppPOC/POCApp/Shared/Models/CultivationPlot.swift	134	beb25cbb8ab28def3e2529dcae63974825b4767a31a51633358f3723cde23df6
AppPOC/POCApp/Shared/Models/Farmer.swift	27	4497a6d1e7caf1af6344e1700d4b52b6dfe05a16432b038054509b1fad589a5d
AppPOC/Packages/TidungCore/Package.swift	3	44dfbfd9d7b80fb3c3995154378836659ea002a57425bb7006c2cb79405a484b
AppPOC/Packages/TidungCore/Sources/TidungCore/Models.swift	112	163620afbf41029cc36b880fdde51076991c69adeeecf8899347bb0c7700f8a5
AppPOC/Packages/TidungCore/Sources/TidungCore/NASAPowerClient.swift	85	166aaf4d7e9a79f0ce49b077b3c93961210c1bab2af08c4535737e19a8aaef48
AppPOC/Packages/TidungCore/Sources/TidungCore/SeriesAnalysis.swift	25	577b6b80d785724744f6c26c4cec939e36363067613b464c8228a4cbbf779342
AppPOC/Packages/TidungCore/Sources/TidungCore/SolarModel.swift	44	2b5c297043fd37d50c141b8edcafaf2a381596c6f452b80ccd87de824b15e41a
AppPOC/Packages/TidungCore/Tests/TidungCoreTests/TidungCoreTests.swift	92	7cd85a87fd93510f3e9e8dc73547a762313fe40eed8e7e12411dc478bddfc72d
AppPOC/README.md	114	7b87e44385e144ae3332357b22ceadb26f89295bcdccaca1444b5e1b4396afc8
AppPOC/TidungMonitor/App/Farms.swift	63	41f88166cc3201e029873c9aa866c685231e70ae2aea25eb1dd4e5cf9b3d5c75
AppPOC/TidungMonitor/App/MonitorViewModel.swift	58	8527556638d36d5ebc6d1710aa0df3192e24c212a76bec4fe6761fecb9d32341
AppPOC/TidungMonitor/App/TidungMonitorApp.swift	89	21d94c9480a177df12654b298ae605a2a69ec8d483af263c052a2d6eb68a9aea
AppPOC/TidungMonitor/App/WaterConditions.swift	114	6cf2c87e96e463862a1558fa813bb747332531a669257e6f5a5407a3f59f697f
AppPOC/TidungMonitor/Views/ConditionCards.swift	250	0bd541537b7b04aa4023ee6da43eb87572060737fe0e266e7aa51e99450fe17b
AppPOC/TidungMonitor/Views/DataSourcesView.swift	35	8673864e9e44548b37d752ca53b468164bc78ae12adae8688a90484392103942
AppPOC/TidungMonitor/Views/HistoryView.swift	127	9ef084f459060ab27f7b8764e144dfbcc0e6e25047d63f7e9ac92f5a40f0aa2f
AppPOC/TidungMonitor/Views/ModelView.swift	32	1ba7dcf9e3286a29b40f5cc82e0808587dcaa634ab06b3015f7ea24297f4abd2
AppPOC/TidungMonitor/Views/OverviewView.swift	266	402c6ffb87943bed0d7775ab10b62cd9c062c44ed7de857a5dcc987cd8b79787
AppPOC/TidungMonitor/Views/PlotDetailView.swift	76	5c9a4650ca71eccd92be843424225b448d5d6731757124d058d82ae309b982ac
AppPOC/UITests/TidungMonitorUITests.swift	73	70b60d71083ec127a51ddff7482f006d12985c66efbbd5ff4e7487e1574c3f7e

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